

Mode Collapse in RLVR & ModeBench

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Abstract

Correctness alone does not reveal how many successful solutions a model can produce. We introduce MODEBENCH, five executable domains that identify task-defined verified modes. Matched replay comparisons improve average held-out correctness and sampled breadth across three model scales, with five-seed MaxRL comparisons complete in every domain; harder-task coverage remains partial. Hosted evaluations also find accurate but concentrated outputs. Categorical analysis explains how binary rewards can amplify common modes and how fixed-bank replay can oppose that pressure under stated assumptions.

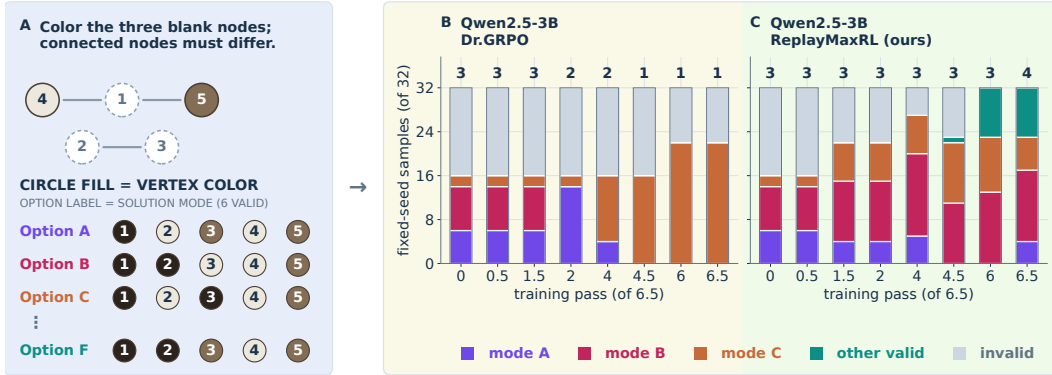


Figure 1: **Correctness can rise while sampled alternatives narrow.** One graph admits six valid colorings. Bars pool four eight-sample draws from matched Qwen2.5-3B runs. At pass 6.5, Dr.GRPO produces 22/32 correct from one mode; ReplayMaxRL produces 32/32 across four. This illustration changes both the fresh objective and replay; Figure 5 separates them.

1 Preserving alternatives for mathematical agents

Mathematical agents can check alternative computations only if the model proposes them [Brown et al., 2024, Snell et al., 2024]. In reinforcement learning with verifiable rewards (RLVR), binary correctness feedback rewards familiar and new successes equally [Guo et al., 2025]. Accuracy can therefore improve while sampled verified outputs narrow (Figure 1).

We store verified outputs by identity and revisit them during training.

2 MODEBENCH makes output identity executable

MODEBENCH assigns each accepted output a canonical key defining its verified *mode* (Figure 2): computation routes in Countdown/MathIR, tested behaviors in Python, and configurations in

18 Graph/PantryPlan. Aliases merge; distinct routes can share an answer (Appendix B). Sampled
 19 modes do not identify total support or latent reasoning diversity.

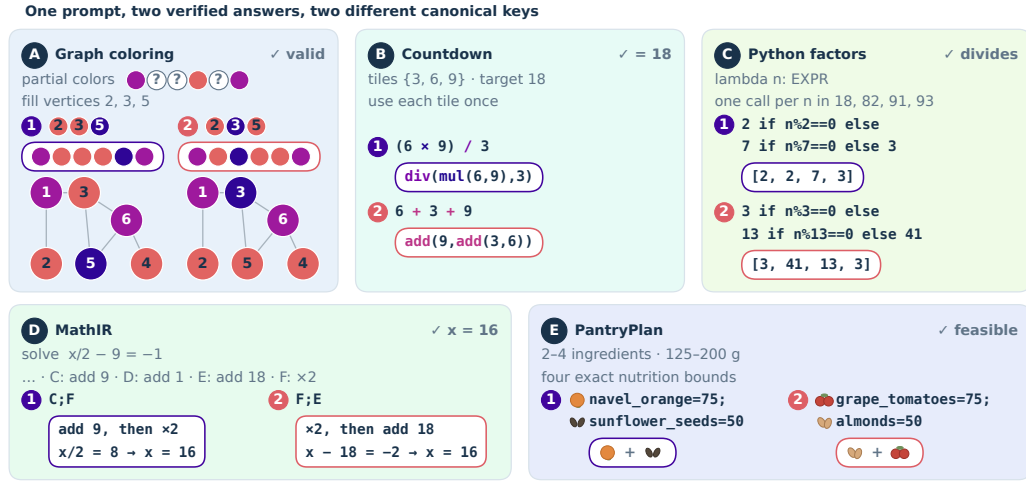


Figure 2: **One prompt, different verified outputs.** Canonical keys identify valid configurations, computation routes and tested behaviors. They merge cosmetic aliases, without identifying latent reasoning.

20 3 ReplayMaxRL: fresh learning with verified memory

21 ReplayMaxRL combines MaxRL’s sampling-aware learning from new samples [Tajwar et al., 2026]
 22 with replay of earlier verified successes (Figure 3).

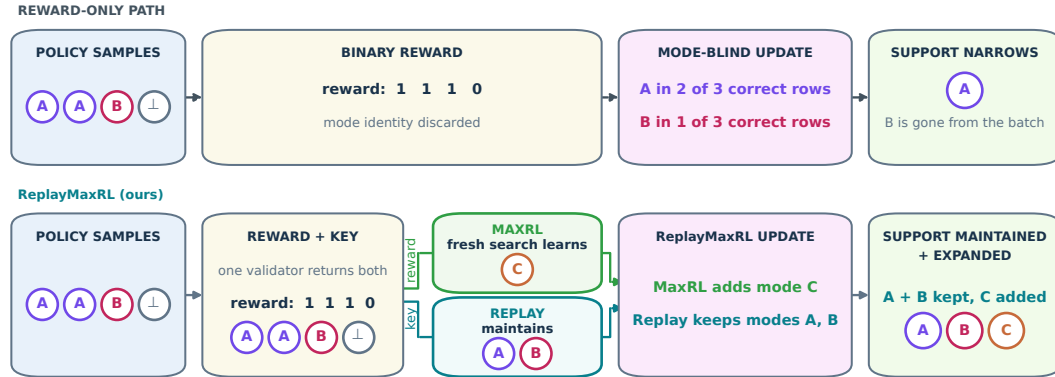


Figure 3: **Discovery and preservation use the same verified samples.** MaxRL learns from fresh rewards; replay revisits discovered modes. Memory cannot protect a mode before it is discovered.

23 For bank $\mathcal{B}_x$ with one policy-generated exemplar e_c per admitted key, replay averages token-
 24 normalized log likelihoods:

$$L_{\text{replay}}(x) = -\frac{1}{|\mathcal{B}_x|} \sum_{c \in \mathcal{B}_x} \frac{\log \pi_{\theta}(e_c | x)}{|e_c|}. \quad (1)$$

25 Equal weights revisit keys regardless of discovery frequency. ReplayDr.GRPO uses Dr.GRPO [Liu
 26 et al., 2025] for fresh updates. Fixed-bank replay bounds probabilities away from zero in categorical
 27 mean flows, not neural training or observable sampling (Appendices L.4, D).

A ReplayDr.GRPO (ours) – Dr.GRPO across three model scales

	Δ pass@8						Δ distinct@8					
	Graph	Countdown	Python	MathIR	Pantry	Average	Graph	Countdown	Python	MathIR	Pantry	Average
Qwen 0.5B	+0.65	+0.19	+0.36	+0.28	+0.21	+0.34	+2.11	+1.15	+0.40	+0.32	+1.02	+1.00
Falcon 1B	+0.15	+0.31 [†]	+0.98	+0.64	+0.12	+0.44 [†]	+0.68	+0.42 [†]	+1.11	+0.66	+0.94	+0.77 [†]
Qwen 3B	+0.15	+0.14	+0.59	+0.41	+0.11	+0.28	+0.58	+0.33	+0.78	+0.43	+0.67	+0.56

B Initial reference + alternatives – terminal Dr.GRPO at Qwen2.5-0.5B

	Δ pass@8						Δ distinct@8					
	Graph	Countdown	Python	MathIR	Pantry	Average	Graph	Countdown	Python	MathIR	Pantry	Average
Before training	+0.13	-0.38	-0.17	-0.29	+0.38	-0.07	+0.24	-0.38	-0.17	-0.29	+1.90	+0.26
ReplayDr.GRPO (ours)	+0.65	+0.19	+0.36	+0.28	+0.21	+0.34	+2.11	+1.15	+0.40	+0.32	+1.02	+1.00
MaxRL (no replay)	+0.21	+0.10	+0.00	-0.07	+0.01	+0.05	+0.35	+0.13	+0.00	-0.07	+0.01	+0.09
UCPO	-0.00	+0.13	+0.00	+0.03	+0.05	+0.04	-0.00	+0.18	+0.00	+0.03	+0.05	+0.05
RLEP-Dr	+0.10	+0.10	+0.00	+0.13	+0.00	+0.07	+0.15	+0.13	+0.00	+0.12	+0.00	+0.08
Semantic-MaxEnt	+0.00	+0.14	+0.00	-0.12	+0.20	+0.04	+0.00	+0.18	+0.00	-0.12	+1.12	+0.24
GRPO	+0.04	+0.16	+0.00	-0.05	+0.02	+0.03	+0.05	+0.17	+0.00	-0.05	+0.02	+0.04

Figure 4: **Replay effects on correctness and sampled breadth across scales.** (A) ReplayDr.GRPO minus Dr.GRPO. (B) Qwen2.5-0.5B alternatives and the initial model minus Dr.GRPO. Daggers indicate four pairs; otherwise five. Averages use shared seeds; intervals: Appendix E. Models: 0.5B, 1B, 3B.

28 4 Does memory preserve verified output support?

29 We report the chance of any correct output (pass@8), mean distinct verified modes (distinct@8),
 30 and their difference, *extra modes* beyond the first success. Counts remain coupled to correct-
 31 ness. Saved-output collision also detects training-induced concentration and replay reductions in
 32 Graph/Pantry (Appendix F.1); finite samples do not identify total support.

33 Five registered seeds train for eight passes. Each domain has 128 held-out prompts with four eight-
 34 sample draws; none enter the bank. Controls use the same bank code with zero replay gradient.
 35 Figure 4 also compares GRPO, UCPO [Lochab et al., 2026], RLEP-Dr success replay, and semantic
 36 entropy (protocols: Appendix B).

37 **Replay improves average correctness and sampled breadth.** ReplayDr.GRPO improves both
 38 domain averages at each scale (Figure 4). Uniform key weighting adds .32 extra modes over frequency
 39 weighting, on average at Qwen2.5-0.5B; its correctness contrast is inconclusive (Appendix E.6).

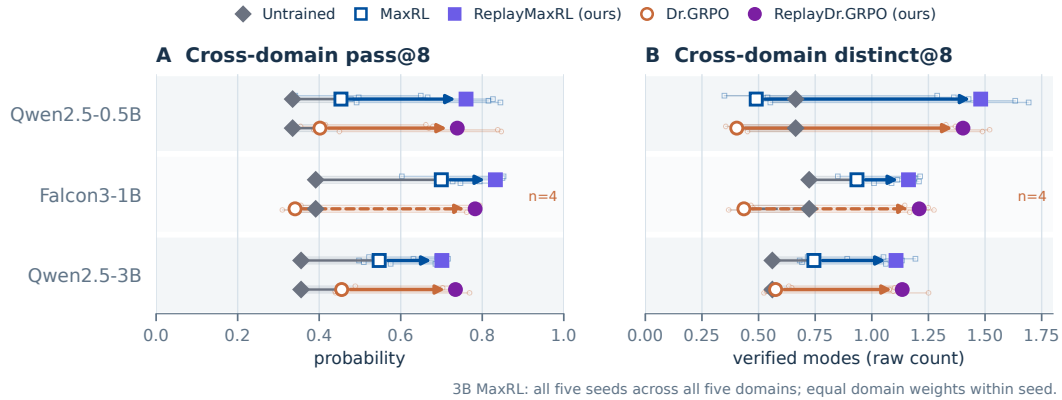


Figure 5: **MaxRL and replay at all three model scales.** Tracks connect initial, trained, and replay models. Complete tracks average within seed. All MaxRL tracks use five paired seeds in every domain, including 3B; Falcon Dr.GRPO uses four common seeds. Archived models.

40 **Replay complements both fresh objectives on average.** Adding replay improves average correctness
41 and sampled breadth under both objectives at all three scales (Figure 5). All 50 Qwen2.5-3B
42 MaxRL endpoints are complete: mean correctness rises $.547 \rightarrow .701$ and modes $.744 \rightarrow 1.107$
43 (Appendix E.2).

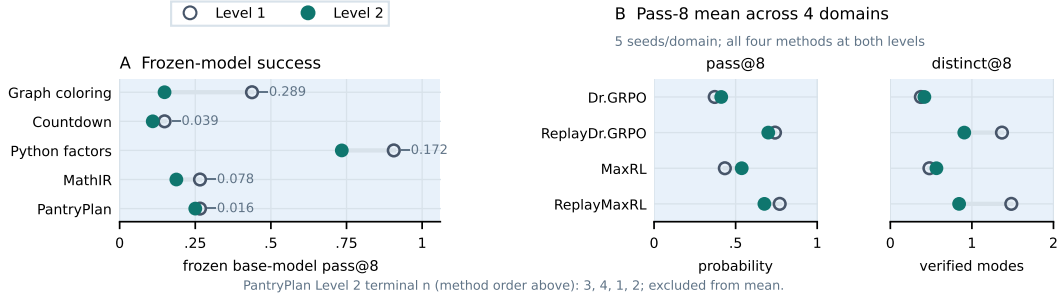


Figure 6: **Replay across difficulty levels with Qwen2.5-0.5B-Instruct.** Open/filled markers: Level 1/2. (A) Frozen-model development admission. (B) Terminal means weight four complete domains equally within five paired seeds, shared across both levels and all methods. Pantry arm counts remain separate (Appendix E.3). Archived models.

44 **Replay gains extend to the harder-level average.** Level 2 retains validators and keys. Its con-
45 struction reserves differ from the native Level-1 terminal tests. Four complete five-seed factorials
46 (80/80) show positive mean replay effects on correctness and sampled breadth within Level 2 (Fig-
47 ure 6); PantryPlan has 10/20 endpoints. Benefits vary: Countdown adds .017 extra modes under
48 ReplayMaxRL, but its correctness interval includes zero (Appendix E.2).

49 **Accurate hosted outputs can remain concentrated.** Seven deployments vary in breadth across five
50 domains and three levels; accurate Graph and MathIR outputs often revisit few modes (Figure 7).
51 A no-reasoning GPT sweep raises aggregate pass@8 and breadth through $T = 1.5$ (Appendix H.3).
52 Prompt and provider settings differ; these observations identify neither training causes nor model-size
53 effects.

54 **Conclusion and limits.** Replay can improve correctness and sampled breadth; only admitted
55 exemplars receive replay. Fixed-bank scores still have a declining tail (Appendix F.4). Exact mode
56 probabilities, causal accuracy mechanisms and benefits under changed constraints remain unverified.

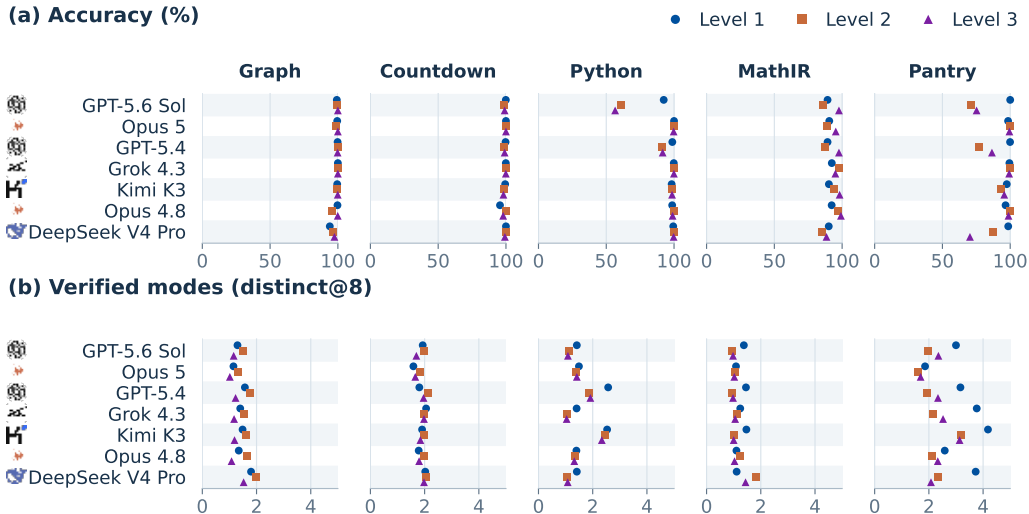


Figure 7: **High accuracy need not imply broad verified outputs.** Each cell uses all eight stateless draws on 128 prompts after frozen formatting normalization. Opus 5 Python uses revised wording without a system message, precluding a common-prompt ranking. Original protocols: Appendix H.

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141 Supplementary Material

142 This supplement gives formal metrics, benchmark and protocol details, the replay algorithm, current
 143 results and training curves, mechanism checks, core proofs, and reproducibility. Model scales and
 144 available paired cohorts match the main figures; detailed histories and theoretical extensions remain
 145 in the research archive.

146 A Formal Metric Definitions

147 For prompt x with specification s_x , the validator maps an output to a canonical valid key or failure:
 148 $V(\cdot, s_x) : \mathcal{Y} \rightarrow \mathcal{C}_x \cup \{\perp\}$. The learner sees keys only for its generated responses, never the full
 149 catalogue $\mathcal{C}_x$ or its size. For $Y \sim \pi_\theta(\cdot | x)$, define

$$P_\theta^+(x) = \Pr[V(Y, s_x) \neq \perp], \quad p_\theta^+(c | x) = \Pr[V(Y, s_x) = c | V(Y, s_x) \neq \perp].$$

150 The conditional distribution is defined only when $P_\theta^+(x) > 0$; otherwise all sampling metrics below
 151 are zero. Mode collapse means concentration onto fewer valid keys, which can coexist with increasing
 152 success probability.

153 For K independent responses, let

$$D_K = |\{V(Y_i, s_x) : i \leq K\} \setminus \{\perp\}|, \quad F_K = K^{-1} \sum_i \mathbf{1}[V(Y_i, s_x) \neq \perp].$$

154 Then $\text{distinct@}K = \mathbb{E}[D_K]$ counts expected unique verified keys in $[0, K]$, whereas $\text{pass@}K =$
 155 $\Pr[D_K > 0]$ and $\text{mean@}K = \mathbb{E}[F_K]$ are probabilities in $[0, 1]$. The samplewise inequality $F_K \leq$
 156 $\mathbf{1}[D_K > 0] \leq D_K$ gives $\text{mean@}K \leq \text{pass@}K \leq \text{distinct@}K$, but changes in the count and
 157 probability have different units.

158 **Lemma A.1** (Success and sampled breadth). *Fix a prompt, a finite or countable valid-key set, and*
 159 *$K \geq 1$ independent draws from the same fixed sampling law. Let $\mu_c = \Pr[V(Y, s_x) = c]$ and*
 160 *$P = \sum_c \mu_c$. Then*

$$S_K := \text{pass@}K = 1 - (1 - P)^K, \quad B_K := \text{distinct@}K = \sum_c [1 - (1 - \mu_c)^K]. \quad (2)$$

161 *For a finite valid-key set of size $m \geq 1$ and $P > 0$, write $q_c = \mu_c/P$. At fixed P , B_K is Schur-*
 162 *concave in q : if q majorizes q' (its decreasingly sorted partial sums are at least those of q'), then*
 163 *$B_K(P, q) \leq B_K(P, q')$. In particular,*

$$1 - (1 - P)^K \leq B_K(P, q) \leq m[1 - (1 - P/m)^K]. \quad (3)$$

164 *For $K \geq 2$, the lower bound is attained only by concentration on one key, and the upper bound only*
 165 *by uniform q . For $K = 1$, $B_1 = P$ for every q . At fixed q , increasing P cannot decrease B_K .*

166 *Proof.* All K draws fail with probability $(1 - P)^K$. For each key c , its appearance indicator has
 167 expectation $1 - (1 - \mu_c)^K$; summing indicators gives the second identity, also for countably many
 168 keys by nonnegativity. For fixed $P > 0$ and $K \geq 2$, the function $f(v) = 1 - (1 - Pv)^K$ is strictly
 169 concave on $[0, 1]$. Transferring mass from a larger coordinate to a smaller one without reversing
 170 their order increases $\sum_c f(q_c)$; successive such transfers give the majorization assertion. The point
 171 mass and uniform vector are its extreme cases, giving the bounds and equality conditions. Finally,
 172 $\partial B_K / \partial P = K \sum_c q_c (1 - Pq_c)^{K-1} \geq 0$, which proves monotonicity in correctness. $\square$

173 These are promptwise identities, before averaging over prompts. In particular, applying the nonlinear
 174 formula to a mean correctness generally changes the result. They explain the shared measurement in
 175 the training and hosted studies: success depends on total correct mass; sampled breadth also depends
 176 on its allocation. More breadth alone proves neither greater conditional diversity nor better success,
 177 and neither metric establishes downstream utility.

178 Raw $\text{distinct@}K$ measures sampled coverage, not total positive-probability support. Uniform
 179 independent sampling from N finite actions, with n_c aliases of valid key c , gives $\sum_c [1 - (1 - n_c/N)^K]$
 180 expected keys; open-ended program spaces have no canonical uniform generator.

181 The terminal co-primary outcomes use $K = 8$: success probability `pass@8` and raw `distinct@8`.
 182 Their difference $\mathbb{E}[D_K - \mathbf{1}[D_K > 0]]$ counts extra verified modes beyond the first. This interpretive
 183 decomposition remains coupled to correctness and is neither a third primary endpoint nor a success-
 184 matched diversity measure.

185 A.1 Why balance within the replay bank

186 Uniform sampling over keys prevents discovery frequency from determining replay frequency. To
 187 explain its relationship to the evaluation metric, we first analyze an equal-weight categorical surrogate,
 188 without length normalization. A stored response’s probability need not equal its key’s total probability.
 189 For $C = |\mathcal{B}_x|$ banked keys, write $p(c)$ for their probabilities and q_B for their total mass. Then

$$-\frac{1}{C} \sum_{c \in \mathcal{B}_x} \log p(c) = -\log q_B + \log C + \text{KL}(U_C \parallel p(c \mid c \in \mathcal{B}_x)). \quad (4)$$

190 The objective rewards greater banked verified mass and a more balanced conditional distribution,
 191 aligning with `pass@K` and `distinct@K`. The combined update need not improve both at every step.
 192 Indeed banked-key coverage contributes $\sum_c [1 - (1 - p(c))^K]$ to expected `distinct@K`; at fixed
 193 q_B and C , concavity makes it largest under uniform allocation, while $1 - (1 - q_B)^K$, the probability
 194 of any banked key, depends only on total mass. This deliberate objective-metric alignment motivates
 195 treating held-out `pass@K` as co-primary. Uniformity also maximizes the smallest conditional banked-
 196 key mass, making equal retention explicit. Banks contain only training-prompt responses; evaluation
 197 responses never enter training.

198 B Benchmark and Evaluation Protocol

199 **MODEBENCH** defines correctness and identity through the same executable validator: invalid outputs
 200 receive reward zero and no key; valid outputs receive reward one and a canonical execution key.
 201 Neither the complete valid-key catalogue nor its size is supplied to training, replay, scheduling, or
 202 stopping. Table 1 specifies the five domains and which surface forms merge. The public dataset
 203 preserves each domain’s split roles and construction identity. This paper’s training comparisons use
 204 Levels 1 and 2; the separate Level-3 release does not imply completed Level-3 training results.

Table 1: **MODEBENCH**. Correctness and identity come from one executable path; training starts without the mode catalogue. *Valid modes* is the mean number of correct execution modes per evaluation prompt, exact over the released multi-answer split. Every domain trains on 384 prompts and evaluates on a fixed 128-prompt split; the exact prompt each policy receives is reproduced verbatim in Appendix C for a direct, byte-level audit of the evaluation interface.

Domain	Generated answer / check	Executed key	Aliases that merge	valid modes
Graph coloring	Complete assignment; check fixed colors and edges	Color vector	Spacing only	6.27
Countdown	Parse operands; execute exactly	Normalized AST	Commutative order	4.52
Python factors	Execute restricted lambda externally	Return vector	Source programs with same behavior	229.44
MathIR	Execute bounded action program	State trajectory	Programs with the same states	5.00
PantryPlan	Exact ingredient allocation; check all bounds	Ingredient support	Quantity and ordering aliases	18.19

205 B.1 Split identities and executable validation

206 Every domain uses 384 training and 128 disjoint evaluation prompts. Python materialization indepen-
 207 dently executes at least two functions with different valid keys for every row; MathIR executes every
 208 certified route through the same interpreter used for model outputs.

Table 2: Deterministic materialization identities. Hashes bind the released rows to their executable checks; the full hashes and construction records accompany the data.

	Python factors	MathIR action menu
Training rows	384	384
Evaluation rows	128	128
Exact modes per prompt	16–3,600	5
Training-row SHA-256	bcfa9acc...37cf7	bfc98362...5d3835
Evaluation-row SHA-256	be0f621c...94d183	df5783dd...328b96
Gold catalogue supplied to policy	no	no
Training-bank initialization	empty	empty

PantryPlan. The deterministic v2 release has 384/64/128 training/development/evaluation prompts across four formulation families. Exact support ranges from 8 to 45; the 128 evaluation prompts contain 2,328 feasible supports, averaging 18.1875. Training and evaluation row hashes are a4883c5e...a757f and 5d81ba8d...5923. The validator checks exact inventory, serving steps, mass, nutrient bounds, and dietary tags; identity is ingredient support. A verified allocation is not a claim of palatability, cookability, or safety.

B.2 Python execution contract

Candidates are stripped, limited to 240 characters and 64 AST nodes, and parsed in expression mode. The only accepted form is `lambda n: EXPR` with exactly one ordinary argument named `n`: no defaults, positional-only, keyword, or variadic arguments. Integer literals have absolute value at most 10,000. The expression grammar permits `n`, `+`, `-`, `*`, floor division, remainder, integer comparisons, Boolean and/or, unary plus/minus/not, and conditionals. Boolean and non-integer literals, calls, imports, attributes, comprehensions, containers, subscripts, assignment, loops, and other identifiers are rejected. Task specifications contain two to eight distinct integers in $[4, 1000]$ and at least two valid behavior vectors.

External execution. The trainer never executes candidate code. A locked parent client launches an isolated Python `-I` process in a fresh session, with a persistent JSON-lines worker accepting only restricted source and trusted task JSON. The worker repeats the AST checks, compiles the expression with empty `__builtins__`, and executes the frozen inputs under a 0.25-second real-time alarm; the parent enforces a separate one-second response deadline. A valid response contains the integer vector and key. Timeouts, empty or malformed responses, broken pipes, non-integer outputs, and failed divisor checks yield $\perp$ and zero reward. Broken pipes, operating-system errors, and malformed responses permit one clean worker restart; a timed-out request remains failed. Worker stdout is never parsed as a candidate answer, and exceptions never produce a key.

Behavioral identity. For every frozen input n_j , the returned d_j must be an integer but not a Boolean, satisfy $1 < d_j < n_j$, and divide n_j exactly. All checks must pass. The key is the ordered vector `python_factor:  $d_1, \dots, d_m$` ; programs merge exactly when their vectors agree. The exact support count $\prod_j |\{d : 1 < d < n_j, d \mid n_j\}|$ is used only for dataset validation.

B.3 Matched Level-2 construction

Level 2 changes problem structure while preserving the verifier and canonicalizer. Its 384/128/128 training/development/evaluation rows have disjoint identities across the Level-1 and Level-2 splits. Training histograms match the native Level-1 training sets. Development and evaluation histograms match designated Level-1 development and confirmation reserves in Graph, Countdown, Python, and MathIR; Pantry uses native references, with its 64-row development histogram doubled to 128 rows.

These construction reserves are not the native Level-1 terminal test sets. The displayed terminal evaluations have different valid-key histograms between levels in Graph, Countdown, and Python. Cross-level changes therefore combine task structure and test population; they do not isolate difficulty at fixed test-set support. Replay comparisons remain paired within each level on the same prompts, seeds, fresh objective, and training budget.

Domain	Level 1	Level 2 structural change
Graph coloring	Three hidden vertices	Four hidden vertices
Countdown	Three operands	Four operands, with values up to 14
Python factors	Four executed cases, all at most 96	Four cases, including a value above 96
MathIR	One-sided linear families	Variable-on-both-sides families
PantryPlan	Base dietary/composition constraints	More active dietary exclusions and tighter composition

Before treatment training, frozen Qwen2.5-0.5B development evaluation admitted a Level-2 domain only when `pass@8` lay in $[.10, .90]$ and was strictly below its construction-reference Level-1 value under identical decoding. The matched-level identity manifest records support histograms, row hashes, construction seeds, disjointness checks, and these admission receipts.

B.4 Training and evaluation settings

Table 3: **Matched run contract.** Settings and systems changes are shared by methods within each model–domain–seed comparison, including the zero-weight replay control.

Component	Setting
Data	384 training and 128 disjoint evaluation prompts per domain and level; no evaluation row enters training or replay.
Fresh rollouts	$G = 16$, temperature 1, top- $p = 1$; one prompt group per update.
Optimization	Eight passes (3,072 updates), one PPO epoch, $\beta_{KL} = 0$; paired initialization, prompt stream, optimizer, hardware class, and checkpoint cadence.
Evaluation	Passes 0, .5, . . . , 8; four fixed temperature-1, top- $p = 1$, $K = 8$ draws per prompt, plus a separate temperature-0 greedy trace. No best-checkpoint selection.
Replay	Capacity $M = 16$ per prompt; one round-robin bank per update; $\lambda_{\text{replay}} = .10$. Controls execute the same code with an exact-zero replay derivative; realized banks and workloads can differ.
Training seeds	Qwen2.5-0.5B: 43–47; Falcon3-1B: 55–59; Qwen2.5-3B: 70–74.
Learning rate	Qwen-0.5B and Falcon-1B: constant 2×10^{-7} , with optimizer-state offload for Falcon. Qwen-3B: cosine decay with 10^{-7} peak, 10^{-8} floor, and 10% warmup.
Response tokens	Graph and Countdown: 192; Python: 192/512/192 at 0.5B/1B/3B; MathIR: 64/128/64; PantryPlan: 8.
Evaluation seeds	Base 610100/610200/610300/610400/76299 for Graph/Countdown/Python/MathIR/PantryPlan; draw d uses base plus d .

Base model	Pinned revision
Qwen2.5-0.5B-Instruct	7ae557604adf67be50417f59c2c2f167def9a775
Falcon3-1B-Instruct	28ba2251970a01dd1edc7ba7dad2eb71216ccfd
Qwen2.5-3B-Instruct	aa8e72537993ba99e69dfaafa59ed015b17504d1

Responses are independent within each evaluation draw. For each training seed, metrics average over all 128 prompts and four draws; raw responses and draw-level records are retained. The co-primary terminal outcomes are `pass@8` and raw `distinct@8`; `mean@8` is diagnostic. Evaluation never supplies training data, bank entries, checkpoint selection, or a stopping rule. Complete paired five-seed blocks receive nominal pointwise 95% Student- t intervals; partial primary intersections report exact counts and descriptive means without intervals. Intervals condition on the frozen prompts and evaluation requests. Domain averages are explicitly labeled, and no missing terminal result is replaced by a shallower checkpoint. Source admission and curve gaps follow Appendix M.1; comparison-specific intersections and the replay-weighting bootstrap are specified with their results.

C Exact Domain Prompts

Each domain uses one system message and one user message, with no few-shot examples. The same template renders training and evaluation for every method. These are the first rows of the frozen evaluation splits, including the exported chat-template wrapper. The checked generator ops/check_

268 paper_domain_prompts.py reproduces them from the datasets. A two-space continuation indent
 269 marks display wrapping: joining it to the previous line with one space recovers the original string.
 270 The PantryPlan listing retains its real newlines.

271 C.1 Graph coloring

272 Template qwen_boxed; 384 training and 128 evaluation
 273 rows in var/data/graph_coloring_modebench_v2.

```
<|im_start|>system
Return only the final answer inside \boxed{}. Do not explain.<|im_end|>
<|im_start|>user
Color this graph with vertices 1 through 6. The undirected edges are: (1,2), (1,3), (3,5),
(3,6), (4,6), (5,6). Assign each vertex one color from {1, 2, 3} so that no edge has the
same color on both endpoints. A partial coloring is 2??1?2, where ? means uncolored.
Keep every shown digit fixed. There are exactly 3 question marks. Return exactly one
final answer and no explanation: exactly 3 digits, each one 1, 2, or 3, for the missing
positions from left to right, inside \boxed{<|im_end|>
<|im_start|>assistant
```

274 C.2 Countdown

275 Template qwen_boxed; 384 training and 128 evaluation
 276 rows in var/data/exact_countdown_easy3_probe.

```
<|im_start|>system
Return only the final answer inside \boxed{}. Do not explain.<|im_end|>
<|im_start|>user
Using the numbers [3, 6, 9], create an arithmetic expression that equals 18. Use each
given number exactly once. You may use +, -, *, /, and parentheses. Return exactly one
final answer and no explanation: the expression inside \boxed{<|im_end|>
<|im_start|>assistant
```

277 C.3 Python factors

278 Template qwen_boxed; 384 training and 128 evaluation
 279 rows in var/data/python_factor_modebench_v1.

```
<|im_start|>system
Return only the final answer inside \boxed{}. Do not explain.<|im_end|>
<|im_start|>user
Write one pure Python function with the exact form lambda n: EXPR. An external Python tool
will call it once for each n in [18, 82, 91, 93]. For every call, return an integer d
satisfying 1 < d < n and n % d == 0. Different valid return vectors are different
solution modes. You may use integer literals, n, +, -, *, //, %, comparisons, Boolean
operators, and conditional expressions; calls, imports, attributes, containers, and
other names are forbidden. Return exactly the one-line lambda inside \boxed{<|im_end|>
<|im_start|>assistant
```

280 C.4 MathIR action menu

281 Template qwen_boxed; 384 training and 128 evaluation
 282 rows in var/data/mathir_action_menu_v1.

```
<|im_start|>system
Return only the final answer inside \boxed{}. Do not explain.<|im_end|>
<|im_start|>user
MathIR linear-menu-v1. Solve for x by choosing an executable action sequence. Every
selected action is applied exactly to BOTH sides of the current equation, followed by
exact simplification. Bindings: a=2, b=-9, c=-1. Initial equation: x/a + b = c. Action
menu: A means div(a); B means add(b); C means sub(b); D means sub(c); E means
sub(mul(a,b)); F means mul(a). Choose 1-4 action IDs. Illegal operations, repeated
states, and paths that do not finish with x isolated are rejected. Return only the IDs
separated by semicolons inside the answer box; for example formatting only: \boxed{A;C}.
Do not return x's numeric value.<|im_end|>
<|im_start|>assistant
```

283 C.5 PantryPlan

284 Template qwen_pantry_support_mask; 384 training and 128 evaluation
 285 rows in var/data/pantry_plan_modebench_v2.

```
<|im_start|>system
Return only six binary digits and no other text. Each digit maps to the corresponding
Pantry row in printed order. A trusted environment will choose exact quantities on
precisely the selected rows.<|im_end|>
<|im_start|>user
Create one feasible high fiber snack from this frozen pantry.
Nutrition values are per 100 g. Quantities are exact integer grams.

Pantry:
- almonds: available=100g, step=25g, minimum-if-used=50g; energy=584kcal, protein=21.5g,
  fiber=10.8g, sodium=0.0mg; tags=tree_nut
- banana: available=125g, step=25g, minimum-if-used=50g; energy=97.0kcal, protein=0.74g,
  fiber=4.62g, sodium=0.0mg; tags=none
- carrots: available=150g, step=25g, minimum-if-used=50g; energy=45.0kcal, protein=0.941g,
  fiber=3.1g, sodium=86.6mg; tags=none
- grape_tomatoes: available=100g, step=25g, minimum-if-used=50g; energy=27.0kcal,
  protein=0.83g, fiber=2.1g, sodium=6.0mg; tags=none
- navel_orange: available=150g, step=25g, minimum-if-used=50g; energy=47.0kcal,
  protein=0.91g, fiber=2.0g, sodium=9.0mg; tags=none
- sunflower_seeds: available=125g, step=25g, minimum-if-used=50g; energy=571kcal,
  protein=18.9g, fiber=7.22g, sodium=0.0mg; tags=none

Use 2 to 4 ingredients.
Forbidden tags: none
Targets:
- mass_g: minimum 125, maximum 200
- energy_kcal: minimum 239.4, maximum 432.25
- protein_g: minimum 6.423
- fiber_g: minimum 3.478
- sodium_mg: maximum 37.6

Choose the ingredient support as a six-bit mask in the exact Pantry row order. Bit 1
includes that row and bit 0 excludes it.<|im_end|>
<|im_start|>assistant
```

286 D Replay Algorithm

287 For each prompt x , $\mathcal{B}_x$ stores at most M verified policy exemplars, one per canonical key; the
 288 serialized cursor τ selects banks for replay. The same update supports either fresh objective.

289 D.1 Fresh task advantages

290 For a group of G fresh responses with verified binary rewards r_i and $m = \sum_i r_i$, Dr.GRPO uses
 291 $A_i = r_i - m/G$, while MaxRL uses

$$A_i^{\text{MaxRL}} = \begin{cases} Gr_i/m - 1, & m > 0, \\ 0, & m = 0. \end{cases}$$

292 The centered MaxRL estimator [Tajwar et al., 2026] gives failures advantage -1 in mixed groups
 293 and weights each success more strongly when successes are scarce. All-correct and all-failure groups
 294 have zero task advantage. Lemma L.2 gives its harmonic pass@ k mean objective. These advantages
 295 apply only to fresh responses; bank exemplars enter the separate verified-likelihood loss.

296 D.2 State and optimizer update

Algorithm 1 One ReplayMaxRL optimizer update

Require: Policy π_θ ; prompt x and verifier specification s_x ; group size G ; fresh objective $o \in \{\text{MaxRL}, \text{Dr.GRPO}\}$; replay weight λ_{replay} ; states $(\mathcal{B}, \tau)$
Ensure: Atomically updated $(\theta', \mathcal{B}', \tau')$

- 1: Draw the on-policy group $\mathcal{G} = \{y_i \sim \pi_\theta(\cdot | x)\}_{i=1}^G$
- 2: **for** $i = 1, \dots, G$ **do**
- 3: Execute $\tilde{c}_i \leftarrow V(y_i, s_x)$ and observe task reward r_i
- 4: Set $c_i \leftarrow \tilde{c}_i$ iff $r_i > 0 \wedge \tilde{c}_i \neq \perp$; otherwise set $c_i \leftarrow \perp$
- 5: **end for**
- 6: Compute fresh task advantages A_i^o from $\{r_j\}_{j=1}^G$; for ReplayMaxRL use the formula in Appendix D.1
- 7: After the full group is validated, deterministically retain one policy exemplar for each new $c_i \neq \perp$ in $\mathcal{B}_x$, up to capacity M
- 8: $(z, \tau') \leftarrow \text{NEXTBANK}(\{\mathcal{B}_u\}_u, \tau)$ by a fixed round-robin cursor
- 9: **if** a replay bank $\mathcal{B}_z$ exists **then**
- 10: For each retained exemplar e_c , compute $s_c = |e_c|^{-1} \log \pi_\theta(e_c | z)$
- 11: $L_{\text{replay}} \leftarrow -|\mathcal{B}_z|^{-1} \sum_{c \in \mathcal{B}_z} s_c$
- 12: **else**
- 13: $L_{\text{replay}} \leftarrow 0$
- 14: **end if**
- 15: $w_{\text{rep}} \leftarrow (G - 1)/G^2$ for one replay pseudo-group per G on-policy rows
- 16: $\theta' \leftarrow \text{UPDATE}(\theta, L_{\text{PPO}}(A^o) + w_{\text{rep}} \lambda_{\text{replay}} L_{\text{replay}})$
- 17: Checkpoint $(\theta', \mathcal{B}', \tau')$ as one atomic state

297 D.3 Admission, capacity, and deterministic scheduling

298 A bank is keyed by the SHA-256 digest of the exact prompt-token sequence. Only loss-active rows
 299 with positive reward and a nonempty validator key qualify. The full group is validated against the
 300 pre-group bank before any new key is committed. For each new key, the lexicographically smallest
 301 normalized response-token tuple is retained; new keys are admitted in sorted-key order. Admission is
 302 therefore invariant to rollout-row order. The bank keeps one exemplar per key, up to $M = 16$, with
 303 no eviction or replacement. Later keys may enter diagnostics but not replay support when capacity is
 304 full. Prompt-hash collisions or changed tokens under an existing hash are hard errors.

305 At each update, the scheduler sorts nonempty prompt-bank hashes and selects one by a fixed
 306 round-robin cursor; singleton banks are eligible. The cursor is independent of reward, evaluation,
 307 desired support, or entropy, and is serialized with the bank. Stored prompt and response tokens are
 308 materialized unchanged into a right-padded teacher-forcing batch. Prompt and padding positions
 309 are masked. Each exemplar contributes its mean response-token log probability, averaged uniformly
 310 across keys in the selected bank before applying λ_{replay} and the exact $(G - 1)/G^2$ rollout scaling in
 311 Algorithm 1. Neither exhaustive support nor evaluation outputs enter this loss.

312 D.4 Matched controls and recoverable state

313 Controls share validation, bank updates, scheduling, materialization, and the score pass, with the
 314 replay score-gradient vector replaced by exact zeros. Treatment activates only that derivative. Policies
 315 can subsequently differ in samples, bank contents, selected banks, exemplar lengths, and realized
 316 computation. Logs retain bank/key counts, replayed token counts, raw and applied losses, score-
 317 gradient summaries, cursor phase, and the exact-zero switch. Model and optimizer state, bank counts
 318 and exemplar tokens, replay and data cursors, and request-stream state are checkpointed atomically;
 319 the loader validates schema, capacity, scheduler and estimator constants, token types, prompt hashes,
 320 key/exemplar consistency, and cursor bounds before accepting a resume.

321 D.5 Fixed Semantic MaxEnt comparator

322 Fixed Semantic MaxEnt is an entropy-inspired fresh-rollout comparator. For validator-positive
 323 canonical outcomes, an empirical predictor combines successful policy history and successful leave-
 324 one-out peers. If K keys have total count N , unit pseudocounts and one reserved unseen bucket

325 give

$$\hat{p}(a) = \frac{n(a) + 1}{N + K + 1}, \quad \hat{p}(\text{UNSEEN}) = \frac{1}{N + K + 1}. \quad (5)$$

326 The added detached advantage on an eligible successful row is

$$A_{i,t}^{\text{sem}} = \eta \frac{\min\{-\log \hat{p}(a_i), C\} - \mathbb{E}_{\hat{p}}[\min\{-\log \hat{p}(a), C\}]}{C}, \quad \eta = .10, C = 5. \quad (6)$$

327 Invalid and unsuccessful rows receive zero. The centered score is bounded by η in magnitude. It
 328 can favor an underrepresented verified outcome present in a fresh rollout, but supplies no direct
 329 reinforcement to an absent remembered exemplar. Its heterogeneous empirical effects therefore
 330 remain a comparison with exploration rather than a retention guarantee.

331 For the true conditional distribution $q_\theta = p_\theta(\cdot | C)$,

$$\nabla_\theta H(q_\theta) = \mathbb{E}_{a \sim q_\theta} [(-\log q_\theta(a)) \nabla_\theta \log q_\theta(a)]. \quad (7)$$

332 This motivates the direction of redistribution, not an unbiased-gradient claim for the empirical
 333 estimator. Hold a detached predictor score vector s and baseline b fixed independently of the scored
 334 action, for example conditional on its leave-one-out peers. Averaging successful rows under the
 335 unconditional policy then gives

$$\mathbb{E}_{A \sim p}[\mathbf{1}\{A \in C\}(s_A - b) \nabla_z \log p_A] = P \mathbb{E}_{c \sim q}[s_c \nabla_z \log q_c] + (\mathbb{E}_q[s] - b) \nabla_z P.$$

336 A historical baseline need not equal the current conditional mean, so this bonus can change correctness
 337 as well as diversity. Even with the exact unclipped score $s_c = -\log q_c$ and baseline $b = H(q)$,
 338 averaging over unconditional rollout rows gives $P \nabla H(q)$. For a single correct key, conditional
 339 entropy is zero, yet the implemented unseen bucket can give that key negative semantic advantage.
 340 The comparator thus carries neither an exact entropy-gradient interpretation nor an individual-mode
 341 preservation guarantee.

342 E Empirical Results and Comparisons

343 E.1 Comparison registry

344 The primary comparisons cross fresh objective (Dr.GRPO or MaxRL) with verified replay, first
 345 across three model scales on Level 1 and then at Qwen2.5-0.5B on Level 2. Table 4 separates these
 346 comparisons from the supporting controls and replay-weighting ablation. Terminal populations
 347 use the September 11 frozen census. The registered Qwen2.5-0.5B weighting aggregate retains its
 348 original frozen population; its endpoint values are unchanged in that census.

Table 4: **Comparison scope and available terminal evidence.** A complete block has five registered paired seeds. Primary partial intersections retain their exact counts and receive no interval. UCPO and sparse RLEP-Dr are reported only at the two smaller model scales; their denominators cover all five benchmark domains and five registered seeds at each selected scale.

Comparison	Scope	Terminal coverage	Question
Dr.GRPO / ReplayDr.GRPO	Level 1, all three scales	74/75 pairs; 14/15 complete blocks	Verified memory beyond fresh reward
MaxRL / ReplayMaxRL	Level 1, all three scales	75/75 pairs; 15/15 complete blocks	Memory beyond a stronger fresh objective
Four-method factorial	Level 2, Qwen-0.5B	4/5 complete domains; Pantry pairs: two Dr.GRPO, one MaxRL	Transfer to harder matched tasks
GRPO	Level 1, all three scales	75/75 runs	Reward-standardization control
UCPO	Level 1, 0.5B and 1B	50/50 runs	On-policy advantage redistribution
Sparse RLEP-Dr	Level 1, 0.5B and 1B	47/50 runs	Success memory without uniform key balancing
Fixed Semantic MaxEnt	Level 1, registered scale/domain blocks	10/15 complete blocks	Fresh-rollout semantic exploration
Uniform / frequency replay	Level 1, nine registered blocks	44/45 treatment cells; 8/9 complete blocks	Contribution of replay-key balancing

E.2 Terminal comparisons across scales

Within each model–domain–seed comparison, methods share initialization, data identities, the eight-pass horizon, evaluation requests, verifier, and replay scheduling code. The replay control runs the same bank and scoring path with zero replay derivative. Resulting policies can build different banks. The admissible intersections are defined in Appendix M.1.

Table 5: Matched results after eight training passes. Columns compare Dr.GRPO with ReplayDr.GRPO after exactly eight training passes. Entries are means over the displayed paired-seed intersection; $n@pass$ gives its size and training pass. P, M, and D denote $pass@8$, $mean@8$, and $distinct@8$. Every available source-admissible terminal pair is included, no shallower checkpoint is carried forward, and no missing result is imputed. Falcon Countdown uses four admissible seeds (Appendix M.1). Pale green marks a ReplayDr.GRPO mean that is strictly higher than its matched Dr.GRPO mean. Archived models: Qwen2.5-0.5B, Falcon3-1B, and Qwen2.5-3B.

Model	Domain	$n@pass$	Dr.GRPO			ReplayDr.GRPO		
			P	M	D	P	M	D
Qwen2.5-0.5B	Graph coloring	5@8.0	.323	.323	.325	.969	.404	2.438
	Countdown	5@8.0	.480	.465	.487	.672	.597	1.637
	Python factors	5@8.0	.172	.172	.172	.528	.507	.571
	MathIR	5@8.0	.512	.464	.513	.796	.688	.829
	PantryPlan	5@8.0	.522	.522	.522	.729	.545	1.539
Falcon3-1B	Graph coloring	5@8.0	.656	.300	1.034	.809	.334	1.717
	Countdown	4@8.0	.325	.267	.415	.632	.343	.838
	Python factors	5@8.0	.000	.000	.000	.979	.722	1.107
	MathIR	5@8.0	.095	.068	.097	.736	.292	.756
	PantryPlan	5@8.0	.633	.633	.633	.751	.560	1.576
Qwen2.5-3B	Graph coloring	5@8.0	.682	.360	1.060	.836	.349	1.637
	Countdown	5@8.0	.609	.569	.628	.746	.505	.962
	Python factors	5@8.0	.069	.069	.069	.659	.603	.852
	MathIR	5@8.0	.278	.168	.279	.686	.580	.706
	PantryPlan	5@8.0	.638	.558	.841	.744	.546	1.514

Figure 8 resolves the domain-specific results behind the main cross-domain display. All five 3B MaxRL domain comparisons are complete: both arms have seeds 70–74 at the registered terminal endpoint (50/50 runs). The Dr.GRPO pair uses five seeds except Falcon Countdown, which uses four. These populations remain separate: a factorial interaction requires the common four-arm intersection, not differences of aggregates with different seeds.

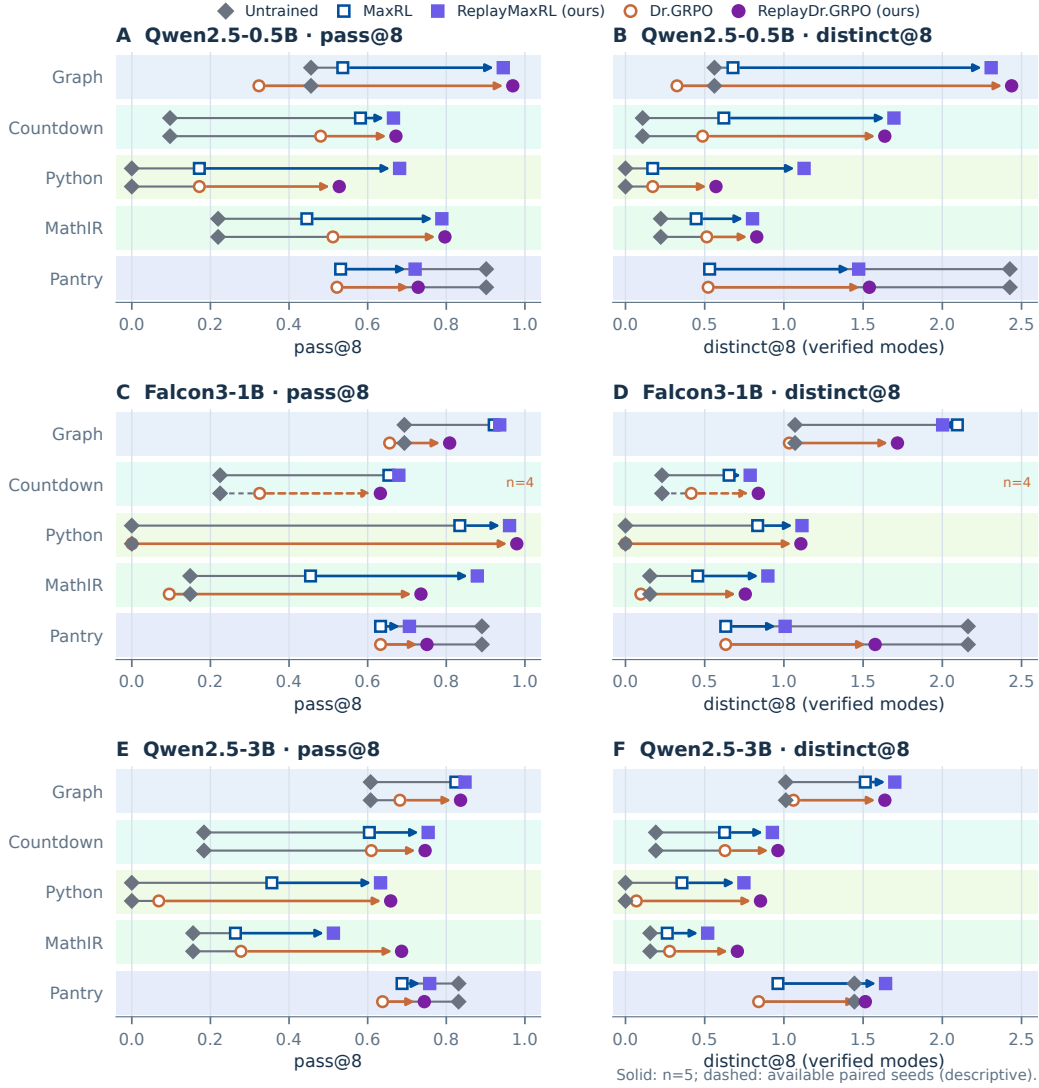


Figure 8: **Per-domain results behind the cross-domain averages in Figure 5.** Rows show Qwen2.5-0.5B, Falcon3-1B, and Qwen2.5-3B; columns show correctness and raw support. MaxRL–ReplayMaxRL pairs retain five terminal seeds throughout the two smaller models and Qwen2.5-3B: every MaxRL domain uses $n = 5$. Solid connectors identify these complete comparisons. Dr.GRPO–ReplayDr.GRPO uses four admissible seeds for Falcon Countdown and five elsewhere. Domain tints match Figure 2. Replay arrows, trained-arm markers, and untrained references use the same encoding as the main figure. The E118 model index groups the archived exports by model and domain.

359 Write $P = \text{pass@8}$, $D = \text{distinct@8}$, and $B = D - P$. The derived B counts verified modes
 360 beyond the first correct outcome in each sampled set and remains coupled to correctness. The endpoint
 361 tables report exact paired terminal effects. A dash denotes no observed pair; partial means retain
 362 $n/5$ without an interval. Complete-block intervals are nominal 95% estimates, without multiple-
 363 comparison or repeated-look adjustment. All underlying arm means and seed effects remain in the
 364 adjacent machine-readable records.

Table 6: **ReplayMaxRL effects across model scales at pass 8.** All effects are ReplayMaxRL minus MaxRL on the displayed paired seeds. Brackets give paired Student- t intervals for ΔB in complete blocks. The machine-readable report also retains both arm means, all per-seed effects, and intervals for P and D .

Model	Domain	Replay arm	$n/5$	ΔP	ΔD	ΔB [95% interval]
Qwen-0.5B	Graph	ReplayMaxRL	5/5	+0.408	+1.630	+1.221 [+0.926, +1.517]
Qwen-0.5B	Countdown	ReplayMaxRL	5/5	+0.084	+1.074	+0.990 [+0.859, +1.121]
Qwen-0.5B	Python	ReplayMaxRL	5/5	+0.509	+0.955	+0.446 [-0.156, +1.048]
Qwen-0.5B	MathIR	ReplayMaxRL	5/5	+0.343	+0.356	+0.013 [+0.008, +0.017]
Qwen-0.5B	PantryPlan	ReplayMaxRL	5/5	+0.189	+0.941	+0.751 [+0.520, +0.982]
Falcon-1B	Graph	ReplayMaxRL	5/5	+0.014	-0.093	-0.107 [-0.220, +0.005]
Falcon-1B	Countdown	ReplayMaxRL	5/5	+0.025	+0.133	+0.108 [+0.082, +0.134]
Falcon-1B	Python	ReplayMaxRL	5/5	+0.127	+0.280	+0.153 [-0.272, +0.578]
Falcon-1B	MathIR	ReplayMaxRL	5/5	+0.424	+0.443	+0.019 [+0.006, +0.031]
Falcon-1B	PantryPlan	ReplayMaxRL	5/5	+0.073	+0.375	+0.302 [+0.034, +0.569]
Qwen-3B	Graph	ReplayMaxRL	5/5	+0.023	+0.186	+0.163 [+0.021, +0.305]
Qwen-3B	Countdown	ReplayMaxRL	5/5	+0.150	+0.302	+0.152 [+0.119, +0.185]
Qwen-3B	Python	ReplayMaxRL	5/5	+0.277	+0.391	+0.115 [+0.080, +0.149]
Qwen-3B	MathIR	ReplayMaxRL	5/5	+0.249	+0.255	+0.006 [+0.001, +0.011]
Qwen-3B	PantryPlan	ReplayMaxRL	5/5	+0.070	+0.679	+0.609 [+0.467, +0.751]

At Qwen2.5-0.5B, ReplayMaxRL adds 1.221, .990, and .751 extra modes on Graph, Countdown, and PantryPlan, respectively; each five-seed interval excludes zero. Falcon Graph instead has a negative mean raw-support effect (-.093, interval [-.229, +.043]), and its correctness effect is inconclusive (+.014, [-.016, +.044]). Positive cross-domain effects therefore do not imply a breadth gain in every domain. All five Qwen2.5-3B blocks are complete. Their extra-mode effects are +.163, +.152, +.115, +.0059, and +.609 for Graph, Countdown, Python, MathIR, and PantryPlan; every nominal interval excludes zero, although MathIR’s breadth gain is small. Correctness intervals exclude zero for Countdown, MathIR, and PantryPlan; Graph and Python remain inconclusive.

E.3 Terminal comparison on the harder level

Figure 6 uses Graph, Countdown, Python, and MathIR: each has all five seeds at step 3,072 for both levels and all four methods. Means average those four domains equally within each seed, then the five seed means. This is a descriptive 20-cell comparison, not a five-domain estimate; the levels have separate held-out prompts, with the population differences in Appendix B.3. At Level 2, PantryPlan has three Dr.GRPO, four ReplayDr.GRPO, one MaxRL, and two ReplayMaxRL terminal endpoints (10/20). Dr.GRPO/replay pairs are seeds 43 and 46; MaxRL/replay has seed 43, the only common four-arm seed. Independently available arm means do not estimate paired effects.

Table 7: **Replay effects on the harder Level-2 tasks at pass 8.** MaxRL and Dr.GRPO identify the fresh objective; each effect is its replay variant minus the corresponding objective without replay. Each row uses its own pairwise intersection, which can exceed the four-method intersection. Brackets give paired Student- t intervals for ΔB only when all five registered pairs are available.

Model	Domain	Objective	$n/5$	ΔP	ΔD	ΔB [95% interval]
Qwen-0.5B	Graph	Dr.GRPO	5/5	+0.561	+1.019	+0.458 [+0.394, +0.522]
Qwen-0.5B	Graph	MaxRL	5/5	+0.305	+0.668	+0.363 [+0.280, +0.446]
Qwen-0.5B	Countdown	Dr.GRPO	5/5	+0.025	+0.045	+0.020 [-0.014, +0.055]
Qwen-0.5B	Countdown	MaxRL	5/5	+0.040	+0.057	+0.017 [+0.001, +0.032]
Qwen-0.5B	Python	Dr.GRPO	5/5	+0.075	+0.400	+0.325 [-0.228, +0.878]
Qwen-0.5B	Python	MaxRL	5/5	+0.037	+0.212	+0.175 [-0.295, +0.645]
Qwen-0.5B	MathIR	Dr.GRPO	5/5	+0.493	+0.493	+0.000 [+0.000, +0.000]
Qwen-0.5B	MathIR	MaxRL	5/5	+0.173	+0.175	+0.002 [-0.003, +0.006]
Qwen-0.5B	PantryPlan	Dr.GRPO	2/5	+0.234	+0.367	+0.133
Qwen-0.5B	PantryPlan	MaxRL	1/5	+0.250	+0.404	+0.154

Level-2 Graph provides the clearest replay effects on correctness and extra modes under both objectives. Countdown’s ReplayMaxRL extra-mode effect is small (+.017, interval [+0.001, +0.032]), while its correctness and raw-support intervals include zero. Python and MathIR have positive mean correctness and raw-support effects with intervals including zero; MathIR’s extra-mode effects are near zero. Pantry’s two-seed Dr.GRPO replay contrast is +.234 on P , +.367 on D , and +.133 on B . Its one-seed MaxRL replay contrast is +.250, +.404, and +.154, respectively. Neither partial contrast receives an interval.

For each common seed and endpoint Y , the replay interaction is

$$I = (Y^{\text{ReplayMaxRL}} - Y^{\text{MaxRL}}) - (Y^{\text{ReplayDr.GRPO}} - Y^{\text{Dr.GRPO}}).$$

Replay can add value under both fresh objectives without a positive interaction. On Level-2 Graph, MaxRL alone raises P relative to Dr.GRPO by +.233 ([+.140, +.325]), while the replay interaction on P is -.256 ([-.329, -.183]). This is an additional replay benefit after stronger fresh optimization, without a superadditive interaction.

Table 8: **Fresh-objective effects and replay interactions on Level 2.** All three contrasts within a domain use the same four-arm seed intersection. The interaction is (ReplayMaxRL - MaxRL) - (ReplayDr.GRPO - Dr.GRPO). Each entry gives its mean and, only for a complete five-seed block, a paired unadjusted 95% Student- t interval. Archived four-arm exports: E119 model index.

Domain	n	Contrast	ΔP	ΔD	ΔB
Graph	5	MaxRL - Dr.GRPO	+.233 [+0.140, +0.325]	+.322 [+0.162, +0.482]	+.089 [+0.019, +0.159]
Graph	5	ReplayMaxRL - ReplayDr.GRPO	-.023 [-0.062, +0.016]	-.029 [-0.198, +0.140]	-.006 [-0.142, +0.130]
Graph	5	Replay $\times$ MaxRL interaction	-.256 [-0.329, -.183]	-.351 [-0.446, -.255]	-.095 [-0.186, -.003]
Countdown	5	MaxRL - Dr.GRPO	-.014 [-0.073, +0.045]	-.014 [-0.073, +0.045]	+0.000 [+0.000, +0.000]
Countdown	5	ReplayMaxRL - ReplayDr.GRPO	+0.002 [-0.008, +0.012]	-.002 [-0.047, +0.044]	-.004 [-0.042, +0.035]
Countdown	5	Replay $\times$ MaxRL interaction	+0.016 [-0.047, +0.078]	+0.012 [-0.068, +0.092]	-.004 [-0.042, +0.035]
Python	5	MaxRL - Dr.GRPO	+0.000 [+0.000, +0.000]	+0.000 [+0.000, +0.000]	+0.000 [+0.000, +0.000]
Python	5	ReplayMaxRL - ReplayDr.GRPO	-.037 [-0.232, +0.157]	-.188 [-0.988, +0.613]	-.150 [-1.014, +0.713]
Python	5	Replay $\times$ MaxRL interaction	-.037 [-0.232, +0.157]	-.188 [-0.988, +0.613]	-.150 [-1.014, +0.713]
MathIR	5	MaxRL - Dr.GRPO	+.283 [-0.193, +0.759]	+.283 [-0.193, +0.759]	+0.000 [+0.000, +0.000]
MathIR	5	ReplayMaxRL - ReplayDr.GRPO	-.036 [-0.158, +0.085]	-.035 [-0.158, +0.088]	+0.002 [-0.003, +0.006]
MathIR	5	Replay $\times$ MaxRL interaction	-.320 [-0.864, +0.225]	-.318 [-0.860, +0.224]	+0.002 [-0.003, +0.006]
PantryPlan	1	MaxRL - Dr.GRPO	+.041	+.061	+.020
PantryPlan	1	ReplayMaxRL - ReplayDr.GRPO	-.021	-.037	-.016
PantryPlan	1	Replay $\times$ MaxRL interaction	-.062	-.098	-.035

E.4 Accuracy and verified modes throughout training

The following curves include every primary method and model scale used in the main paper. Each objective pair uses its domain-specific terminal seed intersection as one fixed cohort throughout

396 training. A checkpoint is plotted only when all members have complete, valid evaluations for both
 397 methods. Missing or conflicting checkpoints leave gaps; seed counts never shrink to extend a mean.
 398 Four fixed-seed draws of eight samples on the same 128 held-out prompts define each evaluation.
 399 Accuracy and mode curves use identical seeds and checkpoints, without smoothing or outcome-based
 400 selection.

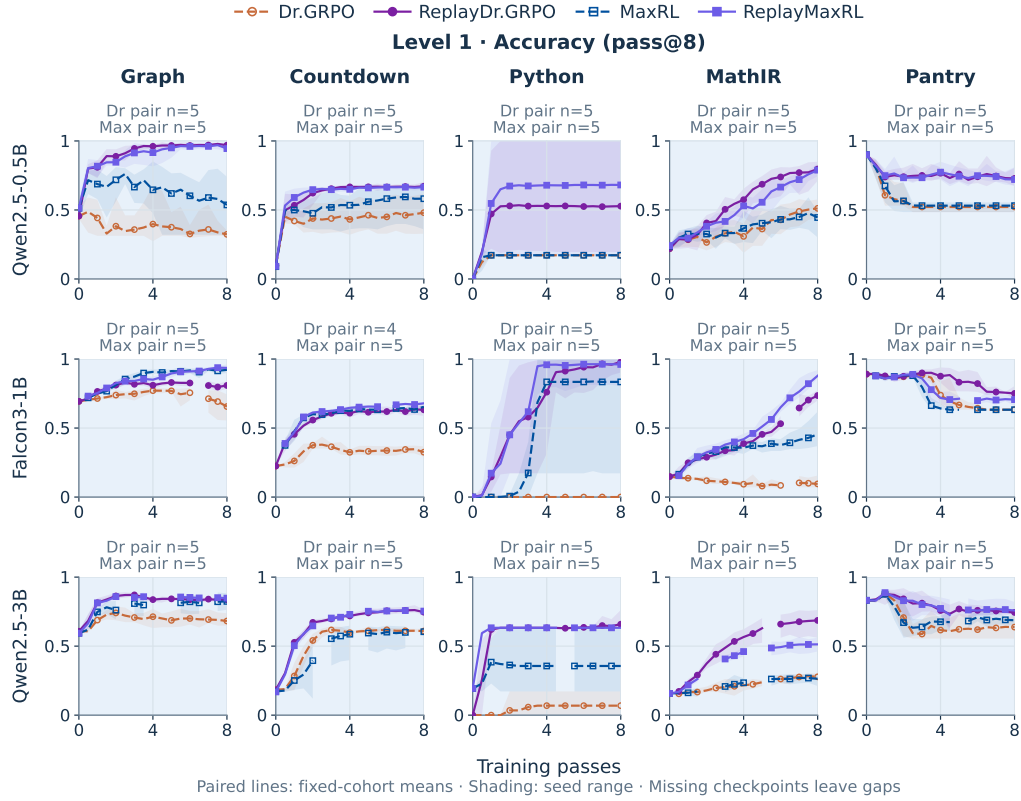


Figure 9: **Accuracy throughout training at all three model scales.** Rows are Qwen2.5-0.5B, Falcon3-1B, and Qwen2.5-3B; columns are the five domains. Dr.GRPO, ReplayDr.GRPO, MaxRL, and ReplayMaxRL use the same method colors as Figure 5. Lines are fixed-cohort means of pass@8; bands are seed ranges, not confidence intervals. Counts identify each objective’s terminal paired seeds. Qwen2.5-3B MaxRL uses all five seeds in every domain; Falcon Countdown Dr.GRPO/replay uses four. Gaps indicate missing or conflicting checkpoints.

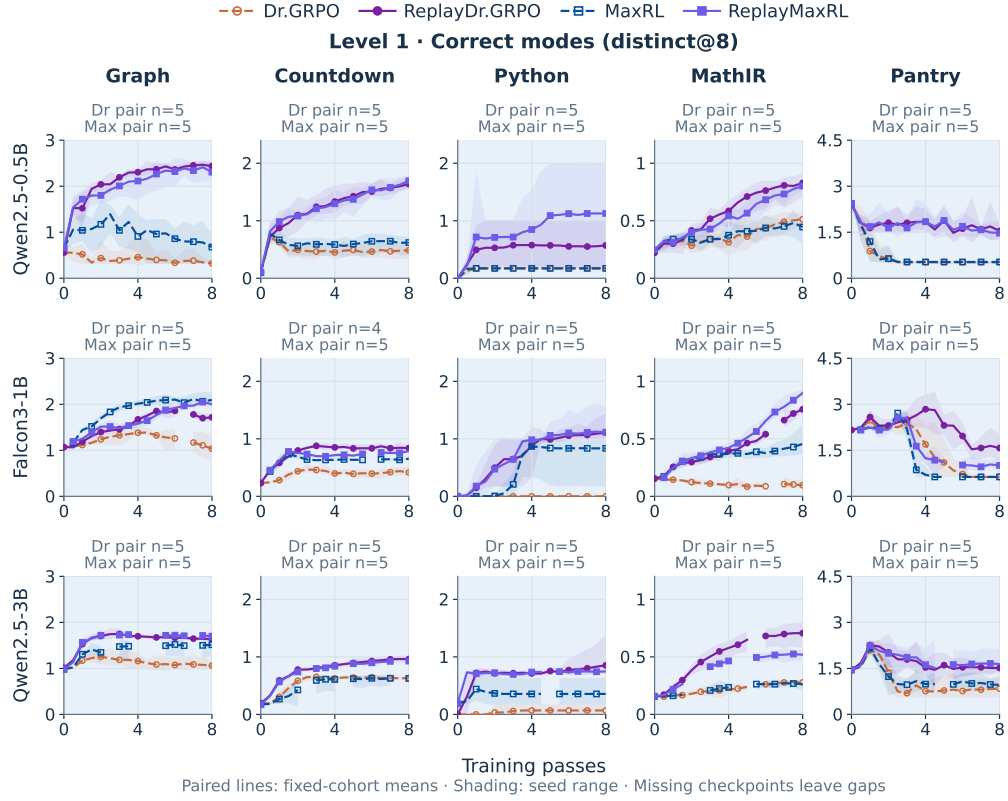


Figure 10: **Verified modes throughout training at all three model scales.** The same model rows, methods, seeds, and checkpoints as Figure 9, now measuring distinct@8. Lines are descriptive fixed-cohort means and bands show seed ranges. A count of one has no band. Domain-specific vertical scales are shared across model rows; absolute heights across different domains should be read against their labeled axes. Missing evaluations are never carried forward or joined across a gap.

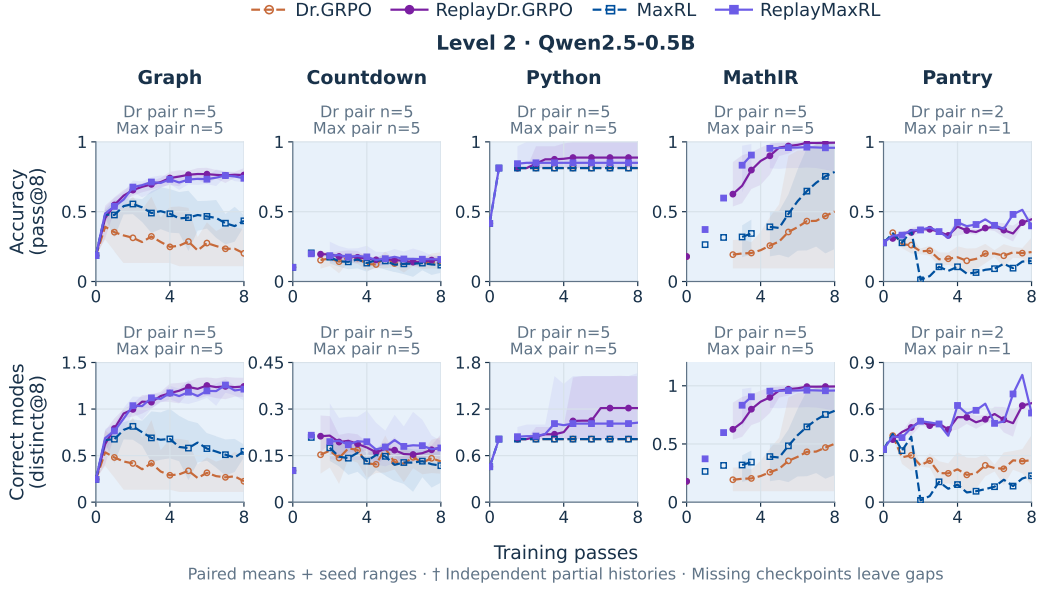


Figure 11: **Accuracy and verified modes during Level-2 training.** Qwen2.5-0.5B is the model tested at this level. Rows show pass@8 and distinct@8 for all four primary methods. Graph, Countdown, Python, and MathIR contain five terminal paired seeds per objective. PantryPlan uses two terminal Dr.GRPO/replay pairs and one MaxRL/replay pair. Additional unpaired histories are marked separately; they do not contribute to paired means. One-seed curves have no band. Both metrics follow the same evaluation rules, with gaps at missing checkpoints and no extrapolation.

401 E.5 Supporting comparators

402 GRPO changes group-reward normalization; UCPO redistributes fresh on-policy advantage; sparse
 403 RLEP-Dr replays verified successes without uniform canonical-key weighting. Fixed Semantic
 404 MaxEnt supplies a fresh-rollout exploration incentive (Appendix D.5). These are controls for
 405 alternative explanations of the retention results.

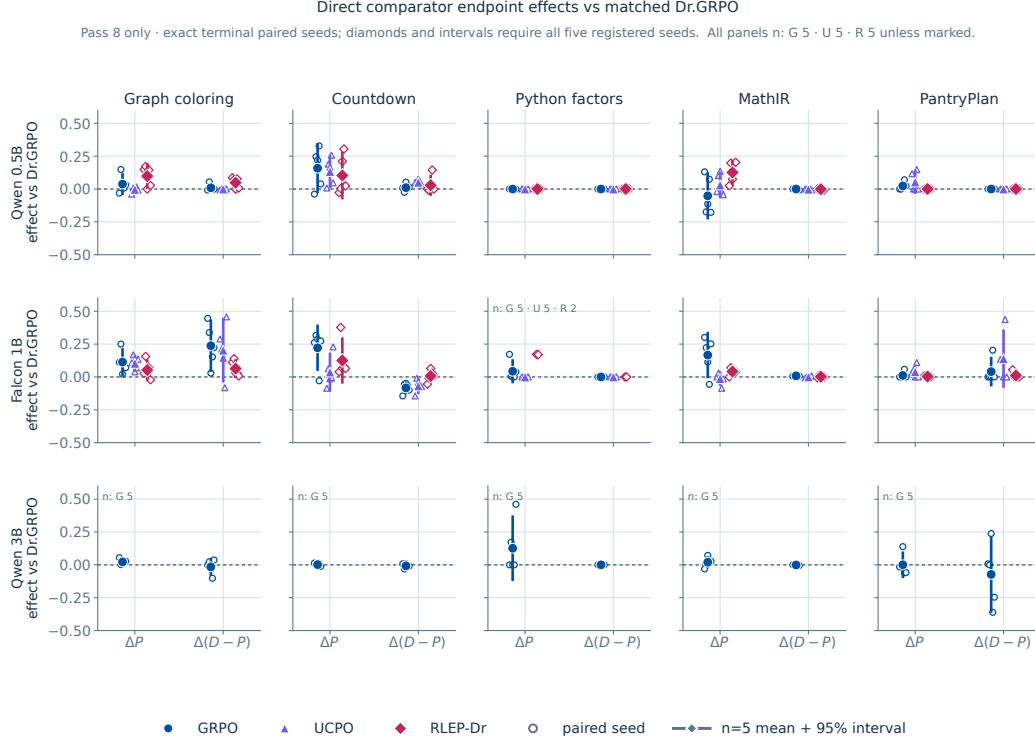


Figure 12: **Direct alternatives produce local rather than uniform final gains.** Each effect subtracts the same-seed Dr.GRPO control from the comparator after eight passes. Effects are shown for pass@8 and extra verified modes beyond the first success. Circles are paired seeds; diamonds and 95% Student-*t* intervals appear only for complete five-seed blocks. Exact *n* is printed, and unsupported comparisons remain blank rather than being extrapolated or imputed at any seed.

406 The alternatives give local rather than uniform benefits. Qwen-0.5B Countdown UCPO raises
 407 *P* by +.133 ([+.005, +.262]) and *B* by +.051 ([+.031, +.071]), whereas its MathIR effects are
 408 inconclusive. MathIR RLEP-Dr raises *P* by +.126 ([+.030, +.222]) with neutral extra modes
 409 (−.002, [−.006, +.003]); its Countdown effects and Falcon Pantry effects are inconclusive. The
 410 full five-seed 3B plain-GRPO blocks likewise have intervals including zero on both correctness and
 411 extra modes. RLEP’s frozen offline pool was collected at temperature .7, top-*p* = .95; all reported
 412 evaluations use the common temperature-1, top-*p* = 1 contract. An offline pool versus an online
 413 bank remains an algorithmic difference.

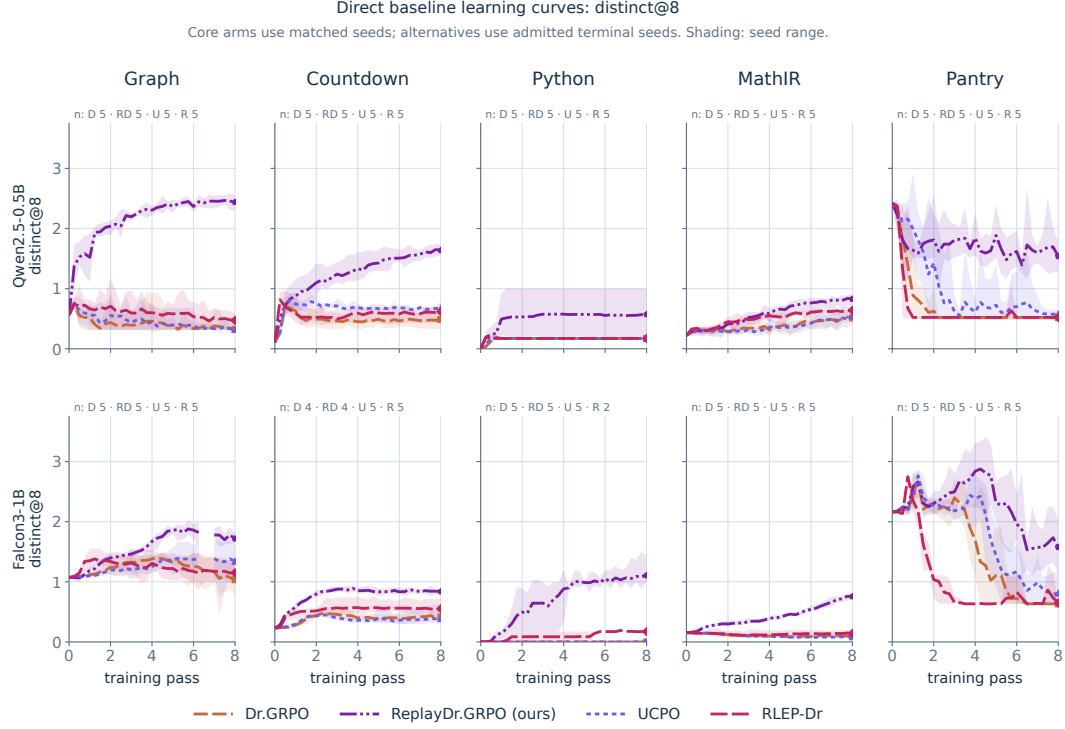


Figure 13: **Verified-mode trajectories for the supporting comparators.** Rows show Qwen2.5-0.5B and Falcon3-1B; columns are the five domains. UCPO and sparse RLEP-Dr are compared with Dr.GRPO and ReplayDr.GRPO at these two smaller scales. Labels give exact method-specific seed counts. Lines are descriptive distinct@8 means and bands are seed ranges. The corresponding accuracy curves use identical populations in Figure 14.

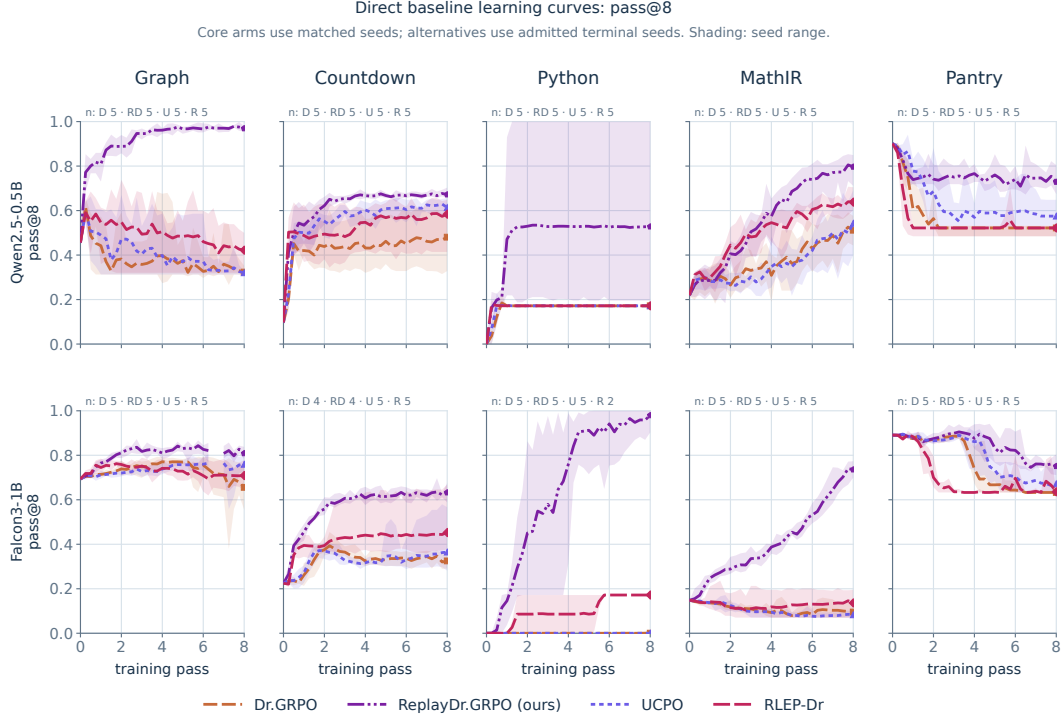


Figure 14: **Accuracy trajectories for the supporting comparators.** The same model rows, domains, methods, terminal seeds, and registered checkpoints as Figure 13, now measuring pass@8. Bands show the seed range and the vertical axes span $[0, 1]$ throughout. MaxRL and ReplayMaxRL at all three scales are compared on both metrics in Figures 9 and 10.

414 E.6 Uniform versus fresh-frequency replay

415 The weighting ablation retains the same bank construction and update code, changing only the within-
 416 bank target: uniform weights over retained keys versus weights proportional to validator-positive
 417 fresh-rollout counts. Policy-dependent bank contents can subsequently differ. Positive contrasts below
 418 favor uniform weighting. The registered co-primary contrasts are ΔB and ΔP ; the five-domain
 419 Qwen2.5-0.5B aggregate first averages domain effects within each matched seed, giving equal domain
 420 weight.

421 Complete-block intervals enumerate all $5^5 = 3,125$ ordered bootstrap resamples of the five paired
 422 training seeds and use linearly interpolated 2.5th and 97.5th percentiles. A resampled seed carries its
 423 complete domain vector when estimating the five-domain aggregate. The endpoints and aggregate
 424 were preregistered; the exact seed-resampling and percentile conventions are analysis choices. These
 425 nominal intervals quantify seed variability within the fixed benchmark, without resampling prompts
 426 or domains or adjusting across comparisons.

Table 9: **Uniform versus fresh-frequency replay at pass 8.** Effects are uniform minus frequency weighting. Brackets give the paired percentile bootstrap interval for ΔB over all 5^5 ordered resamples in complete blocks. Every observed treatment endpoint in this snapshot passes the persisted replay-weight telemetry audit. Partial blocks retain exact seed counts. Paired model exports.

Model	Domain	Weighting	$n/5$	ΔP	ΔD	ΔB [95% interval]
Qwen-0.5B	Graph	Uniform	5/5	+0.066	+0.662	+0.596 [+0.513, +0.687]
Qwen-0.5B	Countdown	Uniform	5/5	+0.022	+0.661	+0.639 [+0.544, +0.733]
Qwen-0.5B	Python	Uniform	5/5	-0.077	-0.040	+0.037 [-0.012, +0.123]
Qwen-0.5B	MathIR	Uniform	5/5	-0.005	+0.014	+0.019 [-0.001, +0.038]
Qwen-0.5B	PantryPlan	Uniform	5/5	+0.043	+0.341	+0.298 [+0.174, +0.400]
Falcon-1B	Graph	Uniform	5/5	+0.045	+0.278	+0.233 [+0.182, +0.320]
Falcon-1B	PantryPlan	Uniform	5/5	-0.021	-0.167	-0.146 [-0.564, +0.414]
Qwen-3B	Graph	Uniform	4/5	-0.006	+0.063	+0.070
Qwen-3B	PantryPlan	Uniform	5/5	+0.011	+0.219	+0.207 [+0.038, +0.377]

The registered Qwen2.5-0.5B five-domain effect is +.318 extra modes per prompt ([+.272, +.358]), with positive within-seed domain averages in all five seeds. Graph, Countdown, and PantryPlan drive this result; Python and MathIR remain inconclusive. The correctness effect is +.010 ([-.119, +.139]), establishing neither a gain nor equivalence. Python’s correctness effects range from -.781 to +.828 across seeds despite small extra-mode effects in four seeds.

Falcon Graph also favors uniform weighting on extra modes (+.233, [+0.182, +.320]), but its mean sampled correctness decreases by -.0152 ([-.0214, -.0090]); its P interval includes zero. Falcon Pantry is inconclusive: $\Delta B = -.146$ ([-.564, +.414]) and $\Delta P = -.021$ ([-.063, +.026]). Qwen2.5-3B Graph has a descriptive extra-mode effect of +.070 at four pairs. Pantry is complete at five pairs, with $\Delta B = +.207$ ([+.038, +.377]) and inconclusive correctness (+.011, [-.020, +.043]). All observed treatment endpoints pass the persisted replay-weight telemetry audit. The evidence supports a contribution from key balancing in the stated domains, not a uniform accuracy or breadth advantage across every model-domain block.

Evidence boundary. The primary effects are within-domain, within-scale paired comparisons. They do not establish a scaling law, a positive factorial interaction, or preservation of every previously discovered mode. Partial domains are reported as observed and do not support complete-block generality.

F Collapse and Replay Mechanism Checks

F.1 Conditional concentration from saved training outputs

This retrospective analysis distinguishes the distribution of correct keys from the probability of producing a correct output. If $P = 0$, C is undefined and $B_2 = 0$. For a fixed prompt with correct mass $P > 0$, write $q_c = \Pr[V(Y, s_x) = c \mid V(Y, s_x) \neq \perp]$ and define

$$C(q) = \sum_c q_c^2, \quad B_2 = 2P - P^2 C(q). \quad (8)$$

The first quantity is the probability that two independent correct outputs share a key. The second identity follows by expanding $B_2 = \sum_c [1 - (1 - Pq_c)^2]$. It makes explicit why sampled breadth can change through correctness or conditional concentration. A decrease in C alone does not imply majorization, larger total support, or recovery of any particular mode. In particular, observed collisions cannot establish that an unobserved key has exactly zero probability under the policy.

Connection to the conditional theory. Under the assumptions of Theorem L.3, the time change in its proof gives $dq_c/d\tau = q_c(q_c - C)$. Consequently,

$$\frac{dC}{d\tau} = 2 \left(\sum_c q_c^3 - C^2 \right) = 2 \operatorname{Var}_{c \sim q}(q_c) \geq 0. \quad (9)$$

The variance vanishes exactly when q is uniform on its positive support. Thus concentration is monotone along this ideal categorical flow, with strict increase whenever the positive coordinates are unequal. The identity provides a theoretical basis for the diagnostic; it does not establish monotonicity for the implemented neural optimizer or a corresponding monotonic decrease under finite-bank replay.

Estimator and identifiable population. For $R \geq 2$ correct samples with per-key counts n_c , use the pair statistic

$$\hat{C} = \frac{\sum_c n_c(n_c - 1)}{R(R - 1)}. \quad (10)$$

For independent samples from one fixed prompt and policy, conditioning on $R = r$ leaves r independent correct labels with distribution q . Every ordered pair therefore agrees with probability $C(q)$, proving $\mathbb{E}[\hat{C} \mid R = r] = C(q)$ for $r \geq 2$. This is a promptwise identity; it does not prove unbiasedness after joint selection using other conditions or dependent prompts. We leave $R < 2$ undefined, rather than assigning a concentration value to insufficient correct observations.

For each paired training seed s , let E_s contain exactly the prompts with a defined estimate in both conditions. The reported seed contrast is $|E_s|^{-1} \sum_{x \in E_s} (\hat{C}_{B,x} - \hat{C}_{A,x})$ when E_s is nonempty. We then weight defined training seeds equally, preserving all registered blocks and their missingness. Before/after contrasts use each method’s own initial checkpoint and step 3,072; replay contrasts compare matched terminal conditions. No other method substitutes for a missing initial evaluation. Five defined seeds receive nominal Student- t intervals; partial blocks are descriptive. These intervals describe the measured seed means and are not simultaneous guarantees or uncertainty over unseen prompts.

The hosted analysis uses the same promptwise collision concept but pools correct pairs. At a common sampling budget, the ratio of expected colliding and correct pair counts is $\sum_x P_x^2 C(q_x) / \sum_x P_x^2$. Its finite-sample ratio is not generally unbiased for that target. The training contrast here weights eligible prompts equally. Changing correctness can change either eligibility or hosted pair weights, so the two aggregate values are not directly interchangeable.

Retrospective protocol and source coverage. The estimator and comparisons were recorded before the initial concentration contrasts, with the sampling-stream amendment also preceding those effects. The paper’s source census was then refreshed during the analysis. A separately recorded extension, declared after inspecting the initial results, includes all thirteen newly available terminal checkpoints and withdraws one newly conflicted initial checkpoint. It preserves the original results and source snapshot. The final collection contains 475 registered runs and 892 available checkpoint evaluations, with 74 Dr.GRPO and 75 MaxRL Level-1 replay pairs before conditional eligibility. Complete terminal coverage does not supply missing initial outputs or make every conditional contrast identifiable. Both analysis phases, exact source-prefix hashes, and the extension’s timing are retained with the result artifact for reproducibility.

Sampling-stream audit. Four saved eight-output groups do not provide 32 independently seeded responses. The recorded parent seeds increment by one; the audited vLLM V0 sampler increments each child’s seed by its output index. The resulting nominal stream ranges overlap, giving eleven distinct identifiers. For each prompt and checkpoint we retain the earliest draw/output position for every identifier, independently of correctness or key identity. Repeated positions sometimes disagree, so identical seed integers are not treated as a proof of identical outputs. Archived actor source is available for 150 of the 400 factorial cells and none of the 75 additional GRPO cells; historical dependency identity is not cryptographically established for every run. The stream mapping rests on the audited implementation and recorded neutral request contract, with the historical-runtime assumption stated explicitly.

The primary eleven-stream comparisons reuse nominal streams across conditions and remain descriptive after common eligibility. Two additional orientations compare the lower five identifiers in condition A with the upper six in B, then reverse the assignment. Each orientation retains its own eligible prompts and seed counts. Shared identifiers also recur across prompts; neither their separation across conditions nor these selected-population means establishes independent prompt observations. Independent fixed-law sampling is an explicit assumption of the estimator identity, not a conclusion of the source audit.

Original mean@8, pass@8, and distinct@8 remain averages over the four intact groups, reconstructed from all saved verified keys. Cross-group overlap affects dependence, not linearity of those marginal averages. The complete result artifact also retains each intact-group collision comparison, a fixed-first-group check, fixed-across-seed prompt intersections, and four-condition change differences. A naive concatenated 32-position collision is retained solely as a reuse diagnostic. These views are not treated as independent replications or substituted for missing cells.

Findings and their limits. In Graph and PantryPlan, all twelve Dr.GRPO/GRPO before/after comparisons across the three scales show increased conditional concentration, with nominal intervals above zero. Correctness also increases on those same eligible prompts. Graph’s increases range from .098 to .274; PantryPlan’s range from .447 to .887. The two disjoint orientations agree in direction in eleven of these twelve blocks; Falcon Graph with GRPO is the exception. The evidence therefore supports observed training-induced concentration in these tasks, with a visible sampling sensitivity, rather than a universal claim about every RLVR run or exact disappearance of modes.

ReplayDr.GRPO lowers concentration relative to Dr.GRPO in all six Graph and PantryPlan comparisons: changes range from $-.756$ to $-.273$ in Graph and $-.433$ to $-.300$ in PantryPlan. Both disjoint orientations agree in direction in all six, and their primary nominal intervals lie below zero. These comparisons also have lower per-sample correctness on their jointly eligible prompts. They establish a change in the conditional correct-key distribution on that observable population; they do not hold correctness fixed or imply that every accuracy measure improves on the same prompts. Full-test pass@8 and sampled breadth remain the separately reported primary outcomes of the replay experiment.

The other domains constrain the interpretation. Dr.GRPO/GRPO Python initial outputs provide no jointly eligible prompts at this budget, so their longitudinal collision contrast is unidentified. Some Countdown initial comparisons use only one or two eligible prompts per seed. Several MathIR conditions already have observed collision one, leaving little detectable change. MaxRL comparisons, incomplete saved-output cohorts, and the two sampling orientations are reported explicitly in the following tables; we do not infer a universal concentration benefit from raw breadth gains.

In the completed 3B MaxRL replay cohort, Countdown, Python, and PantryPlan show lower concentration with nominal intervals below zero; MathIR’s measured contrast is zero. Graph’s primary contrast is negative, but its two disjoint orientations disagree in direction, as they do for Falcon Graph with MaxRL. At Level 2, both Graph replay contrasts are negative with agreement across orientations. Partial PantryPlan comparisons instead show increased collision: $+ .490$ for ReplayDr.GRPO from one defined seed with sixteen eligible prompts, and $+ 1.000$ for ReplayMaxRL from one seed with one eligible prompt. These sparse observations provide no complete-block conclusion and remain visible rather than being dropped from a claim of uniformly beneficial replay.

F.2 Breadth before and after verifier-only training

The problem precheck evaluates unmodified Dr.GRPO and GRPO at passes 0 and 8 on all three models and five domains: 150 complete runs, each with four fixed $K = 8$ draws. The five-domain macro-average of extra verified modes falls by 98.2%–99.4% at Qwen2.5-0.5B and 41.5%–71.6% at larger scales. Correctness falls in 10 of the 30 domain–method–scale comparisons, while macro raw distinct@8 ends above its initial value at Qwen2.5-3B. This distinguishes loss of multiple correct outcomes from the broader question of whether any correct outcome is sampled.

Initial evaluations are measured within each method. Qwen2.5-3B MathIR and Pantry E80-R1 seeds 71–74 carry slightly different initial requests from plain GRPO (.154 versus .162, and .832 versus .828 on P). Those cells therefore have separate initial references.

Graph and PantryPlan contain substantial initial breadth at every scale; the other three domains begin with at most .011 extra modes per prompt. A retained-breadth percentage is therefore reported only when the initial extra-mode count exceeds .05. Raw distinct@8 ends below its initial value in six of the 15 Dr.GRPO domains. ReplayDr.GRPO does so in two available domain summaries, both PantryPlan; Falcon Countdown uses four admissible seeds. These observations concern sampled held-out support, not the survival histories of particular stored modes.

Uniform-action references are meaningful for the enumerable Graph and Pantry representations. They are not defined for open-ended programs or derivations. Pantry’s bit-uniform reference exceeds every

Domain	Objective	n	$\Delta \hat{C}$	nominal 95% interval	$ E_s $	split 0 (n)	split 1 (n)
Graph	Dr.GRPO	5	+0.242	[+0.110, +0.373]	17–40	+0.257 (5)	+0.284 (5)
Graph	GRPO	5	+0.221	[+0.097, +0.344]	15–28	+0.268 (5)	+0.287 (5)
Graph	MaxRL	5	+0.145	[-0.033, +0.324]	18–40	+0.115 (5)	+0.174 (5)
Graph	ReplayDr	5	-0.507	[-0.558, -0.456]	45–49	-0.519 (5)	-0.465 (5)
Graph	ReplayMaxRL	5	-0.399	[-0.458, -0.339]	40–51	-0.436 (5)	-0.348 (5)
Countdown	Dr.GRPO	5	+1.000	[+1.000, +1.000]	1–2	– (0)	+1.000 (5)
Countdown	GRPO	5	+0.964	[+0.863, +1.065]	1–2	– (0)	+0.920 (5)
Countdown	MaxRL	4	+0.977	–	0–2	– (0)	+1.000 (4)
Countdown	ReplayDr	5	+0.467	[+0.117, +0.816]	1–3	– (0)	+0.713 (4)
Countdown	ReplayMaxRL	5	+0.420	[+0.359, +0.480]	1–2	– (0)	+0.493 (5)
Python	Dr.GRPO	0	–	–	0	– (0)	– (0)
Python	GRPO	0	–	–	0	– (0)	– (0)
Python	MaxRL	0	–	–	0	– (0)	– (0)
Python	ReplayDr	0	–	–	0	– (0)	– (0)
Python	ReplayMaxRL	0	–	–	0	– (0)	– (0)
MathIR	Dr.GRPO	5	+0.000	[+0.000, +0.000]	5–8	+0.000 (5)	+0.000 (1)
MathIR	GRPO	5	+0.040	[-0.071, +0.151]	4–10	+0.000 (5)	+0.250 (2)
MathIR	MaxRL	5	+0.022	[-0.039, +0.084]	5–9	+0.000 (5)	+0.067 (5)
MathIR	ReplayDr	5	+0.059	[-0.046, +0.163]	8–11	+0.000 (5)	+0.367 (2)
MathIR	ReplayMaxRL	5	+0.063	[-0.009, +0.134]	9–11	+0.000 (5)	+0.250 (5)
Pantry	Dr.GRPO	5	+0.833	[+0.819, +0.847]	54–64	+1.000 (5)	+1.000 (5)
Pantry	GRPO	5	+0.849	[+0.845, +0.853]	64–65	+1.000 (5)	+1.000 (5)
Pantry	MaxRL	5	+0.859	[+0.836, +0.881]	54–65	+1.000 (5)	+1.000 (5)
Pantry	ReplayDr	5	+0.413	[+0.345, +0.481]	78–89	+0.543 (5)	+0.538 (5)
Pantry	ReplayMaxRL	5	+0.446	[+0.324, +0.567]	80–88	+0.593 (5)	+0.550 (5)

Table 10: **Initial to final: Qwen2.5-0.5B, Level 1.** Positive values indicate greater concentration. Equal prompt weights use $R \geq 2$ correct representatives in both conditions, separately for each contrast; $|E_s|$ is the range of eligible prompt counts out of 128 across available seed pairs. n counts seeds with a defined estimate, from five registered seeds. Each disjoint orientation has its own eligibility and displayed n . Intervals describe measured seed variability, are unadjusted and untruncated, and are omitted unless all five seeds are defined. Dashes retain unavailable or unidentified comparisons. Every registered comparison remains represented.

Domain	Objective	n	$\Delta \hat{C}$	nominal 95% interval	$ E_s $	split 0 (n)	split 1 (n)
Graph	Dr.GRPO	5	+0.233	[+0.124, +0.343]	41–54	+0.059 (5)	+0.428 (5)
Graph	GRPO	5	+0.098	[+0.026, +0.170]	48–56	-0.186 (5)	+0.273 (5)
Graph	MaxRL	3	-0.113	–	48–52	-0.083 (3)	+0.039 (3)
Graph	ReplayDr	5	-0.007	[-0.065, +0.050]	37–45	+0.048 (5)	+0.045 (5)
Graph	ReplayMaxRL	2	-0.062	–	51–52	-0.300 (2)	-0.015 (2)
Countdown	Dr.GRPO	5	+0.022	[-0.038, +0.082]	6–8	+0.000 (5)	– (0)
Countdown	GRPO	5	+0.180	[+0.128, +0.231]	6–11	+0.126 (5)	– (0)
Countdown	MaxRL	2	+0.182	–	11	+0.143 (2)	– (0)
Countdown	ReplayDr	4	+0.035	–	10–11	+0.158 (4)	– (0)
Countdown	ReplayMaxRL	1	+0.135	–	11	+0.143 (1)	– (0)
Python	Dr.GRPO	0	–	–	0	– (0)	– (0)
Python	GRPO	0	–	–	0	– (0)	– (0)
Python	MaxRL	0	–	–	0	– (0)	– (0)
Python	ReplayDr	0	–	–	0	– (0)	– (0)
Python	ReplayMaxRL	0	–	–	–	– (0)	– (0)
MathIR	Dr.GRPO	5	+0.105	[+0.088, +0.122]	6	– (0)	+0.098 (5)
MathIR	GRPO	5	+0.054	[-0.099, +0.206]	3–7	– (0)	+0.233 (5)
MathIR	MaxRL	1	+0.278	–	6	+1.000 (1)	+0.133 (1)
MathIR	ReplayDr	5	+0.086	[+0.026, +0.146]	6–7	– (0)	+0.111 (5)
MathIR	ReplayMaxRL	2	+0.167	–	5–6	+1.000 (2)	+0.000 (2)
Pantry	Dr.GRPO	5	+0.886	[+0.886, +0.886]	76	+1.000 (5)	+1.000 (5)
Pantry	GRPO	5	+0.863	[+0.798, +0.928]	76	+0.945 (5)	+1.000 (5)
Pantry	MaxRL	1	+0.886	–	76	+1.000 (1)	+1.000 (1)
Pantry	ReplayDr	4	+0.437	–	75–83	+0.451 (4)	+0.618 (4)
Pantry	ReplayMaxRL	2	+0.746	–	76–78	+0.740 (2)	+0.978 (2)

Table 11: **Initial to final: Falcon3-1B, Level 1.** Positive values indicate greater concentration. Equal prompt weights use $R \geq 2$ correct representatives in both conditions, separately for each contrast; $|E_s|$ is the range of eligible prompt counts out of 128 across available seed pairs. n counts seeds with a defined estimate, from five registered seeds. Each disjoint orientation has its own eligibility and displayed n . Intervals describe measured seed variability, are unadjusted and untruncated, and are omitted unless all five seeds are defined. Dashes retain unavailable or unidentified comparisons. Every registered comparison remains represented.

Domain	Objective	n	$\Delta\widehat{C}$	nominal 95% interval	$ E_s $	split 0 (n)	split 1 (n)
Graph	Dr.GRPO	5	+0.274	[+0.232, +0.316]	44-47	+0.030 (5)	+0.339 (5)
Graph	GRPO	5	+0.274	[+0.207, +0.341]	46-49	+0.046 (5)	+0.364 (5)
Graph	MaxRL	5	+0.072	[-0.002, +0.145]	45-55	-0.208 (5)	+0.215 (5)
Graph	ReplayDr	5	-0.038	[-0.060, -0.016]	48-53	-0.227 (5)	+0.121 (5)
Graph	ReplayMaxRL	5	-0.145	[-0.198, -0.093]	46-52	-0.220 (5)	+0.165 (5)
Countdown	Dr.GRPO	5	+0.160	[+0.109, +0.211]	11-12	+0.200 (5)	-0.016 (5)
Countdown	GRPO	5	+0.182	[+0.182, +0.182]	11	+0.200 (5)	+0.000 (5)
Countdown	MaxRL	5	+0.056	[-0.046, +0.159]	9-11	+0.050 (5)	+0.000 (5)
Countdown	ReplayDr	5	+0.020	[-0.060, +0.099]	11-15	+0.169 (5)	-0.071 (5)
Countdown	ReplayMaxRL	5	-0.100	[-0.145, -0.055]	10-12	+0.000 (5)	-0.050 (5)
Python	Dr.GRPO	0	-	-	0	- (0)	- (0)
Python	GRPO	0	-	-	0	- (0)	- (0)
Python	MaxRL	5	+0.000	[+0.000, +0.000]	2-9	+0.000 (2)	+0.000 (5)
Python	ReplayDr	0	-	-	0	- (0)	- (0)
Python	ReplayMaxRL	5	-0.023	[-0.050, +0.003]	9	+0.000 (5)	-0.040 (5)
MathIR	Dr.GRPO	5	+0.000	[+0.000, +0.000]	11-14	+0.000 (5)	+0.000 (5)
MathIR	GRPO	5	+0.000	[+0.000, +0.000]	9-11	+0.000 (5)	+0.000 (5)
MathIR	MaxRL	5	+0.000	[+0.000, +0.000]	7-14	+0.000 (5)	+0.000 (5)
MathIR	ReplayDr	4	-0.012	-	14	+0.000 (4)	+0.000 (4)
MathIR	ReplayMaxRL	5	+0.000	[+0.000, +0.000]	12-15	+0.000 (5)	+0.000 (5)
Pantry	Dr.GRPO	5	+0.447	[+0.266, +0.627]	49-67	+0.478 (5)	+0.245 (5)
Pantry	GRPO	5	+0.469	[+0.411, +0.526]	49-62	+0.514 (5)	+0.271 (5)
Pantry	MaxRL	5	+0.431	[+0.297, +0.565]	59-69	+0.394 (5)	+0.246 (5)
Pantry	ReplayDr	5	+0.184	[+0.081, +0.288]	64-71	+0.087 (5)	+0.037 (5)
Pantry	ReplayMaxRL	5	+0.142	[+0.018, +0.266]	68-75	+0.046 (5)	-0.015 (5)

Table 12: **Initial to final: Qwen2.5-3B, Level 1.** Positive values indicate greater concentration. Equal prompt weights use $R \geq 2$ correct representatives in both conditions, separately for each contrast; $|E_s|$ is the range of eligible prompt counts out of 128 across available seed pairs. n counts seeds with a defined estimate, from five registered seeds. Each disjoint orientation has its own eligibility and displayed n . Intervals describe measured seed variability, are unadjusted and untruncated, and are omitted unless all five seeds are defined. Dashes retain unavailable or unidentified comparisons. Every registered comparison remains represented.

Domain	Objective	n	$\Delta\widehat{C}$	nominal 95% interval	$ E_s $	split 0 (n)	split 1 (n)
Graph	Dr.GRPO	5	+0.625	[+0.295, +0.955]	2-10	+0.633 (2)	+0.860 (5)
Graph	MaxRL	5	+0.452	[+0.387, +0.518]	6-10	+0.767 (4)	+0.484 (5)
Graph	ReplayDr	5	-0.204	[-0.282, -0.126]	9-11	+0.033 (5)	-0.372 (5)
Graph	ReplayMaxRL	5	-0.128	[-0.262, +0.006]	9-11	+0.333 (5)	-0.267 (5)
Countdown	Dr.GRPO	1	+0.000	-	0-1	- (0)	- (0)
Countdown	MaxRL	1	+0.000	-	0-1	- (0)	- (0)
Countdown	ReplayDr	5	+0.000	[+0.000, +0.000]	1	- (0)	- (0)
Countdown	ReplayMaxRL	3	+0.000	-	0-1	- (0)	- (0)
Python	Dr.GRPO	5	+0.302	[+0.302, +0.302]	32	+0.000 (5)	+0.378 (5)
Python	MaxRL	5	+0.302	[+0.302, +0.302]	32	+0.000 (5)	+0.378 (5)
Python	ReplayDr	5	+0.094	[-0.259, +0.448]	32	-0.224 (5)	+0.146 (5)
Python	ReplayMaxRL	5	+0.199	[-0.088, +0.485]	32	-0.070 (5)	+0.258 (5)
MathIR	Dr.GRPO	3	+0.000	-	0-5	- (0)	+0.000 (3)
MathIR	MaxRL	4	+0.000	-	0-6	- (0)	+0.000 (4)
MathIR	ReplayDr	5	+0.000	[+0.000, +0.000]	6	- (0)	+0.000 (5)
MathIR	ReplayMaxRL	4	+0.000	-	6	- (0)	+0.000 (4)
Pantry	Dr.GRPO	2	-0.867	-	1-3	-0.500 (1)	-1.000 (1)
Pantry	MaxRL	1	+0.000	-	1	- (0)	- (0)
Pantry	ReplayDr	2	-0.049	-	15-16	-0.075 (2)	-0.050 (2)
Pantry	ReplayMaxRL	1	+0.092	-	19	+0.097 (1)	-0.222 (1)

Table 13: **Initial to final: Qwen2.5-0.5B, Level 2.** Positive values indicate greater concentration. Equal prompt weights use $R \geq 2$ correct representatives in both conditions, separately for each contrast; $|E_s|$ is the range of eligible prompt counts out of 128 across available seed pairs. n counts seeds with a defined estimate, from five registered seeds. Each disjoint orientation has its own eligibility and displayed n . Intervals describe measured seed variability, are unadjusted and untruncated, and are omitted unless all five seeds are defined. Dashes retain unavailable or unidentified comparisons. Every registered comparison remains represented.

Domain	Objective	n	$\Delta\hat{C}$	nominal 95% interval	$ E_s $	split 0 (n)	split 1 (n)
Graph	Dr.GRPO	5	-0.756	[-0.831, -0.680]	37–40	-0.853 (5)	-0.807 (5)
Graph	MaxRL	5	-0.587	[-0.759, -0.414]	41–73	-0.625 (5)	-0.605 (5)
Countdown	Dr.GRPO	5	-0.529	[-0.605, -0.452]	42–78	-0.529 (5)	-0.535 (5)
Countdown	MaxRL	5	-0.547	[-0.635, -0.459]	45–84	-0.578 (5)	-0.530 (5)
Python	Dr.GRPO	5	-0.090	[-0.341, +0.160]	22	-0.115 (5)	-0.062 (5)
Python	MaxRL	5	-0.266	[-0.577, +0.044]	22	-0.298 (5)	-0.272 (5)
MathIR	Dr.GRPO	5	-0.008	[-0.017, +0.001]	44–68	-0.006 (5)	-0.008 (5)
MathIR	MaxRL	5	-0.005	[-0.013, +0.003]	32–56	-0.007 (5)	-0.002 (5)
Pantry	Dr.GRPO	5	-0.433	[-0.518, -0.347]	54–66	-0.442 (5)	-0.494 (5)
Pantry	MaxRL	5	-0.417	[-0.547, -0.288]	53–71	-0.363 (5)	-0.508 (5)

Table 14: **Replay minus control at the final checkpoint: Qwen2.5-0.5B, Level 1.** Positive values indicate greater concentration. Equal prompt weights use $R \geq 2$ correct representatives in both conditions, separately for each contrast; $|E_s|$ is the range of eligible prompt counts out of 128 across available seed pairs. n counts seeds with a defined estimate, from five registered seeds. Each disjoint orientation has its own eligibility and displayed n . Intervals describe measured seed variability, are unadjusted and untruncated, and are omitted unless all five seeds are defined. Dashes retain unavailable or unidentified comparisons. Every registered comparison remains represented.

Domain	Objective	n	$\Delta\hat{C}$	nominal 95% interval	$ E_s $	split 0 (n)	split 1 (n)
Graph	Dr.GRPO	5	-0.309	[-0.523, -0.094]	31–64	-0.129 (5)	-0.452 (5)
Graph	MaxRL	5	-0.021	[-0.040, -0.002]	95–100	+0.100 (5)	-0.013 (5)
Countdown	Dr.GRPO	4	-0.168	–	33–42	-0.139 (4)	-0.233 (4)
Countdown	MaxRL	5	-0.063	[-0.079, -0.047]	78–81	-0.062 (5)	-0.063 (5)
Python	Dr.GRPO	0	–	–	0	– (0)	– (0)
Python	MaxRL	5	-0.093	[-0.350, +0.165]	22–128	-0.039 (5)	-0.090 (5)
MathIR	Dr.GRPO	5	+0.004	[-0.007, +0.015]	9–14	+0.000 (5)	+0.000 (5)
MathIR	MaxRL	5	-0.001	[-0.010, +0.009]	40–72	+0.000 (5)	+0.000 (5)
Pantry	Dr.GRPO	5	-0.404	[-0.600, -0.209]	74–81	-0.435 (5)	-0.352 (5)
Pantry	MaxRL	5	-0.129	[-0.223, -0.035]	81	-0.203 (5)	-0.043 (5)

Table 15: **Replay minus control at the final checkpoint: Falcon3-1B, Level 1.** Positive values indicate greater concentration. Equal prompt weights use $R \geq 2$ correct representatives in both conditions, separately for each contrast; $|E_s|$ is the range of eligible prompt counts out of 128 across available seed pairs. n counts seeds with a defined estimate, from five registered seeds. Each disjoint orientation has its own eligibility and displayed n . Intervals describe measured seed variability, are unadjusted and untruncated, and are omitted unless all five seeds are defined. Dashes retain unavailable or unidentified comparisons. Every registered comparison remains represented.

Domain	Objective	n	$\Delta\hat{C}$	nominal 95% interval	$ E_s $	split 0 (n)	split 1 (n)
Graph	Dr.GRPO	5	-0.273	[-0.299, -0.247]	54–71	-0.363 (5)	-0.151 (5)
Graph	MaxRL	5	-0.162	[-0.261, -0.062]	68–83	-0.169 (5)	+0.059 (5)
Countdown	Dr.GRPO	5	-0.137	[-0.183, -0.090]	70–75	-0.153 (5)	-0.139 (5)
Countdown	MaxRL	5	-0.105	[-0.126, -0.084]	56–76	-0.113 (5)	-0.095 (5)
Python	Dr.GRPO	2	-0.190	–	0–22	-0.221 (2)	-0.141 (2)
Python	MaxRL	5	-0.153	[-0.265, -0.041]	22–81	-0.203 (5)	-0.083 (5)
MathIR	Dr.GRPO	5	-0.010	[-0.022, +0.003]	23–34	-0.017 (5)	+0.003 (5)
MathIR	MaxRL	5	+0.000	[+0.000, +0.000]	21–27	+0.000 (5)	+0.000 (5)
Pantry	Dr.GRPO	5	-0.300	[-0.420, -0.179]	71–83	-0.337 (5)	-0.305 (5)
Pantry	MaxRL	5	-0.292	[-0.319, -0.265]	81–89	-0.355 (5)	-0.237 (5)

Table 16: **Replay minus control at the final checkpoint: Qwen2.5-3B, Level 1.** Positive values indicate greater concentration. Equal prompt weights use $R \geq 2$ correct representatives in both conditions, separately for each contrast; $|E_s|$ is the range of eligible prompt counts out of 128 across available seed pairs. n counts seeds with a defined estimate, from five registered seeds. Each disjoint orientation has its own eligibility and displayed n . Intervals describe measured seed variability, are unadjusted and untruncated, and are omitted unless all five seeds are defined. Dashes retain unavailable or unidentified comparisons. Every registered comparison remains represented.

Domain	Objective	n	$\Delta\hat{C}$	nominal 95% interval	$ E_s $	split 0 (n)	split 1 (n)
Graph	Dr.GRPO	5	-0.577	[-0.736, -0.418]	9–25	-0.599 (5)	-0.544 (5)
Graph	MaxRL	5	-0.405	[-0.472, -0.339]	28–42	-0.262 (5)	-0.416 (5)
Countdown	Dr.GRPO	4	-0.230	–	0–9	-0.281 (4)	-0.292 (4)
Countdown	MaxRL	4	-0.073	–	0–10	+0.000 (3)	-0.118 (4)
Python	Dr.GRPO	5	-0.209	[-0.563, +0.146]	104	-0.219 (5)	-0.234 (5)
Python	MaxRL	5	-0.106	[-0.397, +0.184]	104	-0.076 (5)	-0.121 (5)
MathIR	Dr.GRPO	5	+0.000	[+0.000, +0.000]	12–122	+0.000 (5)	+0.000 (5)
MathIR	MaxRL	5	-0.001	[-0.004, +0.002]	29–125	+0.000 (5)	-0.001 (5)
Pantry	Dr.GRPO	1	+0.490	–	0–16	+0.656 (1)	+0.567 (1)
Pantry	MaxRL	1	+1.000	–	1	– (0)	– (0)

Table 17: **Replay minus control at the final checkpoint: Qwen2.5-0.5B, Level 2.** Positive values indicate greater concentration. Equal prompt weights use $R \geq 2$ correct representatives in both conditions, separately for each contrast; $|E_s|$ is the range of eligible prompt counts out of 128 across available seed pairs. n counts seeds with a defined estimate, from five registered seeds. Each disjoint orientation has its own eligibility and displayed n . Intervals describe measured seed variability, are unadjusted and untruncated, and are omitted unless all five seeds are defined. Dashes retain unavailable or unidentified comparisons. Every registered comparison remains represented.

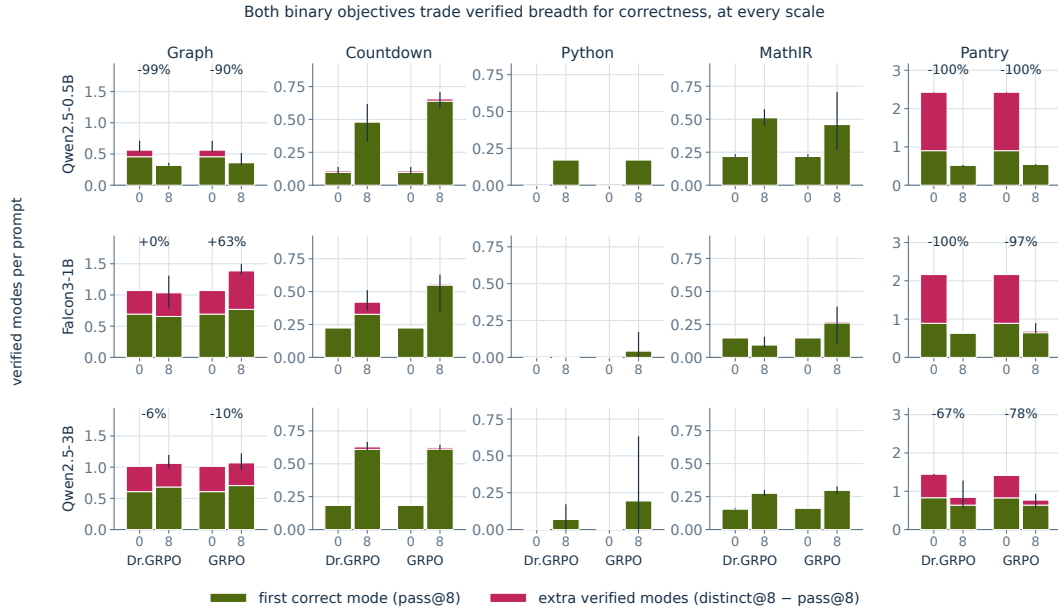


Figure 15: **Verifier-only training reduces average extra verified modes at each tested scale.** Bars split distinct@8 into the first correct mode and every verified mode beyond it, at training pass 0 and pass 8, for unmodified Dr.GRPO then GRPO in each panel. All 150 evaluations are complete at five seeds and four fixed $K = 8$ draws; whiskers are the seed range and pass 0 is measured per method. Badges give the change in extra verified modes, printed only where pass-0 breadth exceeds .05 of them; the other three domains start at .011 or less.

560 final model, so this domain is a support-concentration stress test rather than evidence of complete
561 support coverage.

562 F.3 Fresh-gradient degeneracy and observed output collapse

563 A Dr.GRPO group with identical rewards has exactly zero fresh task gradient. Only mixed groups
564 containing both correct and incorrect responses can supply fresh reward information; optimizer
565 momentum can still move a policy when that task gradient is zero. The fractions of mixed groups
566 over all 3,072 Qwen2.5-0.5B updates, averaged over five seeds, are:

Table 18: **Problem precheck, five-domain macro-averages.** Both verifier-only objectives at pass 0 and pass 8, all three scales, five seeds and four fixed $K = 8$ draws per model–domain–method comparison. “Extra modes” is `distinct@8–pass@8`; “removed” is the fraction of that quantity lost between pass 0 and pass 8. Breadth loss is present at every scale and is far from uniform across scales, and at Falcon3-1B Dr.GRPO macro `pass@8` falls as well.

Model	Dr.GRPO			GRPO		
	pass@8	extra modes	removed	pass@8	extra modes	removed
Qwen2.5-0.5B	0.335 → 0.402	0.329 → 0.002	99.4%	0.335 → 0.435	0.329 → 0.006	98.2%
Falcon3-1B	0.391 → 0.342	0.332 → 0.094	71.6%	0.391 → 0.454	0.332 → 0.135	59.4%
Qwen2.5-3B	0.356 → 0.455	0.205 → 0.120	41.5%	0.356 → 0.490	0.199 → 0.101	49.4%

Table 19: **Absolute references for raw `distinct@8`.** *Uniform* samples the exposed action representation uniformly and is defined only for Graph (27 color strings) and PantryPlan (64 bit masks). Frozen is the matched pass-0 model; the final two columns are pass-8 means. Dashes avoid pretending that an open-ended program or derivation space has a canonical uniform distribution.

Model	Domain	Uniform	Frozen	Dr.GRPO	ReplayDr.GRPO
Qwen2.5-0.5B	Graph coloring	1.633	.561	.325	2.438
	PantryPlan	2.153	2.426	.522	1.539
Falcon3-1B	Graph coloring	1.633	1.070	1.034	1.717
	PantryPlan	2.153	2.162	.633	1.576
Qwen2.5-3B	Graph coloring	1.633	1.012	1.060	1.637
	PantryPlan	2.153	1.445	.841	1.514

Domain	Dr.GRPO	ReplayDr.GRPO
Graph coloring	.144	.954
Countdown	.140	.456
Python factors	.015	.185
MathIR	.364	.453
PantryPlan	.123	.789

567

568 Thus 98.5% of Python control updates carry zero fresh task gradient. Replay supplies a verified-
569 likelihood gradient directly and can also change the policy so that later fresh groups become mixed.

570 A raw-response audit of all five Qwen2.5-0.5B Python control seeds establishes observed output
571 collapse directly. At pass 0, a draw has 7.99 unique raw completions per prompt on average.
572 By steps 288–480, every seed produces 32 byte-identical temperature-1 samples per prompt, also
573 identical to the logged greedy completion, on all 128 prompts. This persists through step 3,072.
574 Terminal draws contain 1.00 unique completion per prompt in the control and 1.97 under replay. A
575 collapsed completion can be incorrect: this is one observed completion per prompt, not one correct
576 key everywhere. Aggregate endpoint equality or zero draw spread alone would not establish that
577 conclusion, nor does this finite audit rule out low-probability tails.

578 These checks explain a vulnerability of the registered sparse binary-reward recipe. They do not
579 establish dominance over methods that keep fresh on-policy signal active. Dynamic mixed-group
580 sampling, larger groups, reference KL, token-entropy regularization, and wider-decoding evaluations
581 remain relevant controls beyond the registered factorial.

582 F.4 Fixed-bank exemplar score trajectories

583 The fixed-bank study tracks Qwen2.5-0.5B ReplayDr.GRPO on Graph for seeds 43–47. Before step
584 384’s fresh group, bank membership and fresh counts are frozen; admission then stops while uniform
585 replay continues through 3,072 updates under the same training recipe. Every frozen exemplar is
586 identified by prompt and canonical-key fingerprints, without selection by its score trajectory. Each
587 scheduled replay visit records mean token log probability s and sequence log probability ℓ . Changes
588 compare an identity’s first and final post-freeze visits; observation steps can differ across identities.

589 All 3,406 frozen identities across 1,575 seed–prompt pairs have finite, aligned scores at every
590 scheduled visit: 8–9 visits per identity and 29,041 observations in total. The pooled median change
591 is $+0.038$ nat/token (95% interval $[+.025, +.048]$), but the 10th percentile is $-.256$ and 2.61% of
592 identities lose more than $.5$ nat/token ($[1.54, 3.77]\%$). The worst intermediate change is -1.328
593 nat/token. On the sequence scale, the median is $+0.154$ nat/sequence ($[+.104, +.198]$), while 21.99%
594 lose more than $.5$ nat/sequence. Lengths range from 4 to 100 tokens, so the two threshold percentages
595 have different meanings.

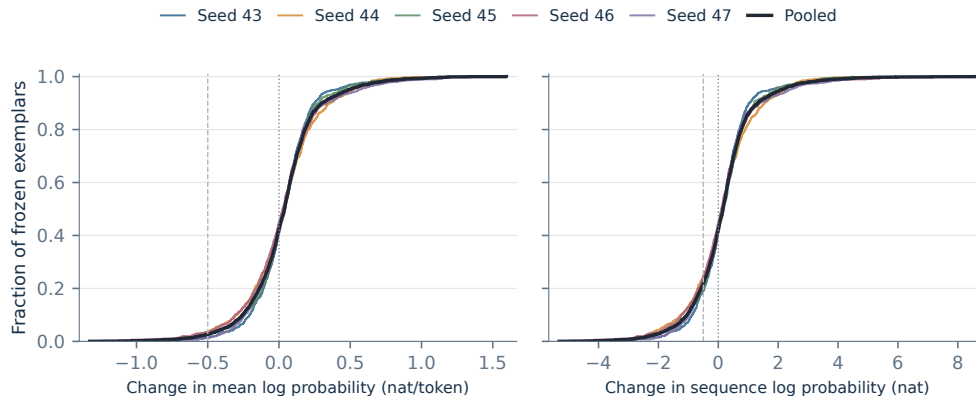


Figure 16: **Fixed-exemplar scores improve centrally but retain a declining tail.** Empirical cumulative distributions show first-to-final post-freeze score changes for every frozen identity, with one curve per seed and the pooled identity distribution. Mean token and sequence scores are separate panels; positive values indicate increased teacher-forced likelihood. Neither panel measures the exact probability of generating a canonical mode during policy sampling.

596 Pooled summaries weight identities equally. Descriptive 95% intervals use 10,000 hierarchical
597 bootstrap draws (random seed 121), resampling prompts within each original seed and then five
598 seeds with replacement; repeated seeds reuse their within-draw prompt sample. Linear empirical
599 quantiles and percentile intervals are analysis choices. The artifact `paper/results/e121_fixed_`
600 `bank_survival.json` retains all seed summaries, identity trajectories, source hashes, and audit
601 diagnostics.

602 These are teacher-forced score surrogates, not exact probabilities of sampling canonical modes. Visits
603 do not cover every intervening update, and the declining tails exclude a claim that every exemplar
604 improves. Without a matched no-replay fixed-bank arm, this study does not identify a causal replay
605 effect. It neither numerically validates Theorem L.6’s probability floor nor certifies a finite-network
606 guarantee. It adds longitudinal measurements of fixed exemplars to the held-out support evidence.

607 G Inference-Only Concentration in a Hosted Model

608 This exploratory evaluation measures the current output distribution of GPT-5.6 Sol under the
609 inherited MODEBENCH interface. It is separate from the matched training interventions: it adds no
610 frontier-model training, replay comparison, or controlled model-size contrast. Only the completed,
611 audited deployment enters these tables.

612 **Protocol and provenance.** The September 11, 2026 run uses the Azure deployment `gpt-5.6-sol`;
613 saved provider headers identify snapshot `gpt-5.6-sol-2026-07-09`. Each of the five domains
614 contributes 128 held-out prompts at each of three levels. Eight stateless requests per prompt yield
615 15,360 responses. Requests use medium reasoning, an 8,192-token output limit, no tools and no
616 conversation state. The service returns temperature 1.0 and top- p .98; neither setting was overridden.
617 These settings and the hosted natural-language interface differ from the local training evaluation’s
618 token-level grammar constraints.

619 Level 1 uses the native E118 evaluation splits retained in E124. Level 2 uses the frozen matched
620 r5 construction; Level 3 uses the matched v3 construction admitted by the completed canonical v3

confirmation. Level 3 is an exploratory test extension here, not a third level of the paper’s training intervention. All cells use the multi-answer split. The Level-2/Level-3 support histograms match a separate fixed-reference Level-1 confirmation reserve, which differs from the native Level-1 split in Graph, Countdown and Python. Prompts differ across levels and are not paired by row index. Calibration against small-model references does not guarantee increasing difficulty for this hosted model.

Exact prompts, dataset identities and hashes, requests, provider-exposed responses, grades, retry receipts, usage and verifier snapshots are retained in the experiment’s evidence directory. All 15,360 intended samples completed, with distinct response identifiers; 107 failed API attempts were retained and recovered by retries. Provider random-number independence and hidden reasoning contents cannot be audited from the exposed responses.

Table 20: **Hosted correctness and concentration under strict grading.** GPT-5.6 Sol, with medium reasoning. Each level weights five domains equally. M and P are mean@8 and pass@8 (%); D is raw distinct@8. C is the percentage of within-prompt correct pairs sharing a mode, pair weighted within each domain. Brackets give pointwise 95% intervals from whole-prompt resampling.

Level	M	P	D	C (95% interval)
1	83.0	95.8	1.71	74.0 [71.9, 76.1]
2	62.9	77.5	1.16	82.3 [80.3, 84.4]
3	68.7	80.3	1.18	84.5 [82.8, 86.2]

Metrics and uncertainty. Per-response correctness is mean@8 in this paper (called pass@1 in the hosted report). Raw distinct@8 counts verified keys across eight draws, including zero for unsolved prompts. For prompt x , let c_x be its number of correct draws and n_{xc} its count for correct key c . The within-cell correct-pair collision estimate is

$$\hat{C} = \frac{\sum_x \sum_c \binom{n_{xc}}{2}}{\sum_x \binom{c_x}{2}}.$$

It describes concentration conditional on successful draws, with prompts weighted by their number of correct pairs. Level summaries average the five domain estimates equally. When the certified correct-key count m_x is available, a uniform-correct-key reference replaces the numerator by $\sum_x \binom{c_x}{2} / m_x$. This differs from a uniform-action baseline and is a descriptive reference, not a normative sampling requirement. Countdown’s verifier accepts modes beyond its enumerated binary-expression library, including unary negation, so its total support is unknown and this reference is omitted. Intervals use 2,000 whole-prompt bootstrap replicates, seed 20260911, resampling within each domain and level and keeping all eight draws together. They are pointwise descriptive intervals.

Formatting sensitivity. Strict grades remain primary. A secondary typography normalizer was introduced after inspecting the first 15 responses. It fixes a frozen set of presentation conventions, including escaped Python/LaTeX syntax, and reuses the original executable validators without repairing calculations or program logic. It recovers 2,580 additional verified outputs; strict successes and their keys are retained. This is a post hoc diagnostic, not a replacement benchmark. All 3,072 primary Python responses passed an independent serial recheck with no grade changes. One cold-start timeout in the secondary grading cache was reconciled with repeated serial verification; the original cache and correction audit remain preserved.

Concentration and level. The strict five-domain collision average is 74.0%, 82.3% and 84.5% at Levels 1–3; the normalized diagnostic gives 74.1%, 81.2% and 83.6%. This aggregate pattern is not a monotonic law within domains. Graph collision is 89.7%, 83.3%, then 95.1%; normalized PantryPlan collision is 46.9%, 64.0%, then 53.0%. Normalized overall correctness also rebounds from 83.2% at Level 2 to 85.7% at Level 3. Thus a higher level label is not equivalent to a measured increase in difficulty for this model, and raw distinct-mode counts also change with correctness and support.

Level-3 Graph separates concentration from failed responses particularly clearly: all 1,024 outputs verify, eight draws produce 1.16 distinct keys on average, and 3,407 of 3,584 correct pairs share

Table 21: **GPT-5.6 Sol under strict executable grading.** Every cell contains 128 prompts and 1,024 responses. M and P are mean@8 and pass@8 (%); D is distinct@8. C is correct-pair collision (% with 95% prompt-bootstrap interval); U is the uniform-correct-key collision reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	99.2	100.0	1.30	89.7 [86.3, 92.8]	18.3
1	Countdown	86.1	98.4	1.77	71.2 [66.1, 75.9]	–
1	Python	40.4	90.6	1.11	81.8 [76.9, 87.2]	0.8
1	MathIR	89.3	89.8	1.38	80.3 [75.4, 85.0]	20.0
1	PantryPlan	100.0	100.0	3.00	46.9 [42.2, 51.5]	6.5
2	Graph	99.3	100.0	1.48	83.3 [79.6, 87.0]	18.9
2	Countdown	65.4	86.7	1.57	64.7 [59.0, 70.6]	–
2	Python	8.5	11.7	0.12	100.0 [100.0, 100.0]	0.7
2	MathIR	86.2	93.0	0.93	100.0 [100.0, 100.0]	20.0
2	PantryPlan	55.3	96.1	1.72	63.6 [56.8, 70.3]	6.9
3	Graph	100.0	100.0	1.16	95.1 [92.4, 97.3]	19.0
3	Countdown	71.2	89.8	1.38	76.4 [71.1, 81.8]	–
3	Python	9.4	14.1	0.14	100.0 [100.0, 100.0]	0.9
3	MathIR	97.7	97.7	0.98	100.0 [100.0, 100.0]	20.0
3	PantryPlan	65.3	100.0	2.22	51.0 [45.3, 57.3]	6.8

Table 22: **Post hoc formatting-normalized diagnostic.** Same samples and metric definitions as Table 21, with fixed presentation normalization before executable verification. The strict table remains primary; normalization was designed after the first 15 responses.

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	99.2	100.0	1.30	89.7 [86.3, 92.8]	18.3
1	Countdown	99.9	100.0	1.93	67.9 [63.3, 72.4]	–
1	Python	92.4	100.0	1.41	85.6 [82.3, 88.8]	1.4
1	MathIR	89.3	89.8	1.38	80.3 [75.4, 85.0]	20.0
1	PantryPlan	100.0	100.0	3.00	46.9 [42.2, 51.5]	6.5
2	Graph	99.3	100.0	1.48	83.3 [79.6, 87.0]	18.9
2	Countdown	98.5	100.0	1.99	63.9 [59.5, 68.4]	–
2	Python	60.7	99.2	1.12	94.8 [91.9, 97.2]	1.1
2	MathIR	86.2	93.0	0.93	100.0 [100.0, 100.0]	20.0
2	PantryPlan	71.3	100.0	1.96	64.0 [58.1, 69.6]	6.8
3	Graph	100.0	100.0	1.16	95.1 [92.4, 97.3]	19.0
3	Countdown	99.0	100.0	1.70	75.7 [71.2, 80.1]	–
3	Python	56.5	98.4	1.09	94.3 [90.8, 97.2]	1.0
3	MathIR	97.7	97.7	0.98	100.0 [100.0, 100.0]	20.0
3	PantryPlan	75.2	100.0	2.35	53.0 [47.3, 58.9]	6.7

661 a key (95.1%). The uniform-correct-key reference is 19.0%; the excess is 76.1 percentage points
 662 (95% interval [73.5, 78.3]). Graph was selected after the GPT run for this illustration because its
 663 correctness is high, support is certified and formatting normalization changes no scores; both tables
 664 report all domains.

665 **Interface and claim limits.** Level-2/Level-3 MathIR instructions prescribe moving the right-side
 666 variable term left, removing the constant and then dividing. Python instructions suggest a nested small-
 667 divisor conditional chain, and PantryPlan instructions prefer particular ingredient families. These
 668 preferences can concentrate the very keys being measured. In addition, Level-1 PantryPlan requests
 669 a support mask whose quantities are supplied by a trusted projection; Levels 2–3 ask the model to
 670 provide quantities. MathIR verifies symbolic action compliance, which can reject a numerically valid
 671 shortcut that omits a required symbolic step. Neither these failures nor the cross-level differences
 672 isolate numerical reasoning ability. Canonical keys describe executed outputs, not hidden reasoning
 673 strategies. Eight draws cannot establish that an unseen mode has zero probability. The evidence
 674 supports concentration under the specified prompts and deployment configuration; it identifies neither
 675 a training cause nor a causal difficulty or model-size effect.

676 **Reproduction record.** The source record is artifacts/frontier_modebench_gpt56sol_
 677 20260911/summary.json. Its hash, complete cell statistics and run settings are copied into
 678 results/frontier_hosted_20260911.json; adjacent generated table bodies supply the dis-
 679 played values. The evidence directory also retains completion_audit.json, prompt_contract_
 680 audit.json, the original requests and responses, and interpretation_notes.md. Hosted-model
 681 results are separate from the training-checkpoint archive and its matched-seed comparisons.

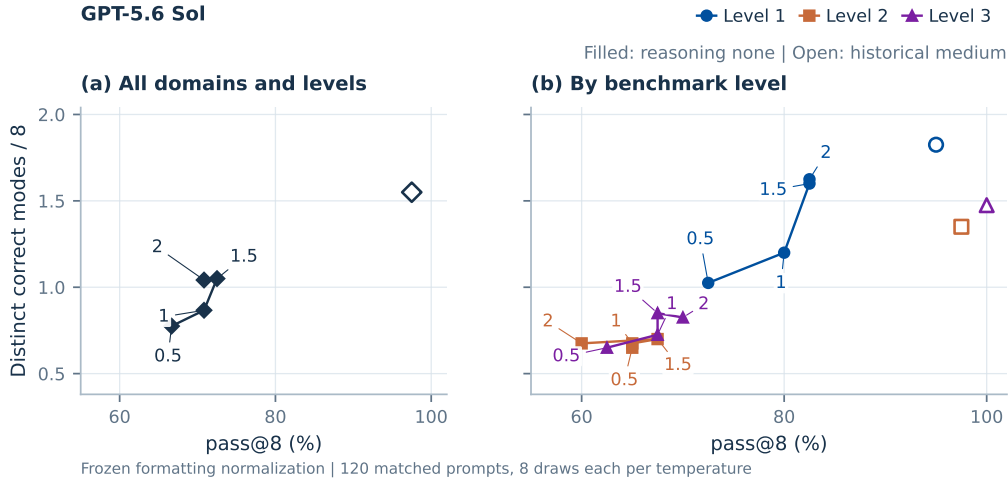


Figure 17: **Temperature traces pass@8 and verified breadth.** GPT-5.6 Sol, with 120 fixed prompts and eight draws each at four temperatures. Left: all domains and levels; right: separate levels, averaging five domains equally. Empirical pass@8 is the fraction with at least one correct draw; distinct@8 averages unique correct modes, including zeroes. Connected points use reasoning none and the frozen formatting normalizer. Open markers show the historical medium-reasoning reference on the same prompts (returned $T = 1$), unconnected because its configuration differs. This does not test temperature robustness with medium reasoning. Appendix H.3 gives pointwise uncertainty intervals for every level and all four measured temperatures.

H Comparison across Hosted Deployments

Figure 7 compares verified output breadth across hosted deployments. Its main display applies the frozen formatting normalizer to every deployment and uses the complete revised-wording Opus 5 Python cohort at all three levels. Every response enters its accuracy and mode-count metrics; outputs are not selected by correctness or provider outcome. The other cells use their original prompts. This is a descriptive comparison across the stated task formulations, rather than a common-prompt model ranking.

The tables and Graph figure below preserve every original-protocol result, including failed draws and native refusals. The separate Python comparison reports the original and revised conditions under both grading rules. These records expose the effect of the prompt change without replacing the original evidence. Level 3 remains an exploratory test extension; no hosted-model training or replay intervention is performed. Original deployments enter the comparison only with a passing integrity audit and all 15,360 responses.

Completed hosted comparison. The admitted deployments are GPT-5.6 Sol, Claude Opus 5, GPT-5.4, Grok 4.3, Kimi K3, Claude Opus 4.8, DeepSeek V4 Pro: 7 complete runs, 107,520 responses, and the same 128 held-out prompts per domain and level with eight stateless samples per prompt. Only runs with a passing completion audit and the identical frozen prompt digest enter these tables. No training, model-side tools or conversation history is added.

Provider configurations. GPT-5.6 Sol: hosted API, requested reasoning setting medium, native output limit 8192. Claude Opus 5: Anthropic Messages, requested reasoning setting medium (requested, provider-specific), native output limit 8192. GPT-5.4: hosted API, requested reasoning setting medium, native output limit 8192. Grok 4.3: hosted API, requested reasoning setting medium, native output limit 8192. Kimi K3: hosted API, requested reasoning setting medium, native output limit 8192. Claude Opus 4.8: Anthropic Messages, requested reasoning setting medium (requested, provider-specific), native output limit 8192. DeepSeek V4 Pro: hosted API, requested reasoning setting medium, native output limit 8192.

Provider effort labels do not establish equal compute, and native token accounting differs. The saved native request and returned usage determine each configuration; provider defaults remain unspecified whenever native responses omit them.

The appendix retains strict executable grades alongside formatting-normalized scores. The main display uses the same frozen formatting rule for every cell and the complete separately collected revised Opus 5 Python condition; all other displayed cells retain the original request wording. The shared typography diagnostic was developed after the first 15 GPT-5.6 Sol responses and frozen before the prospective comparison runs. It retains its post hoc status for GPT-5.6 Sol. Inherited route preferences and the Level-1 PantryPlan interface change limit cross-level interpretation. Countdown has no certified finite-total-support uniform reference. These data establish neither a training cause nor a causal model-size or difficulty effect. All pair-collision intervals are pointwise; overlapping or nonoverlapping cell intervals are not a paired model-difference test.

Collection interruption: DeepSeek V4 Pro. 13 HTTP-200 bodies had no terminal finish reason or visible answer and were preserved as provider protocol failures. A later storage interruption left 192 logged physical request starts without a saved outcome or usage; their outcome and possible billing remain unknown. The audit registers these attempts separately from the 15,360 terminal draws. Only missing terminal draws were recollected; saved terminal responses, including truncations, were retained. Reported token totals are subtotals of known usage, not zero-cost claims for failed or interrupted attempts. Source-bound recovery sidecars preserve the original receipts and the exact interruption accounting.

H.1 A separate Python prompt sensitivity test

After the original Opus 5 Python refusals, synthetic tasks were used to select a direct arithmetic-expression formulation with no system message. We then collected eight independent draws for every original Python test task: all 384 prompts across three levels, with unchanged adaptive thinking, medium effort, 8,192 output tokens, allowed operators, public inputs and frozen graders. The revised

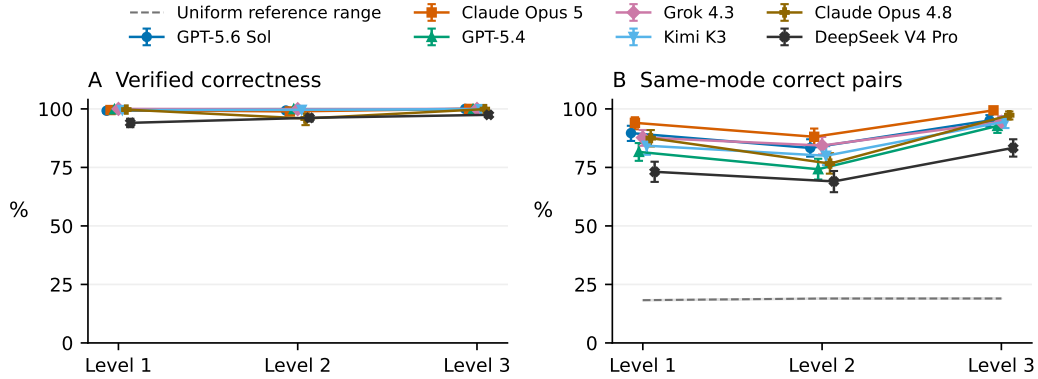


Figure 18: **Verified correctness and concentration in hosted Graph outputs.** Each deployment receives the same 128 prompts per level, with eight stateless samples per prompt. Intervals resample whole prompts. The gray band spans the models’ uniform-correct-key references, conditional on their observed correct draws. Graph was selected after the GPT-5.6 Sol run and before the prospective comparison runs. Level labels do not guarantee increasing difficulty for each model.

733 user instruction asks for a one-line boxed Python lambda returning any proper divisor for each listed
 734 input. It does not prescribe a preferred divisor. Both task wording and system-message presence
 735 change; collection time also differs. This exploratory follow-up therefore supports a practical prompt
 736 repair without isolating its cause. Original refusals and all three residual refusals are retained. The
 737 original-protocol appendix tables retain all original Python responses; the main display uses this
 738 complete revised condition with an explicit caption note.

Table 23: **Opus 5 Python: original versus plain task wording.** Each row has 1,024 responses. R counts native refusals; A is per-response accuracy (%), D is distinct@8, and C is correct-pair collision (%). Strict and normalized results use the same frozen grader and typography rule. Original Level-2/3 collision is undefined because no prompt has two correct draws.

Condition	L	R	Strict			Normalized		
			A	D	C	A	D	C
Original	1	409	38.1	0.91	93.0	60.0	1.12	90.4
	2	1021	0.3	0.02	–	0.3	0.02	–
	3	1024	0.0	0.00	–	0.0	0.00	–
Plain user, no system	1	0	74.9	1.42	80.2	100.0	1.49	79.9
	2	0	71.8	1.31	84.7	100.0	1.38	85.7
	3	3	76.2	1.38	82.7	99.7	1.41	84.1

739 The new condition has no API failures or token-limit truncations. Strict grading accepts 2,282
 740 responses; the frozen rule rescues 787 more by converting escaped percent signs (%) to Python
 741 modulo (%). Thus every non-refused answer passes after this formatting-only correction. Normalized
 742 collision is 79.9/85.7/84.1%, versus conditional uniform references of 1.43/1.15/1.15%. The residual
 743 three refusals occur on two prompts whose other identical requests are accepted. No retry-until-
 744 accepted selection is used.

745 The paired analysis resamples whole eight-draw prompt groups across conditions (20,000 replicates).
 746 Normalized accuracy gains at Levels 1/2/3 are 40.0 [34.7, 45.5], 99.7 [99.3, 100.0], and 99.7 [99.2,
 747 100.0] percentage points (pointwise 95% intervals). Output draws themselves are independent, and
 748 the different correct-pair populations preclude interpreting a raw collision difference as a matched-
 749 population prompt effect.

750 The saved sensitivity record binds the original and revised native receipts, request payloads, strict
 751 regrade audit and frozen normalization outputs.

Separate first-valid retry diagnostic. After completing the fixed 3,072-response revised-wording cohort, a separate condition retries only its three remaining refused slots, retaining the first verified-valid response. All three succeed on their first additional request, and all reproduce a mode already seen for that prompt. This uses 3,075 requests to select 3,072 valid responses; its 100% accepted-set accuracy follows from selection and is not unconditional model accuracy. The selected set’s `distinct@8` is unchanged at 1.492, 1.375, 1.414 at Levels 1–3. Level-3 correct-pair collision changes from 84.0909% to 84.1518%; the other levels are unchanged. All initial and retry responses remain preserved. The main figure continues to use the complete fixed eight-draw condition, including its three refusals; these selected-validity results are only a separate diagnostic with variable request effort.

H.2 A matched requested-temperature sensitivity test

A separate fixed subset contains eight prompts from each of the five domains at each of three levels: 120 prompts, with eight independent draws per prompt and temperature. Grok 4.3 and Kimi K3 each receive requested temperatures $T = 1.0$ and $T = 1.5$, giving 960 draws per condition and 3,840 in total. The two conditions share task rows and all non-temperature request settings; both temperatures are collected concurrently within each deployment. Every response is retained. This subset is a separate experiment and does not replace any full-cohort result in the main hosted display.

Provider API contracts support the requested controls and the deployments accept the requests, but native responses do not report effective sampling temperature. DeepSeek is excluded from this contrast because its documented thinking-mode interface ignores the temperature setting. These contrasts therefore concern the tested requested settings, not arbitrary temperatures or all deployments.

Tables average all five domains equally within each level; All equally averages all 15 domain–level cells. Confidence intervals use 20,000 paired whole-prompt bootstrap resamples stratified by domain and level, retaining each eight-draw group. They are pointwise and have no multiplicity adjustment. Correct-pair collision remains conditional on correct draws and can change its eligible prompt population; an undefined cell makes the corresponding macro undefined.

Observed sensitivity. For Grok, the observed breadth change is +0.050 [-0.058, +0.158] modes, and the collision change is -0.78 [-3.86, +2.12] percentage points. These intervals do not show a clear broadening at the tested setting. The per-response accuracy change is -2.29 [-3.85, -0.94] percentage points. This does not establish invariance to temperature in general.

Kimi returns 696/960 token-limited outputs at $T = 1.5$, compared with 0 at $T = 1.0$. Strict accuracy falls from 96.56% to 21.25%, while raw breadth falls from 1.933 to 0.983 modes. This is a severe token-limit and accuracy failure, not a clean increase in correct-output concentration. The five-domain collision mean is undefined at $T = 1.5$ because some cells have no eligible correct pairs. Neither model has provider-declared refusals in this experiment.

† marks an omitted interval when fewer than all 20,000 bootstrap replicates define the contrast. For Kimi collision, only 11,815 and 11,616 replicates are defined at Levels 1 and 2; the raw conditional percentile summaries remain in the source record. We do not present them as ordinary 95% intervals. Undefined cells are never replaced by zero or silently omitted.

Full domain–level estimates, counts, native outcomes and collision eligibility remain in the source-bound temperature analysis. Countdown has no finite-total-support uniform reference; finite-support sensitivity summaries use the other four domains. The paper record authenticates every included condition summary, native completion audit, frozen normalizer, paired analysis source and sampling-control review, with the complete fixed response cohort retained throughout.

H.3 GPT-5.6 Sol: temperature, pass@8 and verified modes

Figure 17 plots empirical `pass@8` against `distinct@8` for $T \in \{0.5, 1.0, 1.5, 2.0\}$, with reasoning none. Each condition retains the same 120 held-out prompts (eight per domain–level cell), with eight stateless responses per prompt: 960 responses per temperature and 3,840 in total. A prompt contributes one to `pass@8` if at least one of its eight answers is verified correct, and contributes its number of distinct correct canonical keys to `distinct@8`. Zero-success groups and all unsuccessful

Table 24: **Requested-temperature sensitivity: Strict executable grades.** A is per-response accuracy (%), D is distinct@8, and C is correct-pair collision (%). D differences are mode counts; A/C differences are percentage points. Brackets give paired pointwise 95% intervals for the paired change in requested temperature. Every fixed draw remains in its original denominator.

Deployment	Level	Metric	$T = 1.0$	$T = 1.5$	Difference [95% interval]
Grok 4.3	1	A	98.12	95.62	-2.50 [-4.06, -0.94]
		D	1.900	2.000	+0.100 [-0.100, +0.300]
		C	72.89	73.16	+0.27 [-6.31, +6.37]
	2	A	99.06	99.38	+0.31 [-0.94, +1.56]
		D	1.425	1.425	+0.000 [-0.175, +0.200]
		C	86.61	85.20	-1.41 [-6.26, +3.30]
	3	A	98.75	94.06	-4.69 [-9.06, -0.94]
		D	1.500	1.550	+0.050 [-0.125, +0.225]
		C	84.04	82.85	-1.20 [-5.48, +2.79]
	All	A	98.65	96.35	-2.29 [-3.85, -0.94]
		D	1.608	1.658	+0.050 [-0.058, +0.158]
		C	81.18	80.40	-0.78 [-3.86, +2.12]
Kimi K3	1	A	92.50	22.50	-70.00 [-75.62, -63.75]
		D	2.200	1.075	-1.125 [-1.475, -0.775]
		C	66.23	53.12	-13.11 [-] [†]
	2	A	98.12	20.31	-77.81 [-82.50, -72.81]
		D	1.850	0.900	-0.950 [-1.300, -0.625]
		C	72.18	66.55	-5.63 [-] [†]
	3	A	99.06	20.94	-78.12 [-83.75, -72.81]
		D	1.750	0.975	-0.775 [-1.025, -0.525]
		C	78.33	—	—
	All	A	96.56	21.25	-75.31 [-78.44, -72.19]
		D	1.933	0.983	-0.950 [-1.133, -0.767]
		C	72.25	—	—

Table 25: **Requested-temperature sensitivity: Frozen formatting-normalized grades.** A is per-response accuracy (%), D is distinct@8, and C is correct-pair collision (%). D differences are mode counts; A/C differences are percentage points. Brackets give paired pointwise 95% intervals for the paired change in requested temperature. Every fixed draw remains in its original denominator.

Deployment	Level	Metric	$T = 1.0$	$T = 1.5$	Difference [95% interval]
Grok 4.3	1	A	98.12	95.62	-2.50 [-4.06, -0.94]
		D	1.900	2.000	+0.100 [-0.100, +0.300]
		C	72.89	73.16	+0.27 [-6.31, +6.37]
	2	A	99.06	99.38	+0.31 [-0.94, +1.56]
		D	1.425	1.425	+0.000 [-0.175, +0.200]
		C	86.61	85.20	-1.41 [-6.26, +3.30]
	3	A	98.75	94.06	-4.69 [-9.06, -0.94]
		D	1.500	1.550	+0.050 [-0.125, +0.225]
		C	84.04	82.85	-1.20 [-5.48, +2.79]
	All	A	98.65	96.35	-2.29 [-3.85, -0.94]
		D	1.608	1.658	+0.050 [-0.058, +0.158]
		C	81.18	80.40	-0.78 [-3.86, +2.12]
Kimi K3	1	A	92.50	22.81	-69.69 [-75.31, -63.44]
		D	2.200	1.075	-1.125 [-1.475, -0.775]
		C	66.23	53.61	-12.62 [-] [†]
	2	A	98.12	20.31	-77.81 [-82.50, -72.81]
		D	1.850	0.900	-0.950 [-1.300, -0.625]
		C	72.18	66.55	-5.63 [-] [†]
	3	A	99.38	20.94	-78.44 [-84.06, -73.12]
		D	1.750	0.975	-0.775 [-1.025, -0.525]
		C	78.18	—	—
	All	A	96.67	21.35	-75.31 [-78.44, -72.19]
		D	1.933	0.983	-0.950 [-1.133, -0.767]
		C	72.20	—	—

draws remain in the denominator. We compute success directly from the saved groups, not as $1 - (1 - p)^8$ from pooled per-response accuracy; task difficulty varies across prompts.

The deployment rejected non-default temperatures with medium reasoning. The connected curves therefore use the supported reasoning `none` condition. The separate historical medium reference uses the same 120 prompts and eight draws; its requests omitted temperature and receipts returned $T = 1$. It remains unconnected because both reasoning setting and collection time differ. This does not test temperature robustness of the medium condition.

Each level averages five domains equally; All averages all 15 cells. Pointwise 95 percent intervals use 20,000 paired whole-prompt bootstrap resamples stratified by domain and level, retaining all eight draws. There is no multiplicity adjustment. A zero-width empirical interval reflects constant observed groups, not certain success on unseen prompts.

Table 26: **GPT temperature frontier: Strict executable grades.** P is solved prompts (% , pass@8); D is distinct@8; A is per-response accuracy (%), retained as a separate diagnostic. Brackets give pointwise 95% intervals. M is the historical medium reference; numeric temperatures all use reasoning `none` and the same fixed eight-draw sampling budget.

T	Level	P [95% interval]	D [95% interval]	A [95% interval]
0.5	1	62.5 [50.0, 75.0]	0.925 [0.675, 1.200]	52.5 [41.2, 63.4]
	2	45.0 [37.5, 55.0]	0.450 [0.375, 0.550]	29.7 [22.5, 36.2]
	3	47.5 [35.0, 60.0]	0.475 [0.350, 0.600]	40.0 [27.8, 52.2]
	All	51.7 [45.0, 58.3]	0.617 [0.517, 0.717]	40.7 [34.8, 46.7]
1.0	1	77.5 [65.0, 90.0]	1.175 [0.950, 1.400]	53.1 [42.2, 63.7]
	2	45.0 [35.0, 55.0]	0.475 [0.350, 0.600]	29.4 [22.2, 36.6]
	3	52.5 [40.0, 65.0]	0.550 [0.400, 0.700]	38.8 [27.8, 50.0]
	All	58.3 [51.7, 65.0]	0.733 [0.642, 0.833]	40.4 [34.7, 46.0]
1.5	1	82.5 [70.0, 92.5]	1.525 [1.250, 1.825]	54.4 [44.1, 64.7]
	2	47.5 [37.5, 57.5]	0.475 [0.375, 0.575]	27.8 [22.2, 33.1]
	3	55.0 [42.5, 67.5]	0.675 [0.475, 0.875]	42.2 [31.2, 53.4]
	All	61.7 [55.0, 68.3]	0.892 [0.775, 1.008]	41.5 [36.0, 46.8]
2.0	1	82.5 [70.0, 92.5]	1.600 [1.325, 1.900]	58.4 [48.1, 68.4]
	2	40.0 [32.5, 47.5]	0.425 [0.350, 0.500]	26.6 [21.2, 31.9]
	3	55.0 [42.5, 67.5]	0.625 [0.475, 0.800]	34.7 [25.6, 44.4]
	All	59.2 [53.3, 65.0]	0.883 [0.767, 1.000]	39.9 [34.9, 44.8]
M	1	95.0 [87.5, 100.0]	1.675 [1.425, 1.925]	79.4 [71.6, 86.6]
	2	75.0 [67.5, 80.0]	1.025 [0.825, 1.225]	59.4 [50.3, 67.2]
	3	85.0 [80.0, 92.5]	1.225 [1.000, 1.475]	71.2 [63.1, 79.7]
	All	85.0 [81.7, 88.3]	1.308 [1.183, 1.442]	70.0 [65.2, 74.6]

Observed temperature frontier. From $T = 0.5$ to $T = 1.5$, normalized pass@8 rises from 66.67% to 72.50%, while distinct@8 rises from 0.775 to 1.050. At $T = 2.0$, the corresponding values are 70.83% and 1.042. Thus $T = 1.5$ is the best observed aggregate point on both axes among these four temperatures. This finite exploratory sweep does not establish a population optimum or a continuous Pareto boundary, and level-specific curves differ. In particular, Level 1 continues gaining modes at the highest temperature even though the overall point declines slightly.

Per-response accuracy nevertheless decreases from 55.52% to 50.21% between the endpoints. That diagnostic is distinct from pass@8: success within eight draws can improve while an individual draw becomes less reliable on these same prompts.

The paired $T = 2.0$ minus $T = 0.5$ contrast is 4.2 [-0.8, 9.2] pass@8 percentage points and 0.267 [0.183, 0.350] distinct correct modes, with pointwise 95% intervals. All prompt-level records, domain estimates, original per-response and collision analyses, and paired comparisons remain in the retained reports for both strict and normalized grading.

Retained outcomes. At $T = 0.5$, 0 of 960 outputs are token-limited and 0 are provider-declared refusals. At $T = 1.0$, 0 of 960 outputs are token-limited and 0 are provider-declared refusals. At $T = 1.5$, 0 of 960 outputs are token-limited and 0 are provider-declared refusals. At $T = 2.0$, 0 of 960 outputs are token-limited and 0 are provider-declared refusals. These outcome counts retain every

Table 27: **GPT temperature frontier: Frozen formatting-normalized grades.** P is solved prompts (% pass@8); D is distinct@8; A is per-response accuracy (%), retained as a separate diagnostic. Brackets give pointwise 95% intervals. M is the historical medium reference; numeric temperatures all use reasoning none and the same fixed eight-draw sampling budget.

<i>T</i>	Level	P [95% interval]	D [95% interval]	A [95% interval]
0.5	1	72.5 [60.0, 85.0]	1.025 [0.775, 1.300]	67.8 [55.9, 79.1]
	2	65.0 [57.5, 75.0]	0.650 [0.575, 0.750]	46.2 [38.1, 53.8]
	3	62.5 [50.0, 75.0]	0.650 [0.525, 0.775]	52.5 [40.3, 64.4]
	All	66.7 [60.0, 73.3]	0.775 [0.683, 0.875]	55.5 [49.3, 61.7]
1.0	1	80.0 [67.5, 90.0]	1.200 [0.975, 1.425]	67.2 [55.3, 78.4]
	2	65.0 [55.0, 75.0]	0.675 [0.550, 0.800]	44.4 [36.6, 52.2]
	3	67.5 [55.0, 80.0]	0.725 [0.600, 0.875]	50.0 [39.4, 60.9]
	All	70.8 [64.2, 77.5]	0.867 [0.775, 0.967]	53.9 [47.9, 59.6]
1.5	1	82.5 [70.0, 92.5]	1.600 [1.325, 1.900]	67.2 [55.9, 78.1]
	2	67.5 [57.5, 77.5]	0.700 [0.600, 0.800]	39.1 [32.5, 45.6]
	3	67.5 [55.0, 80.0]	0.850 [0.650, 1.050]	50.9 [40.0, 61.9]
	All	72.5 [65.8, 79.2]	1.050 [0.925, 1.175]	52.4 [46.8, 57.9]
2.0	1	82.5 [70.0, 92.5]	1.625 [1.325, 1.925]	68.4 [57.5, 78.8]
	2	60.0 [52.5, 67.5]	0.675 [0.575, 0.775]	37.5 [31.6, 43.1]
	3	70.0 [57.5, 82.5]	0.825 [0.650, 1.000]	44.7 [34.7, 54.7]
	All	70.8 [65.0, 76.7]	1.042 [0.925, 1.158]	50.2 [44.9, 55.3]
M	1	95.0 [87.5, 100.0]	1.825 [1.575, 2.075]	93.4 [86.6, 98.8]
	2	97.5 [92.5, 100.0]	1.350 [1.175, 1.550]	80.9 [73.8, 87.5]
	3	100.0 [100.0, 100.0]	1.475 [1.250, 1.700]	85.0 [80.0, 90.0]
	All	97.5 [95.0, 100.0]	1.550 [1.425, 1.683]	86.5 [82.8, 89.9]

830 registered draw and are separate from the verifier-based correctness and canonical-key measurements
831 above.

832 H.4 Native provider outcomes and undefined concentration

833 Provider-declared refusals and filtering are deployment outcomes, not evidence that the underlying
834 model cannot solve the mathematical task. Empty refusals remain failed draws under the frozen
835 verifier; no requests or grades are overwritten. Classification uses native metadata, not a semantic
836 detector of refusals written as ordinary answer text.

837 Correct-pair collision is undefined when no prompt supplies two correct draws. The five-domain
838 mean is therefore undefined if any domain has no eligible pairs; such entries remain – rather than
839 zero or a mean over fewer domains.

840 GPT-5.6 Sol native completion states: completed: 15,360. Empty visible answers are reported
841 separately from provider refusals.

842 Claude Opus 5 native completion states: end_turn: 12,838, refusal: 2,522. Empty visible answers are
843 reported separately from provider refusals.

844 Claude Opus 5 returns provider-declared refusals for 409, 1021, 1024 of the 1,024 Python requests at
845 Levels 1, 2 and 3, respectively. The native refusal categories on these Python draws are cyber: 2454.
846 These are benign factor-selection tasks; the reported deployment policy must be distinguished from
847 concentration among correct Python outputs.

848 GPT-5.4 native completion states: completed: 15,355, incomplete: 5. Empty visible answers are
849 reported separately from provider refusals.

850 Grok 4.3 native completion states: stop: 15,360. Empty visible answers are reported separately from
851 provider refusals.

852 Kimi K3 native completion states: length: 15, stop: 15,345. Empty visible answers are reported
853 separately from provider refusals.

854 Claude Opus 4.8 native completion states: end_turn: 15,355, max_tokens: 5. Empty visible answers
855 are reported separately from provider refusals.

Table 28: **Native service outcomes: GPT-5.6 Sol.** Every row contains 1,024 responses. R counts provider-declared refusals, F content-filtered stops, and E empty visible answers. Counters may overlap; they do not replace the frozen verifier grades.

Level	Domain	R	F	E
1	Graph	0	0	0
1	Countdown	0	0	0
1	Python	0	0	0
1	MathIR	0	0	0
1	PantryPlan	0	0	0
2	Graph	0	0	0
2	Countdown	0	0	0
2	Python	0	0	0
2	MathIR	0	0	0
2	PantryPlan	0	0	0
3	Graph	0	0	0
3	Countdown	0	0	0
3	Python	0	0	0
3	MathIR	0	0	0
3	PantryPlan	0	0	0

Table 29: **Native service outcomes: Claude Opus 5.** Every row contains 1,024 responses. R counts provider-declared refusals, F content-filtered stops, and E empty visible answers. Counters may overlap; they do not replace the frozen verifier grades.

Level	Domain	R	F	E
1	Graph	0	0	0
1	Countdown	0	0	0
1	Python	409	0	409
1	MathIR	0	0	0
1	PantryPlan	0	0	0
2	Graph	0	0	0
2	Countdown	0	0	0
2	Python	1021	0	1021
2	MathIR	35	0	35
2	PantryPlan	0	0	0
3	Graph	0	0	0
3	Countdown	0	0	0
3	Python	1024	0	1024
3	MathIR	33	0	32
3	PantryPlan	0	0	0

856 Claude Opus 4.8 returns provider-declared refusals for 0, 0, 0 of the 1,024 Python requests at Levels
857 1, 2 and 3, respectively. The native refusal categories on these Python draws are not supplied.
858 These are benign factor-selection tasks; the reported deployment policy must be distinguished from
859 concentration among correct Python outputs.

860 DeepSeek V4 Pro native completion states: length: 632, stop: 14,728. Empty visible answers are
861 reported separately from provider refusals.

862 The result record `results/frontier_comparison_20260911.json` binds every source summary,
863 completion audit, frozen prompt digest, raw-sample digest and primary graded sample digest. Pending
864 deployments are omitted from both the figure and numerical comparisons. The figure was selected
865 after the reference GPT run for its high strict correctness, certified support and lack of formatting
866 corrections, and before collecting the prospective comparison runs. All five domains are retained in
867 the tables, including disagreement across models.

Table 30: **Native service outcomes: GPT-5.4.** Every row contains 1,024 responses. R counts provider-declared refusals, F content-filtered stops, and E empty visible answers. Counters may overlap; they do not replace the frozen verifier grades.

Level	Domain	R	F	E
1	Graph	0	0	0
1	Countdown	0	0	0
1	Python	0	0	0
1	MathIR	0	0	0
1	PantryPlan	0	0	1
2	Graph	0	0	0
2	Countdown	0	0	1
2	Python	0	0	0
2	MathIR	0	0	0
2	PantryPlan	0	0	0
3	Graph	0	0	0
3	Countdown	0	0	1
3	Python	0	0	0
3	MathIR	0	0	0
3	PantryPlan	0	0	2

Table 31: **Native service outcomes: Grok 4.3.** Every row contains 1,024 responses. R counts provider-declared refusals, F content-filtered stops, and E empty visible answers. Counters may overlap; they do not replace the frozen verifier grades.

Level	Domain	R	F	E
1	Graph	0	0	0
1	Countdown	0	0	0
1	Python	0	0	0
1	MathIR	0	0	0
1	PantryPlan	0	0	0
2	Graph	0	0	0
2	Countdown	0	0	0
2	Python	0	0	0
2	MathIR	0	0	0
2	PantryPlan	0	0	0
3	Graph	0	0	0
3	Countdown	0	0	0
3	Python	0	0	0
3	MathIR	0	0	0
3	PantryPlan	0	0	0

Table 32: **Native service outcomes: Kimi K3.** Every row contains 1,024 responses. R counts provider-declared refusals, F content-filtered stops, and E empty visible answers. Counters may overlap; they do not replace the frozen verifier grades.

Level	Domain	R	F	E
1	Graph	0	0	0
1	Countdown	0	0	0
1	Python	0	0	0
1	MathIR	0	0	0
1	PantryPlan	0	0	0
2	Graph	0	0	0
2	Countdown	0	0	0
2	Python	0	0	0
2	MathIR	0	0	0
2	PantryPlan	0	0	0
3	Graph	0	0	0
3	Countdown	0	0	0
3	Python	0	0	0
3	MathIR	0	0	0
3	PantryPlan	0	0	0

Table 33: **Native service outcomes: Claude Opus 4.8.** Every row contains 1,024 responses. R counts provider-declared refusals, F content-filtered stops, and E empty visible answers. Counters may overlap; they do not replace the frozen verifier grades.

Level	Domain	R	F	E
1	Graph	0	0	0
1	Countdown	0	0	0
1	Python	0	0	0
1	MathIR	0	0	0
1	PantryPlan	0	0	0
2	Graph	0	0	0
2	Countdown	0	0	0
2	Python	0	0	0
2	MathIR	0	0	0
2	PantryPlan	0	0	0
3	Graph	0	0	0
3	Countdown	0	0	3
3	Python	0	0	0
3	MathIR	0	0	0
3	PantryPlan	0	0	1

Table 34: **Native service outcomes: DeepSeek V4 Pro.** Every row contains 1,024 responses. R counts provider-declared refusals, F content-filtered stops, and E empty visible answers. Counters may overlap; they do not replace the frozen verifier grades.

Level	Domain	R	F	E
1	Graph	0	0	59
1	Countdown	0	0	0
1	Python	0	0	0
1	MathIR	0	0	4
1	PantryPlan	0	0	13
2	Graph	0	0	38
2	Countdown	0	0	1
2	Python	0	0	0
2	MathIR	0	0	59
2	PantryPlan	0	0	80
3	Graph	0	0	24
3	Countdown	0	0	7
3	Python	0	0	0
3	MathIR	0	0	100
3	PantryPlan	0	0	246

Table 35: **Strict level summaries for completed hosted deployments.** Each level weights the five domains equally. M and P are mean@8 and pass@8 (%); D is raw distinct@8. C is correct-pair collision (%; with pointwise 95% whole-prompt intervals), pair weighted within each domain. Provider settings differ; these are descriptive outputs.

Deployment	Level	M	P	D	C (95% interval)
GPT-5.6 Sol	1	83.0	95.8	1.71	74.0 [71.9, 76.1]
GPT-5.6 Sol	2	62.9	77.5	1.16	82.3 [80.3, 84.4]
GPT-5.6 Sol	3	68.7	80.3	1.18	84.5 [82.8, 86.2]
Claude Opus 5	1	85.3	94.5	1.32	86.3 [84.5, 88.0]
Claude Opus 5	2	76.6	80.2	1.16	—
Claude Opus 5	3	77.8	79.8	1.07	—
GPT-5.4	1	93.7	98.1	2.09	67.2 [65.1, 69.3]
GPT-5.4	2	80.6	97.7	1.61	74.7 [72.4, 76.8]
GPT-5.4	3	83.3	97.8	1.57	78.8 [76.9, 80.7]
Grok 4.3	1	98.3	99.5	1.97	72.4 [70.7, 74.0]
Grok 4.3	2	99.3	100.0	1.57	82.1 [80.6, 83.7]
Grok 4.3	3	98.8	100.0	1.55	83.0 [81.5, 84.6]
Kimi K3	1	96.9	98.4	2.31	62.5 [60.6, 64.4]
Kimi K3	2	96.8	100.0	2.05	69.4 [67.6, 71.2]
Kimi K3	3	97.8	100.0	1.90	74.6 [73.1, 76.2]
Claude Opus 4.8	1	96.4	98.8	1.64	78.4 [76.6, 80.3]
Claude Opus 4.8	2	98.5	100.0	1.67	77.2 [75.4, 79.1]
Claude Opus 4.8	3	98.9	99.8	1.51	83.4 [81.8, 84.9]
DeepSeek V4 Pro	1	96.4	98.4	2.01	71.0 [69.4, 72.7]
DeepSeek V4 Pro	2	93.7	99.1	1.84	68.7 [67.0, 70.3]
DeepSeek V4 Pro	3	91.0	99.7	1.62	78.8 [76.9, 80.9]

Table 36: **GPT-5.6 Sol: Strict executable grades.** Each cell has 128 prompts and 1,024 responses. M/P denote mean@8/pass@8 (%); D is distinct@8. C is correct-pair collision (%) with pointwise 95% prompt-bootstrap intervals. U is its conditional uniform reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	99.2	100.0	1.30	89.7 [86.3, 92.8]	18.3
1	Countdown	86.1	98.4	1.77	71.2 [66.1, 75.9]	–
1	Python	40.4	90.6	1.11	81.8 [76.9, 87.2]	0.8
1	MathIR	89.3	89.8	1.38	80.3 [75.4, 85.0]	20.0
1	PantryPlan	100.0	100.0	3.00	46.9 [42.2, 51.5]	6.5
2	Graph	99.3	100.0	1.48	83.3 [79.6, 87.0]	18.9
2	Countdown	65.4	86.7	1.57	64.7 [59.0, 70.6]	–
2	Python	8.5	11.7	0.12	100.0 [100.0, 100.0]	0.7
2	MathIR	86.2	93.0	0.93	100.0 [100.0, 100.0]	20.0
2	PantryPlan	55.3	96.1	1.72	63.6 [56.8, 70.3]	6.9
3	Graph	100.0	100.0	1.16	95.1 [92.4, 97.3]	19.0
3	Countdown	71.2	89.8	1.38	76.4 [71.1, 81.8]	–
3	Python	9.4	14.1	0.14	100.0 [100.0, 100.0]	0.9
3	MathIR	97.7	97.7	0.98	100.0 [100.0, 100.0]	20.0
3	PantryPlan	65.3	100.0	2.22	51.0 [45.3, 57.3]	6.8

Table 37: **Claude Opus 5: Strict executable grades.** Each cell has 128 prompts and 1,024 responses. M/P denote mean@8/pass@8 (%); D is distinct@8. C is correct-pair collision (%) with pointwise 95% prompt-bootstrap intervals. U is its conditional uniform reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	99.5	100.0	1.15	94.0 [91.3, 96.4]	18.3
1	Countdown	100.0	100.0	1.59	78.8 [74.6, 83.0]	–
1	Python	38.1	81.2	0.91	93.0 [88.7, 96.6]	0.7
1	MathIR	90.5	91.4	1.09	92.2 [88.8, 95.4]	20.0
1	PantryPlan	98.4	100.0	1.86	73.5 [68.5, 78.2]	6.5
2	Graph	98.9	100.0	1.32	88.1 [84.5, 91.6]	18.9
2	Countdown	100.0	100.0	1.81	69.3 [65.2, 73.6]	–
2	Python	0.3	2.3	0.02	–	–
2	MathIR	89.1	98.4	1.06	96.5 [93.9, 98.6]	20.0
2	PantryPlan	94.5	100.0	1.56	80.7 [76.0, 85.3]	6.4
3	Graph	100.0	100.0	1.02	99.4 [98.3, 100.0]	19.0
3	Countdown	99.9	100.0	1.66	76.2 [71.7, 80.7]	–
3	Python	0.0	0.0	0.00	–	–
3	MathIR	95.3	99.2	1.02	98.8 [97.5, 99.8]	20.0
3	PantryPlan	93.6	100.0	1.63	78.8 [74.2, 83.4]	6.5

Table 38: **GPT-5.4: Strict executable grades.** Each cell has 128 prompts and 1,024 responses. M/P denote mean@8/pass@8 (%); D is distinct@8. C is correct-pair collision (%) with pointwise 95% prompt-bootstrap intervals. U is its conditional uniform reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	99.8	100.0	1.57	81.6 [77.8, 85.4]	18.3
1	Countdown	91.8	100.0	1.73	76.6 [71.2, 81.3]	–
1	Python	87.6	100.0	2.52	56.4 [51.3, 61.5]	1.4
1	MathIR	89.3	90.6	1.45	78.6 [74.0, 83.4]	20.0
1	PantryPlan	99.9	100.0	3.16	42.8 [38.4, 47.4]	6.5
2	Graph	99.9	100.0	1.74	74.2 [69.8, 78.7]	19.0
2	Countdown	91.1	100.0	1.98	63.0 [58.6, 67.5]	–
2	Python	47.9	96.1	1.47	72.9 [64.6, 80.2]	1.0
2	MathIR	87.0	92.2	0.92	100.0 [100.0, 100.0]	20.0
2	PantryPlan	77.0	100.0	1.93	63.2 [58.4, 68.6]	6.8
3	Graph	99.8	100.0	1.23	92.7 [89.7, 95.4]	19.0
3	Countdown	82.2	93.8	1.72	69.1 [64.4, 73.5]	–
3	Python	50.1	97.7	1.58	74.4 [68.6, 79.8]	1.1
3	MathIR	97.7	97.7	0.98	100.0 [100.0, 100.0]	20.0
3	PantryPlan	86.5	100.0	2.34	57.8 [52.8, 63.4]	6.6

Table 39: **Grok 4.3: Strict executable grades.** Each cell has 128 prompts and 1,024 responses. M/P denote mean@8/pass@8 (%); D is distinct@8. C is correct-pair collision (%) with pointwise 95% prompt-bootstrap intervals. U is its conditional uniform reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	100.0	100.0	1.40	87.8 [84.2, 91.0]	18.3
1	Countdown	99.9	100.0	2.05	64.8 [60.1, 69.5]	–
1	Python	99.8	100.0	1.41	88.0 [84.8, 91.0]	1.4
1	MathIR	92.4	97.7	1.24	90.1 [86.9, 93.1]	20.0
1	PantryPlan	99.4	100.0	3.77	31.4 [27.8, 35.1]	6.5
2	Graph	99.9	100.0	1.52	84.3 [80.5, 88.0]	19.0
2	Countdown	99.8	100.0	1.99	65.8 [61.8, 70.2]	–
2	Python	99.8	100.0	1.05	98.8 [97.8, 99.6]	1.1
2	MathIR	97.5	100.0	1.12	94.9 [92.4, 97.2]	20.0
2	PantryPlan	99.5	100.0	2.16	66.6 [61.2, 71.4]	6.5
3	Graph	100.0	100.0	1.17	94.5 [92.0, 96.8]	19.0
3	Countdown	99.9	100.0	1.98	66.3 [61.7, 71.0]	–
3	Python	99.9	100.0	1.04	98.6 [97.2, 99.6]	1.2
3	MathIR	95.0	100.0	1.05	97.8 [95.9, 99.3]	20.0
3	PantryPlan	99.2	100.0	2.52	58.0 [53.1, 63.5]	6.5

Table 40: **Kimi K3: Strict executable grades.** Each cell has 128 prompts and 1,024 responses. M/P denote mean@8/pass@8 (%); D is distinct@8. C is correct-pair collision (%) with pointwise 95% prompt-bootstrap intervals. U is its conditional uniform reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	99.5	100.0	1.48	84.3 [80.4, 87.8]	18.3
1	Countdown	99.1	100.0	1.91	68.4 [63.8, 73.3]	–
1	Python	97.9	100.0	2.52	57.7 [53.5, 61.9]	1.4
1	MathIR	90.2	92.2	1.47	75.8 [70.7, 80.7]	20.0
1	PantryPlan	97.5	100.0	4.18	26.2 [23.1, 29.2]	6.5
2	Graph	99.6	100.0	1.62	80.0 [76.0, 84.1]	19.0
2	Countdown	98.4	100.0	1.98	64.4 [59.9, 68.7]	–
2	Python	98.2	100.0	2.44	61.9 [57.6, 66.2]	1.1
2	MathIR	94.3	100.0	1.02	99.8 [99.3, 100.0]	20.0
2	PantryPlan	93.3	100.0	3.20	41.3 [36.6, 45.9]	6.6
3	Graph	99.9	100.0	1.19	94.6 [91.8, 96.9]	19.0
3	Countdown	97.5	100.0	1.84	70.6 [66.1, 74.8]	–
3	Python	98.0	100.0	2.34	64.7 [60.1, 69.0]	1.2
3	MathIR	98.1	100.0	1.00	100.0 [100.0, 100.0]	20.0
3	PantryPlan	95.6	100.0	3.13	43.4 [38.6, 48.3]	6.5

Table 41: **Claude Opus 4.8: Strict executable grades.** Each cell has 128 prompts and 1,024 responses. M/P denote mean@8/pass@8 (%); D is distinct@8. C is correct-pair collision (%) with pointwise 95% prompt-bootstrap intervals. U is its conditional uniform reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	99.7	100.0	1.34	87.5 [83.9, 91.0]	18.3
1	Countdown	95.6	99.2	1.78	69.8 [64.9, 74.5]	–
1	Python	98.0	100.0	1.40	85.7 [81.7, 89.4]	1.4
1	MathIR	92.3	94.5	1.10	93.7 [91.1, 96.4]	20.0
1	PantryPlan	96.6	100.0	2.59	55.3 [50.3, 60.1]	6.5
2	Graph	95.9	100.0	1.66	76.6 [72.3, 81.1]	18.8
2	Countdown	99.9	100.0	1.99	65.3 [60.9, 69.9]	–
2	Python	100.0	100.0	1.34	86.0 [82.3, 89.6]	1.2
2	MathIR	97.0	100.0	1.24	90.1 [86.7, 93.4]	20.0
2	PantryPlan	99.5	100.0	2.11	67.9 [62.8, 73.1]	6.5
3	Graph	99.9	100.0	1.08	97.3 [95.5, 98.9]	19.0
3	Countdown	97.9	99.2	1.80	72.5 [68.2, 76.6]	–
3	Python	98.9	100.0	1.32	86.4 [82.5, 90.1]	1.2
3	MathIR	99.1	100.0	1.03	99.1 [98.1, 99.8]	20.0
3	PantryPlan	98.5	100.0	2.33	61.6 [56.7, 66.6]	6.5

Table 42: **DeepSeek V4 Pro: Strict executable grades.** Each cell has 128 prompts and 1,024 responses. M/P denote mean@8/pass@8 (%); D is distinct@8. C is correct-pair collision (%) with pointwise 95% prompt-bootstrap intervals. U is its conditional uniform reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	94.0	100.0	1.80	73.1 [68.8, 77.4]	18.1
1	Countdown	100.0	100.0	2.02	65.8 [61.4, 70.3]	–
1	Python	99.3	100.0	1.41	88.8 [86.1, 91.4]	1.4
1	MathIR	90.1	92.2	1.11	93.9 [91.2, 96.2]	20.0
1	PantryPlan	98.5	100.0	3.73	33.5 [29.8, 37.2]	6.5
2	Graph	96.2	100.0	1.97	69.0 [64.4, 73.5]	19.1
2	Countdown	99.9	100.0	2.04	63.4 [59.2, 67.7]	–
2	Python	99.6	100.0	1.05	98.4 [97.1, 99.4]	1.2
2	MathIR	85.4	95.3	1.82	51.7 [49.8, 53.9]	20.0
2	PantryPlan	87.1	100.0	2.34	61.1 [56.0, 66.2]	6.5
3	Graph	97.5	100.0	1.52	83.4 [79.6, 87.0]	18.9
3	Countdown	99.2	100.0	1.98	67.5 [63.1, 72.2]	–
3	Python	99.6	100.0	1.08	97.8 [96.3, 99.0]	1.2
3	MathIR	88.5	100.0	1.44	83.6 [79.2, 87.6]	20.0
3	PantryPlan	70.3	98.4	2.08	61.9 [55.9, 68.2]	6.5

Table 43: **GPT-5.6 Sol: Post hoc formatting diagnostic.** Each cell has 128 prompts and 1,024 responses. M/P denote mean@8/pass@8 (%); D is distinct@8. C is correct-pair collision (%) with pointwise 95% prompt-bootstrap intervals. U is its conditional uniform reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	99.2	100.0	1.30	89.7 [86.3, 92.8]	18.3
1	Countdown	99.9	100.0	1.93	67.9 [63.3, 72.4]	–
1	Python	92.4	100.0	1.41	85.6 [82.3, 88.8]	1.4
1	MathIR	89.3	89.8	1.38	80.3 [75.4, 85.0]	20.0
1	PantryPlan	100.0	100.0	3.00	46.9 [42.2, 51.5]	6.5
2	Graph	99.3	100.0	1.48	83.3 [79.6, 87.0]	18.9
2	Countdown	98.5	100.0	1.99	63.9 [59.5, 68.4]	–
2	Python	60.7	99.2	1.12	94.8 [91.9, 97.2]	1.1
2	MathIR	86.2	93.0	0.93	100.0 [100.0, 100.0]	20.0
2	PantryPlan	71.3	100.0	1.96	64.0 [58.1, 69.6]	6.8
3	Graph	100.0	100.0	1.16	95.1 [92.4, 97.3]	19.0
3	Countdown	99.0	100.0	1.70	75.7 [71.2, 80.1]	–
3	Python	56.5	98.4	1.09	94.3 [90.8, 97.2]	1.0
3	MathIR	97.7	97.7	0.98	100.0 [100.0, 100.0]	20.0
3	PantryPlan	75.2	100.0	2.35	53.0 [47.3, 58.9]	6.7

Table 44: **Claude Opus 5: Post hoc formatting diagnostic.** Each cell has 128 prompts and 1,024 responses. M/P denote mean@8/pass@8 (%); D is distinct@8. C is correct-pair collision (%) with pointwise 95% prompt-bootstrap intervals. U is its conditional uniform reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	99.5	100.0	1.15	94.0 [91.3, 96.4]	18.3
1	Countdown	100.0	100.0	1.59	78.8 [74.6, 83.0]	–
1	Python	60.0	94.5	1.12	90.4 [86.2, 94.3]	0.9
1	MathIR	90.5	91.4	1.09	92.2 [88.8, 95.4]	20.0
1	PantryPlan	98.4	100.0	1.86	73.5 [68.5, 78.2]	6.5
2	Graph	98.9	100.0	1.32	88.1 [84.5, 91.6]	18.9
2	Countdown	100.0	100.0	1.81	69.3 [65.2, 73.6]	–
2	Python	0.3	2.3	0.02	–	–
2	MathIR	89.1	98.4	1.06	96.5 [93.9, 98.6]	20.0
2	PantryPlan	99.9	100.0	1.61	79.7 [75.0, 84.3]	6.5
3	Graph	100.0	100.0	1.02	99.4 [98.3, 100.0]	19.0
3	Countdown	100.0	100.0	1.66	76.1 [71.7, 80.7]	–
3	Python	0.0	0.0	0.00	–	–
3	MathIR	95.3	99.2	1.02	98.8 [97.5, 99.8]	20.0
3	PantryPlan	99.4	100.0	1.70	77.3 [72.6, 82.0]	6.5

Table 45: **GPT-5.4: Post hoc formatting diagnostic.** Each cell has 128 prompts and 1,024 responses. M/P denote mean@8/pass@8 (%); D is distinct@8. C is correct-pair collision (%) with pointwise 95% prompt-bootstrap intervals. U is its conditional uniform reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	99.8	100.0	1.57	81.6 [77.8, 85.4]	18.3
1	Countdown	99.7	100.0	1.80	73.6 [68.5, 78.3]	–
1	Python	98.7	100.0	2.57	56.8 [51.9, 61.6]	1.4
1	MathIR	89.3	90.6	1.45	78.6 [74.0, 83.4]	20.0
1	PantryPlan	99.9	100.0	3.16	42.8 [38.4, 47.4]	6.5
2	Graph	99.9	100.0	1.74	74.2 [69.8, 78.7]	19.0
2	Countdown	98.4	100.0	2.12	61.1 [57.0, 65.1]	–
2	Python	91.3	100.0	1.87	74.0 [69.4, 78.1]	1.1
2	MathIR	87.0	92.2	0.92	100.0 [100.0, 100.0]	20.0
2	PantryPlan	77.0	100.0	1.93	63.2 [58.4, 68.6]	6.8
3	Graph	99.8	100.0	1.23	92.7 [89.7, 95.4]	19.0
3	Countdown	99.0	100.0	1.96	68.4 [64.2, 72.7]	–
3	Python	91.6	100.0	1.91	72.6 [68.3, 77.1]	1.2
3	MathIR	97.7	97.7	0.98	100.0 [100.0, 100.0]	20.0
3	PantryPlan	86.5	100.0	2.34	57.8 [52.8, 63.4]	6.6

Table 46: **Grok 4.3: Post hoc formatting diagnostic.** Each cell has 128 prompts and 1,024 responses. M/P denote mean@8/pass@8 (%); D is distinct@8. C is correct-pair collision (%) with pointwise 95% prompt-bootstrap intervals. U is its conditional uniform reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	100.0	100.0	1.40	87.8 [84.2, 91.0]	18.3
1	Countdown	99.9	100.0	2.05	64.8 [60.1, 69.5]	–
1	Python	99.8	100.0	1.41	88.0 [84.8, 91.0]	1.4
1	MathIR	92.4	97.7	1.24	90.1 [86.9, 93.1]	20.0
1	PantryPlan	99.4	100.0	3.77	31.4 [27.8, 35.1]	6.5
2	Graph	99.9	100.0	1.52	84.3 [80.5, 88.0]	19.0
2	Countdown	99.8	100.0	1.99	65.8 [61.8, 70.2]	–
2	Python	99.8	100.0	1.05	98.8 [97.8, 99.6]	1.1
2	MathIR	97.5	100.0	1.12	94.9 [92.4, 97.2]	20.0
2	PantryPlan	99.5	100.0	2.16	66.6 [61.2, 71.4]	6.5
3	Graph	100.0	100.0	1.17	94.5 [92.0, 96.8]	19.0
3	Countdown	99.9	100.0	1.98	66.3 [61.7, 71.0]	–
3	Python	99.9	100.0	1.04	98.6 [97.2, 99.6]	1.2
3	MathIR	95.0	100.0	1.05	97.8 [95.9, 99.3]	20.0
3	PantryPlan	99.2	100.0	2.52	58.0 [53.1, 63.5]	6.5

Table 47: **Kimi K3: Post hoc formatting diagnostic.** Each cell has 128 prompts and 1,024 responses. M/P denote mean@8/pass@8 (%); D is distinct@8. C is correct-pair collision (%) with pointwise 95% prompt-bootstrap intervals. U is its conditional uniform reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	99.5	100.0	1.48	84.3 [80.4, 87.8]	18.3
1	Countdown	99.4	100.0	1.91	68.4 [63.9, 73.3]	–
1	Python	98.2	100.0	2.53	57.7 [53.5, 61.9]	1.4
1	MathIR	90.2	92.2	1.47	75.8 [70.7, 80.7]	20.0
1	PantryPlan	97.5	100.0	4.18	26.2 [23.1, 29.2]	6.5
2	Graph	99.6	100.0	1.62	80.0 [76.0, 84.1]	19.0
2	Countdown	98.7	100.0	1.98	64.5 [60.1, 68.9]	–
2	Python	98.3	100.0	2.44	61.9 [57.6, 66.2]	1.1
2	MathIR	94.3	100.0	1.02	99.8 [99.3, 100.0]	20.0
2	PantryPlan	93.4	100.0	3.20	41.4 [36.7, 46.0]	6.6
3	Graph	99.9	100.0	1.19	94.6 [91.8, 96.9]	19.0
3	Countdown	98.1	100.0	1.84	70.3 [65.8, 74.5]	–
3	Python	98.5	100.0	2.34	64.9 [60.3, 69.2]	1.2
3	MathIR	98.1	100.0	1.00	100.0 [100.0, 100.0]	20.0
3	PantryPlan	95.6	100.0	3.13	43.4 [38.6, 48.3]	6.5

Table 48: **Claude Opus 4.8: Post hoc formatting diagnostic.** Each cell has 128 prompts and 1,024 responses. M/P denote mean@8/pass@8 (%); D is distinct@8. C is correct-pair collision (%) with pointwise 95% prompt-bootstrap intervals. U is its conditional uniform reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	99.7	100.0	1.34	87.5 [83.9, 91.0]	18.3
1	Countdown	95.6	99.2	1.78	69.8 [64.9, 74.5]	–
1	Python	98.6	100.0	1.40	85.8 [81.8, 89.5]	1.5
1	MathIR	92.3	94.5	1.10	93.7 [91.1, 96.4]	20.0
1	PantryPlan	96.6	100.0	2.59	55.3 [50.3, 60.1]	6.5
2	Graph	95.9	100.0	1.66	76.6 [72.3, 81.1]	18.8
2	Countdown	99.9	100.0	1.99	65.3 [60.9, 69.9]	–
2	Python	100.0	100.0	1.34	86.0 [82.3, 89.6]	1.2
2	MathIR	97.0	100.0	1.24	90.1 [86.7, 93.4]	20.0
2	PantryPlan	99.5	100.0	2.11	67.9 [62.8, 73.1]	6.5
3	Graph	99.9	100.0	1.08	97.3 [95.5, 98.9]	19.0
3	Countdown	97.9	99.2	1.80	72.5 [68.2, 76.6]	–
3	Python	98.9	100.0	1.32	86.4 [82.5, 90.1]	1.2
3	MathIR	99.1	100.0	1.03	99.1 [98.1, 99.8]	20.0
3	PantryPlan	98.7	100.0	2.33	61.7 [56.7, 66.6]	6.5

Table 49: **DeepSeek V4 Pro: Post hoc formatting diagnostic.** Each cell has 128 prompts and 1,024 responses. M/P denote mean@8/pass@8 (%); D is distinct@8. C is correct-pair collision (%) with pointwise 95% prompt-bootstrap intervals. U is its conditional uniform reference (%).

Level	Domain	M	P	D	C (95% interval)	U
1	Graph	94.0	100.0	1.80	73.1 [68.8, 77.4]	18.1
1	Countdown	100.0	100.0	2.02	65.8 [61.4, 70.3]	–
1	Python	99.4	100.0	1.41	88.8 [86.1, 91.4]	1.4
1	MathIR	90.1	92.2	1.11	93.9 [91.2, 96.2]	20.0
1	PantryPlan	98.5	100.0	3.73	33.5 [29.8, 37.2]	6.5
2	Graph	96.2	100.0	1.97	69.0 [64.4, 73.5]	19.1
2	Countdown	99.9	100.0	2.04	63.4 [59.2, 67.7]	–
2	Python	99.7	100.0	1.05	98.4 [97.1, 99.4]	1.2
2	MathIR	85.4	95.3	1.82	51.7 [49.8, 53.9]	20.0
2	PantryPlan	87.1	100.0	2.34	61.1 [56.0, 66.2]	6.5
3	Graph	97.5	100.0	1.52	83.4 [79.6, 87.0]	18.9
3	Countdown	99.2	100.0	1.98	67.5 [63.1, 72.2]	–
3	Python	99.7	100.0	1.08	97.8 [96.3, 99.0]	1.2
3	MathIR	88.5	100.0	1.44	83.6 [79.2, 87.6]	20.0
3	PantryPlan	70.3	98.4	2.08	61.9 [55.9, 68.2]	6.5

I Matched Prompt-Hint Ablation

I.1 Removing strategy hints from unchanged problems

We freeze a paired follow-up on Python factors, MathIR, and Pantry at Levels 2 and 3. Within each of these six cells, the 32 smallest SHA-256 hashes of (20260911, level, domain, row index) select problems from the 128-row evaluation split, without consulting outcomes. Both arms generate eight fresh draws per identical problem. The original arm retains the published prompt. The neutral arm removes the Python suggestions to test small divisors with nested conditionals or dispatch on listed values, the MathIR ordered algebraic-isolation suggestion, and the Pantry preference for high-energy/protein, low-sodium ingredients such as seeds or oats. MathIR changes “those operations” to “the operations” to preserve the menu-ID instruction. The user problem, answer specification, executable verifier, and remaining format requirements are unchanged.

We report the paired neutral-minus-original effects on empirical $P_8 = \text{pass@8}$ and $D_8 = \text{distinct@8}$, with $B_8 = D_8 - P_8$ as additional verified breadth. All returned answers, including incorrect, empty, and truncated outputs, remain in their eight-draw groups; only verifier-accepted outputs create a verified key. Strict verification is primary; one unchanged, previously frozen formatting normalizer is a sensitivity analysis and preserves every strict success and canonical key. Python outputs in both arms undergo the same serial, warmed-verifier recheck, with original grades and any corrections retained in separate receipts.

Intervals use 20,000 whole-prompt bootstrap resamples, paired across arms and stratified by domain and level (seed 20260911). Cells contribute equally to reported macros. These are pointwise exploratory intervals for 32 problems per cell; crossing zero does not establish equivalence or absence of a prompt effect. A zero-width bootstrap interval records zero observed resampled variation, including all-tied P_8 outcomes; it is not precise evidence of population equivalence. Correct-pair collision and the certified-support uniform reference are secondary and retain their correct-pair denominators; the machine-readable report includes joint eligibility counts and undefined cells. Eight sampled draws do not identify total unseen support.

This appendix reports the complete local panel only. The separately registered frontier panel is not included here, and the overall local-plus-frontier experiment is incomplete.

For local models, we evaluate the initial Qwen2.5-0.5B model and fixed Level-2-trained DrGRPO and Replay DrGRPO checkpoints. Python and MathIR use matched training seeds 43–47; Pantry uses seeds 43 and 46. Level 3 is transfer evaluation. Seed means retain the same selected problem set across checkpoints. Five-seed intervals resample whole training seeds and paired prompts; Pantry reports both seed effects and intervals conditional on the two checkpoints. The native local syntax constraints and 192-token budget differ from hosted inference, so effects are interpreted within each model and interface.

Table 50: Local Qwen2.5-0.5B checkpoints under strict verification. Each arrow gives original $\rightarrow$ neutral. P_8 is empirical pass@8 , D_8 is verified distinct@8 , and $B_8 = D_8 - P_8$. Effects are neutral minus original; brackets are pointwise 95% intervals. Local rows average the displayed number of training seeds. Five-seed intervals include seed and paired-prompt resampling; two-seed Pantry intervals condition on the two fixed checkpoints.

Model	Cell	P_8 : O $\rightarrow$ N	D_8 : O $\rightarrow$ N	ΔP_8 [95%]	ΔD_8 [95%]	ΔB_8
Initial Qwen 0.5B ($s = 1$)	Python factors L2	0.656 $\rightarrow$ 0.125	0.69 $\rightarrow$ 0.12	-0.531 [-0.719, -0.344]	-0.56 [-0.75, -0.38]	-0.03
Initial Qwen 0.5B ($s = 1$)	MathIR L2	0.250 $\rightarrow$ 0.281	0.25 $\rightarrow$ 0.28	+0.031 [+0.000, +0.094]	+0.03 [+0.00, +0.09]	+0.00
Initial Qwen 0.5B ($s = 1$)	Pantry L2	0.156 $\rightarrow$ 0.188	0.19 $\rightarrow$ 0.22	+0.031 [+0.000, +0.094]	+0.03 [+0.00, +0.09]	+0.00
Initial Qwen 0.5B ($s = 1$)	Python factors L3	0.719 $\rightarrow$ 0.031	0.81 $\rightarrow$ 0.03	-0.688 [-0.844, -0.531]	-0.78 [-0.97, -0.59]	-0.09
Initial Qwen 0.5B ($s = 1$)	MathIR L3	0.219 $\rightarrow$ 0.219	0.22 $\rightarrow$ 0.22	+0.000 [-0.125, +0.125]	+0.00 [-0.12, +0.12]	+0.00
Initial Qwen 0.5B ($s = 1$)	Pantry L3	0.281 $\rightarrow$ 0.312	0.66 $\rightarrow$ 0.69	+0.031 [+0.000, +0.094]	+0.03 [-0.06, +0.12]	+0.00
DrGRPO ($s = 5$)	MathIR L2	0.531 $\rightarrow$ 0.506	0.53 $\rightarrow$ 0.51	-0.025 [-0.100, +0.031]	-0.03 [-0.10, +0.03]	+0.00
DrGRPO ($s = 5$)	MathIR L3	0.194 $\rightarrow$ 0.156	0.19 $\rightarrow$ 0.16	-0.037 [-0.113, +0.019]	-0.04 [-0.11, +0.02]	+0.00
DrGRPO ($s = 2$)	Pantry L2	0.219 $\rightarrow$ 0.203	0.28 $\rightarrow$ 0.25	-0.016 [-0.156, +0.125]	-0.03 [-0.19, +0.12]	-0.02
DrGRPO ($s = 2$)	Pantry L3	0.172 $\rightarrow$ 0.172	0.27 $\rightarrow$ 0.25	+0.000 [-0.078, +0.078]	-0.02 [-0.16, +0.11]	-0.02
DrGRPO ($s = 5$)	Python factors L2	0.812 $\rightarrow$ 0.000	0.81 $\rightarrow$ 0.00	-0.812 [-0.938, -0.656]	-0.81 [-0.94, -0.66]	+0.00
DrGRPO ($s = 5$)	Python factors L3	1.000 $\rightarrow$ 0.000	1.00 $\rightarrow$ 0.00	-1.000 [-1.000, -1.000]	-1.00 [-1.00, -1.00]	+0.00
Replay DrGRPO ($s = 5$)	MathIR L2	0.994 $\rightarrow$ 0.975	0.99 $\rightarrow$ 0.97	-0.019 [-0.075, +0.000]	-0.02 [-0.07, +0.00]	+0.00
Replay DrGRPO ($s = 5$)	MathIR L3	0.312 $\rightarrow$ 0.275	0.31 $\rightarrow$ 0.28	-0.037 [-0.138, +0.044]	-0.04 [-0.14, +0.04]	+0.00
Replay DrGRPO ($s = 2$)	Pantry L2	0.516 $\rightarrow$ 0.500	0.75 $\rightarrow$ 0.73	-0.016 [-0.109, +0.078]	-0.02 [-0.20, +0.17]	+0.00
Replay DrGRPO ($s = 2$)	Pantry L3	0.234 $\rightarrow$ 0.281	0.39 $\rightarrow$ 0.50	+0.047 [+0.000, +0.109]	+0.11 [+0.05, +0.19]	+0.06
Replay DrGRPO ($s = 5$)	Python factors L2	0.887 $\rightarrow$ 0.556	1.24 $\rightarrow$ 0.68	-0.331 [-0.525, -0.150]	-0.57 [-0.85, -0.32]	-0.24
Replay DrGRPO ($s = 5$)	Python factors L3	1.000 $\rightarrow$ 0.494	1.42 $\rightarrow$ 0.66	-0.506 [-0.825, -0.194]	-0.76 [-1.07, -0.38]	-0.26

Table 51: Local Qwen2.5-0.5B checkpoints under frozen formatting normalization. Each arrow gives original $\rightarrow$ neutral. P_8 is empirical pass@8, D_8 is verified distinct@8, and $B_8 = D_8 - P_8$. Effects are neutral minus original; brackets are pointwise 95% intervals. Local rows average the displayed number of training seeds. Five-seed intervals include seed and paired-prompt resampling; two-seed Pantry intervals condition on the two fixed checkpoints.

Model	Cell	P_8 : O $\rightarrow$ N	D_8 : O $\rightarrow$ N	ΔP_8 [95%]	ΔD_8 [95%]	ΔB_8
Initial Qwen 0.5B ($s = 1$)	Python factors L2	0.656 $\rightarrow$ 0.125	0.69 $\rightarrow$ 0.12	-0.531 [-0.719, -0.344]	-0.56 [-0.75, -0.38]	-0.03
Initial Qwen 0.5B ($s = 1$)	MathIR L2	0.250 $\rightarrow$ 0.281	0.25 $\rightarrow$ 0.28	+0.031 [+0.000, +0.094]	+0.03 [+0.00, +0.09]	+0.00
Initial Qwen 0.5B ($s = 1$)	Pantry L2	0.156 $\rightarrow$ 0.188	0.19 $\rightarrow$ 0.22	+0.031 [+0.000, +0.094]	+0.03 [+0.00, +0.09]	+0.00
Initial Qwen 0.5B ($s = 1$)	Python factors L3	0.719 $\rightarrow$ 0.031	0.81 $\rightarrow$ 0.03	-0.688 [-0.844, -0.531]	-0.78 [-0.97, -0.59]	-0.09
Initial Qwen 0.5B ($s = 1$)	MathIR L3	0.219 $\rightarrow$ 0.219	0.22 $\rightarrow$ 0.22	+0.000 [-0.125, +0.125]	+0.00 [-0.12, +0.12]	+0.00
Initial Qwen 0.5B ($s = 1$)	Pantry L3	0.281 $\rightarrow$ 0.312	0.66 $\rightarrow$ 0.69	+0.031 [+0.000, +0.094]	+0.03 [-0.06, +0.12]	+0.00
DrGRPO ($s = 5$)	MathIR L2	0.531 $\rightarrow$ 0.506	0.53 $\rightarrow$ 0.51	-0.025 [-0.100, +0.031]	-0.03 [-0.10, +0.03]	+0.00
DrGRPO ($s = 5$)	MathIR L3	0.194 $\rightarrow$ 0.156	0.19 $\rightarrow$ 0.16	-0.037 [-0.113, +0.019]	-0.04 [-0.11, +0.02]	+0.00
DrGRPO ($s = 2$)	Pantry L2	0.219 $\rightarrow$ 0.203	0.28 $\rightarrow$ 0.25	-0.016 [-0.156, +0.125]	-0.03 [-0.19, +0.12]	-0.02
DrGRPO ($s = 2$)	Pantry L3	0.172 $\rightarrow$ 0.172	0.27 $\rightarrow$ 0.25	+0.000 [-0.078, +0.078]	-0.02 [-0.16, +0.11]	-0.02
DrGRPO ($s = 5$)	Python factors L2	0.812 $\rightarrow$ 0.000	0.81 $\rightarrow$ 0.00	-0.812 [-0.938, -0.656]	-0.81 [-0.94, -0.66]	+0.00
DrGRPO ($s = 5$)	Python factors L3	1.000 $\rightarrow$ 0.000	1.00 $\rightarrow$ 0.00	-1.000 [-1.000, -1.000]	-1.00 [-1.00, -1.00]	+0.00
Replay DrGRPO ($s = 5$)	MathIR L2	0.994 $\rightarrow$ 0.975	0.99 $\rightarrow$ 0.97	-0.019 [-0.075, +0.000]	-0.02 [-0.07, +0.00]	+0.00
Replay DrGRPO ($s = 5$)	MathIR L3	0.312 $\rightarrow$ 0.275	0.31 $\rightarrow$ 0.28	-0.037 [-0.138, +0.044]	-0.04 [-0.14, +0.04]	+0.00
Replay DrGRPO ($s = 2$)	Pantry L2	0.516 $\rightarrow$ 0.500	0.75 $\rightarrow$ 0.73	-0.016 [-0.109, +0.078]	-0.02 [-0.20, +0.17]	+0.00
Replay DrGRPO ($s = 2$)	Pantry L3	0.234 $\rightarrow$ 0.281	0.39 $\rightarrow$ 0.50	+0.047 [+0.000, +0.109]	+0.11 [+0.05, +0.19]	+0.06
Replay DrGRPO ($s = 5$)	Python factors L2	0.887 $\rightarrow$ 0.556	1.24 $\rightarrow$ 0.68	-0.331 [-0.525, -0.150]	-0.57 [-0.85, -0.32]	-0.24
Replay DrGRPO ($s = 5$)	Python factors L3	1.000 $\rightarrow$ 0.494	1.42 $\rightarrow$ 0.66	-0.506 [-0.825, -0.194]	-0.76 [-1.07, -0.38]	-0.26

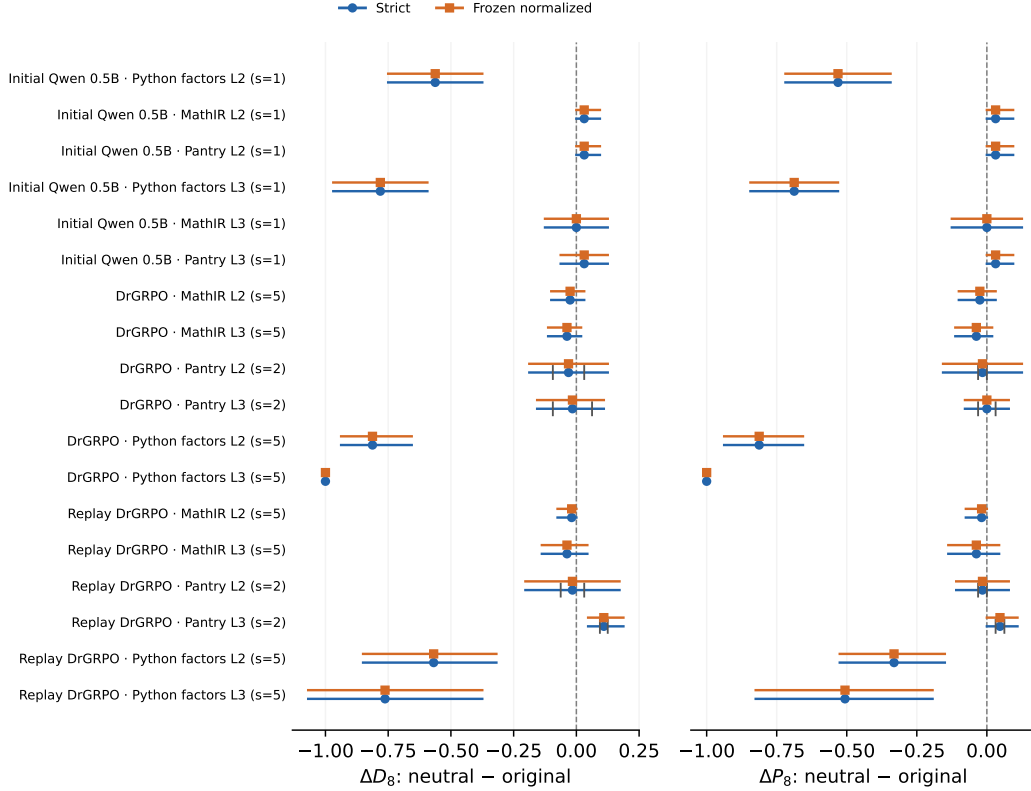


Figure 19: Prompt-hint removal effects for local checkpoints, preserving every registered domain-level cell. Blue circles use strict verification; orange squares use the same frozen formatting normalizer. Horizontal bars show the intervals described in the tables; zero is the vertical reference. Gray endpoint marks show the two individual Pantry seed effects.

903 **Two-seed Pantry sensitivity.** Two checkpoints cannot support a stable estimate of training-seed
 904 variability. The following lists both strict prompt effects; the intervals in the local table condition
 905 on these two checkpoints. DrGRPO L2 (seed 43: $\Delta P_8 = +0.000$, $\Delta D_8 = +0.03$; seed 46:
 906 $\Delta P_8 = -0.031$, $\Delta D_8 = -0.09$); DrGRPO L3 (seed 43: $\Delta P_8 = +0.031$, $\Delta D_8 = +0.06$; seed 46:
 907 $\Delta P_8 = -0.031$, $\Delta D_8 = -0.09$); Replay DrGRPO L2 (seed 43: $\Delta P_8 = -0.031$, $\Delta D_8 = -0.06$;
 908 seed 46: $\Delta P_8 = +0.000$, $\Delta D_8 = +0.03$); Replay DrGRPO L3 (seed 43: $\Delta P_8 = +0.031$,
 909 $\Delta D_8 = +0.12$; seed 46: $\Delta P_8 = +0.062$, $\Delta D_8 = +0.09$).

910 **Interpreting the intervention.** The Python intervention removes an executable nested-conditional
 911 example along with divisor-testing and dispatch guidance; it also shortens the prompt. Its paired
 912 effect concerns this combined removal and does not isolate a preference cue from example or length
 913 effects. A reduction in `distinct08` accompanied by lower `pass08` is not by itself evidence of greater
 914 concentration among correct outputs. Here the initial model is the single fixed Qwen2.5-0.5B-Instruct
 915 checkpoint before the reported RL training; it is already instruction-tuned and has no fine-tuning-seed
 916 replication in this ablation.

917 **Observed prompt dependence.** Python shows a large loss of correctness after guidance removal.
 918 Across the five DrGRPO seeds, mean P_8 falls from .8125 to zero at Level 2 and from 1.0 to zero at
 919 Level 3; D_8 falls identically and $B_8 = 0$ in both arms. With no correct neutral DrGRPO outputs, its
 920 conditional-correct concentration is undefined. Replay DrGRPO retains neutral $P_8 = .55625$ and
 921 $.49375$ and $D_8 = .675$ and $.65625$ at Levels 2 and 3, respectively, while losing additional breadth:
 922 $\Delta B_8 = -.2375$ and $-.25625$. The initial checkpoint also loses Python correctness (Table 50). These
 923 observations identify substantial dependence on the combined Python wording intervention; a raw
 924 mode-count decline cannot be interpreted solely as a redistribution among correct answers.

925 **Post-hoc Python failure audit.** A descriptive audit of all 5,120 saved DrGRPO Python draws (five
926 seeds, two levels, both wordings) finds that all 2,560 original outputs extract the same lambda, whose
927 body matches the conditional example in the prompt:

```
928 lambda n: 2 if n % 2 == 0 else 3 if n % 3 == 0 else 5
```

929 Of the 2,560 neutral draws, 1,816 (70.94%) are invalid Python expressions, 59 violate other restricted-
930 language rules, 376 return non-integers, 276 fail the proper-divisor condition, and 33 lack an extracted
931 candidate after reaching the token limit. Thus most neutral failures involve syntax; truncation
932 accounts for only 1.29%. The unchanged extractor, parser, and executor reproduce every stored
933 grade and successful canonical key. This taxonomy was developed after the paired results were
934 known and does not identify an internal training mechanism. Counts, deterministic paired examples,
935 and source hashes are retained in `results/modebench_prompt_ablation_python_failure_`
936 `diagnostic_20260911.json`.

937 **Concentration after hint removal.** Every trained MathIR checkpoint has $B_8 = 0$ under both
938 wordings at both levels. Within each problem’s eight-draw group, every observed pair of correct
939 outputs shares a canonical key, in every trained checkpoint–level cell: correct-pair collision is 1.0,
940 versus the certified-support uniform reference of .2. For example, at Level 2, Replay DrGRPO
941 retains mean $P_8 = .975$ without the order hint, compared with .99375 originally. Across its five
942 checkpoints, the neutral arm has 4,179 colliding pairs out of 4,179 correct pairs, from 154 eligible
943 checkpoint–problem groups; the original arm has 4,382 out of 4,382 pairs from 157 such groups. Both
944 arms evaluate the same 32 problems at each checkpoint. This preserves the observed concentration
945 despite removing the specified order hint; it does not measure all unseen modes or remove the other
946 interface constraints. Pantry effects are mixed. Replay DrGRPO at Level 3 gains .109375 in D_8 ,
947 comprising .046875 in P_8 and .0625 in B_8 , but this comparison has only two training seeds. These
948 local findings supply no completed frontier-model prompt control; that separately registered panel
949 remains pending.

950 J Related Work

951 Repeated sampling and verifier-guided selection depend on the proposal distribution [Brown et al.,
952 2024, Snell et al., 2024]. Prior work studies diversity loss in alignment and reasoning [Kirk et al.,
953 2024, Dang et al., 2025], entropy dynamics [Cui et al., 2025], and high-budget coverage [Yue et al.,
954 2025]. Semantic clustering [Farquhar et al., 2024] addresses open-ended outputs; MODEBENCH
955 instead uses executable identity keys. Sampling-aware objectives promote useful proposal sets [Song
956 et al., 2025, Chen et al., 2025, Li et al., 2025, 2026a,b, Lochab et al., 2026]. MaxRL optimizes
957 best-of- k success [Tajwar et al., 2026], which equals $\text{pass}@k$ for binary rewards. Replay has a
958 long history in reinforcement and continual learning [Lin, 1992, Rolnick et al., 2019]; RLEP reuses
959 successful reasoning traces [Zhang et al., 2025]. Our bank retains one policy exemplar per verified key
960 and weights keys uniformly, connecting persistent success memory to novelty and quality-diversity
961 search [Lehman and Stanley, 2011, Mouret and Clune, 2015].

962 K Claims and Their Evidential Basis

963 The training and hosted studies share a measurement question: how is correct probability allocated
964 among verified alternatives? They supply different kinds of evidence. The table states the basis of
965 each central claim and the inference it supports; motivating downstream benefits remain hypotheses.

Claim	Identifiable basis	Scope of the inference
Success and breadth differ	Identity and concavity: Lemma A.1	At fixed correctness, concentration in majorization order lowers expected breadth. More correct samples can also raise the raw mode count.
Uniform categorical replay balances mass	Identity: Equation (4)	Likelihood replay already contains a balancing term. This neither requires a separate balance loss nor equates exemplar scores with key probabilities.
Binary reward can induce collapse	Conditional theorem: Theorem L.3	Independent canonical-outcome logits, common length normalization, Euclidean mean flow, and a unique initial winner; no neural-optimizer guarantee.
Replay remains available when fresh gradients vanish	Conditional score calculation: Appendix L.3	Equal-reward groups have zero centered task update; bank scoring need not. An available gradient does not ensure improvement after a shared-parameter update.
Persistent replay protects banked modes	Conditional theorem and corollary: Theorem L.6, Corollary L.7	Fixed bank, fixed positive dose, and the stated mean flow yield probability floors. Uniformity needs full coverage and equal categorical weights; unbanked modes may vanish.
Exemplar loss can bound key probability	Conditional lemma: Lemma L.8	A bounded joint potential and complete-response likelihood under the same sampling law protect those prompts. Neither held-out transfer nor visibility in eight draws follows.
Training can concentrate correct-key outputs	Retrospective longitudinal comparison: Appendix F.1	Graph/Pantry Dr.GRPO and GRPO show increased collision at each scale. Eligibility and stream sensitivities limit the observed population; exact support extinction is not identified.
Replay can reduce conditional concentration	Retrospective matched replay comparison: Appendix F.1	ReplayDr.GRPO reduces collision in Graph/Pantry at each scale. Correctness can also change on those eligible prompts; this is not an all-prompt effect.
Replay improves measured endpoints	Controlled comparisons: Figures 4 and 5	Paired replay additions isolate the implemented intervention within the reported model/domain cohorts. All five 3B MaxRL blocks use five paired seeds; individual-mode retention is not identified.
Equal key weights can aid breadth	Controlled weighting comparison: Appendix E.6	Uniform versus fresh-frequency weighting changes extra sampled modes in the reported cohorts; the average correctness contrast is inconclusive.
Replay can help on Level 2	Controlled within-level comparisons: Figure 6	Four complete domains support within-level replay effects. Separate terminal test splits prevent a pure cross-level difficulty-effect interpretation.
Fixed exemplars show heterogeneous score changes	Exploratory longitudinal finding: Appendix F.4	Central scores improve while a tail declines. No matched no-replay bank arm identifies a causal retention effect or exact key probabilities.
Accurate hosted outputs can be concentrated	Exploratory deployment comparison: Appendix H	Observed correctness and breadth describe the stated protocols. They do not identify RLVR as the cause, a model-size effect, or training-time collapse.
Temperature changes observed breadth	Exploratory matched-prompt comparison: Appendix H.3	Four temperature settings describe one deployment and reasoning profile. The best observed point does not establish an optimal sampling temperature in general.

Table 52: **What each central claim rests on.** Identities concern the sampling law, theorems specify conditional dynamics, and experiments support the stated intervention or exploratory observation. All reported intervals are nominal and unadjusted.

966 The before/after verifier-training precheck is a longitudinal checkpoint comparison (Appendix F.2).
967 Its extra-mode declines are measured changes in the registered runs, not proof of zero probability for
968 lost keys or a universal decrease in raw breadth across domains.

969 L Mode Collapse and Verified Replay

970 Under the explicit categorical assumptions below, centered binary-reward estimators amplify a pre-
 971 existing imbalance among correct modes, whereas persistent verified replay supplies a loss barrier for
 972 banked modes. The complete-response argument then states what would be needed to transfer that
 973 protection to a language model. These results concern idealized update rules; they do not certify the
 974 implemented AdamW/PPO trajectories.

975 L.1 Categorical abstraction and mean updates

976 Fix one isolated prompt and a finite category set $\mathcal{A} = \mathcal{C} \dot{\cup} \mathcal{N}$, with $m \geq 2$ correct execution modes
 977 and $n \geq 1$ incorrect categories. Assign one independently optimized logit to each category, including
 978 one logit per canonical correct mode, and set $p_a = \text{softmax}(z)_a$, $P = \sum_{c \in \mathcal{C}} p_c$, and $q_c = p_c/P$.
 979 Thus P measures correctness and q the conditional distribution among correct modes. Logits are
 980 finite initially, $0 < P(0) < 1$, and explicit reference KL is zero. The collapse result uses Euclidean
 981 infinitesimal expected updates in these logits, without optimizer preconditioning, momentum, or
 982 another regularizer.

983 A language model induces probabilities over canonical outcomes, but this identification of probabili-
 984 ties does not identify its update geometry. Aggregating several response strings into a mode does
 985 not generally commute with gradient flow in response logits. One independent logit per canonical
 986 outcome is therefore an explicit abstraction, not a reduction proved for the neural policy or its
 987 validator equivalence classes.

988 Draw a group $A_1, \dots, A_G \stackrel{\text{iid}}{\sim} p$ with $G \geq 2$, write $R_i = \mathbf{1}[A_i \in \mathcal{C}]$, and let $M = \sum_i R_i$. At the
 989 on-policy point where the PPO ratio equals one, consider the common-length estimator

$$\hat{g}_G = \frac{1}{G} \sum_{i=1}^G w(M) \left(R_i - \frac{M}{G} \right) \nabla_z \log p_{A_i}. \quad (11)$$

990 For the Dr.GRPO abstraction, $w(M) = 1$ after absorbing the shared positive length scale. Reward-
 991 standardized GRPO with equal or common response-length normalization takes $w(M)$ to be the
 992 reciprocal group reward standard deviation on mixed groups (with a fixed numerical stabilizer if
 993 used). Standard variable-length GRPO also weights each response by its own inverse length [Shao
 994 et al., 2024, Liu et al., 2025]; that factor cannot generally be absorbed into $w(M)$ and is outside the
 995 equivalence proved here. In either included case, $0 < w(\ell) < \infty$ for $\ell \in \{1, \dots, G-1\}$. We define
 996 the entire update as zero on all-equal groups without evaluating an undefined reciprocal standard
 997 deviation; likewise, $w(M)M(G-M)$ below is defined as zero when $M = 0$ or $M = G$. All
 998 advantages and normalization weights are detached in the score estimator.

999 **Lemma L.1** (A group baseline does not remove frequency amplification). *For $0 < P < 1$, the*
 1000 *expected update in Equation (11) is*

$$\mathbb{E}[\hat{g}_G] = c_G(P) \nabla_z P, \quad c_G(P) = \frac{\mathbb{E}[w(M)M(G-M)]}{G^2 P(1-P)} > 0. \quad (12)$$

1001 *For the Dr.GRPO abstraction, $c_G(P) = (G-1)/G$. Group centering and reward standardization*
 1002 *rescale the expected gradient without changing its direction under these assumptions.*

1003 *Proof.* Condition on the complete binary reward vector, whose sum is M . The score identities give

$$\mathbb{E}[\nabla_z \log p_{A_i} \mid R_i = 1] = \nabla_z \log P, \quad \mathbb{E}[\nabla_z \log p_{A_i} \mid R_i = 0] = \nabla_z \log(1-P).$$

1004 There are M correct rows with centered reward $(G-M)/G$ and $G-M$ incorrect rows with centered
 1005 reward $-M/G$. Therefore the conditional expectation of the unaveraged sum in (11) is

$$w(M) \frac{M(G-M)}{G} (\nabla_z \log P - \nabla_z \log(1-P)) = w(M) \frac{M(G-M)}{GP(1-P)} \nabla_z P.$$

1006 The outer factor $1/G$ and expectation over $M \sim \text{Binomial}(G, P)$ yield (12). For $w \equiv 1$, $\mathbb{E}[M(G-M)]$
 1007 $= G(G-1)P(1-P)$, proving the Dr.GRPO value. $\square$

1008 The binomial identity also gives the denominator-free expression

$$c_G(P) = \frac{G-1}{G} \sum_{j=0}^{G-2} \binom{G-2}{j} w(j+1) P^j (1-P)^{G-2-j}.$$

1009 The binomial weights sum to one, so c_G lies between $(G-1) \min_j w(j+1)/G$ and $(G-1) \max_j w(j+1)/G$. This establishes the smooth, positive extension to $P = 0, 1$ used in the
1010 subsequent flow and convergence arguments.
1011

1012 The next lemma specializes the practical-estimator analysis of Tajwar et al. [2026, Section 4.3 and
1013 Appendix D, Theorem 5] to our categorical notation; we include the derivation for completeness.

1014 **Lemma L.2** (Binary MaxRL has the same correctness-gradient direction). *Under the same cate-*
1015 *gorical and common-length assumptions, let MaxRL assign $A_i^{\text{MaxRL}} = GR_i/M - 1$ when $M > 0$*
1016 *and zero when $M = 0$, as in Section 3. For $\hat{g}_G^{\text{MaxRL}} = G^{-1} \sum_i A_i^{\text{MaxRL}} \nabla_z \log p_{A_i}$ at the on-policy*
1017 *point, the expected update is*

$$\mathbb{E}[\hat{g}_G^{\text{MaxRL}}] = c_G^{\text{MaxRL}}(P) \nabla_z P, \quad c_G^{\text{MaxRL}}(P) = \frac{1 - (1-P)^{G-1}}{P} > 0, \quad (13)$$

1018 with continuous endpoint values $c_G^{\text{MaxRL}}(0) = G-1$ and $c_G^{\text{MaxRL}}(1) = 1$.

1019 *Proof.* Condition on $M = m > 0$. Each successful row has advantage $(G-m)/m$ and each failed
1020 row has advantage -1 . Using the same conditional score identities as above, the expected unaveraged
1021 sum is

$$(G-m)(\nabla_z \log P - \nabla_z \log(1-P)) = \frac{G-m}{P(1-P)} \nabla_z P.$$

1022 After the outer factor $1/G$ and averaging over $M \sim \text{Binomial}(G, P)$,

$$\mathbb{E}[(G-M)\mathbf{1}\{M > 0\}] = G(1-P) - G \Pr(M=0) = G[(1-P) - (1-P)^G].$$

1023 Substitution gives Equation (13). Its endpoint values follow by continuity (or the finite geometric-
1024 series identity), which also gives the stated continuous extension at both boundaries. $\square$

1025 The same finite-series identity gives the potential used below:

$$\Psi_{\text{MaxRL}}(P) := \int_0^P c_G^{\text{MaxRL}}(s) ds = \sum_{k=1}^{G-1} \frac{1 - (1-P)^k}{k}. \quad (14)$$

1026 In particular, $\Psi_{\text{MaxRL}}(1) = \sum_{k=1}^{G-1} 1/k$. For binary rewards, $\text{best}@k$ equals $\text{pass}@k$, so this is
1027 a harmonic mixture of sampling objectives for $k = 1, \dots, G-1$. The upper limit is $G-1$, not
1028 G : the implemented centered estimator drops the entire update on all-failure groups. Subtracting
1029 an unconditional score baseline even on those groups instead gives the order- G estimator. This
1030 distinction concerns the exact finite-group expectation at the on-policy point; it does not assert
1031 unbiasedness of subsequent clipped or adaptive-optimizer updates.

1032 L.2 Winner-take-all collapse in the categorical mean flow

1033 Consider $\dot{z} = c_G(P) \nabla_z P$, with any of the smooth positive coefficients established above: Dr.GRPO,
1034 common-length reward-standardized GRPO, or the specified centered binary MaxRL estimator.
1035 This infinitesimal expected-update model removes sampling noise and retains independently trained
1036 Euclidean logits. The conditional dynamics reduce to the standard frequency-dependent replicator
1037 system with fitness $f_c(q) = q_c$ [Hofbauer and Sigmund, 1998]. The result is a specialization to
1038 the group estimators, not a new general theorem about replicator dynamics. Related correctness-
1039 indifferent collapse analyses include Sinha et al. [2026], Lochab et al. [2026].

1040 **Theorem L.3** (Standard replicator specialization for binary-reward RL). *Under the independent-*
1041 *logit mean-flow assumptions above, start from finite logits with $0 < P(0) < 1$. If the conditional*
1042 *correct-mode distribution $q(0)$ has a unique largest coordinate c^* , then*

$$P(t) \longrightarrow 1, \quad q_{c^*}(t) \longrightarrow 1, \quad q_c(t) \longrightarrow 0 \quad (c \neq c^*).$$

1043 *Thus these categorical mean flows converge to a single correct execution mode for generic initial*
 1044 *conditions, even though every mode in $\mathcal{C}$ has exactly the same reward. This is an asymptotic statement*
 1045 *within the specified update geometry, not a claim of finite-time support loss or a convergence*
 1046 *guarantee for the stochastic neural optimizers used in our experiments.*

1047 *Proof.* The coefficient c_G is smooth and bounded on $[0, 1]$, and the softmax reward gradient is
 1048 bounded. Thus the logit flow exists for all finite times and cannot develop infinite logits at finite time.
 1049 For a correct mode c and incorrect category a ,

$$\dot{z}_c = c_G(P)p_c(1 - P), \quad \dot{z}_a = -c_G(P)p_aP.$$

1050 Also $\dot{P} = c_G(P)\|\nabla_z P\|_2^2 > 0$. If P converged to an interior value, continuity and positivity of c_G ,
 1051 together with

$$\|\nabla_z P\|_2^2 \geq P^2(1 - P)^2 \left(\frac{1}{m} + \frac{1}{n} \right),$$

1052 would keep $\dot{P}$ bounded away from zero, a contradiction. Hence $P \rightarrow 1$.

1053 Define the effective optimization time

$$\tau(t) = \int_0^t c_G(P(s))P(s)(1 - P(s)) ds.$$

1054 If $\tau(\infty) < \infty$, every correct and incorrect logit would have finite total variation, because the absolute
 1055 velocity of each is at most $\dot{\tau}$. All logits would then converge to finite values, contradicting $P \rightarrow 1$.
 1056 Therefore $\tau(t) \rightarrow \infty$, leaving infinite effective time for the correct-simplex dynamics below.

1057 On the correct simplex, changing from t to τ gives the replicator flow

$$\frac{dq_c}{d\tau} = q_c \left(q_c - \sum_{d \in \mathcal{C}} q_d^2 \right), \quad \frac{d}{d\tau} \log \frac{q_c}{q_d} = q_c - q_d. \quad (15)$$

1058 The initially unique maximum remains the unique maximum. For $c \neq c^*$, set $\xi_c = q_c/q_{c^*}$, so
 1059 $\xi_c(0) < 1$ and ξ_c is nonincreasing. Since $q_{c^*} \geq 1/m$,

$$\frac{d \log \xi_c}{d\tau} = -q_{c^*}(1 - \xi_c) \leq -\frac{1 - \xi_c(0)}{m}.$$

1060 Hence $\xi_c(\tau) \leq \xi_c(0) \exp[-(1 - \xi_c(0))\tau/m] \rightarrow 0$. Every competing mode vanishes relative to c^* ;
 1061 normalization then yields $q_{c^*} \rightarrow 1$, establishing the claimed single-mode limit. $\square$

1062 If several modes tie for the largest initial probability, symmetry preserves the tie; the same ratio
 1063 argument eliminates strictly smaller coordinates and leaves the tied maxima uniform. An exactly
 1064 uniform correct distribution is therefore a special initial condition. The theorem concerns limiting
 1065 concentration: finite softmax logits already give positive probabilities at every finite time, so finite-
 1066 time positivity alone is not a retention result.

1067 **Dependence on optimization geometry.** For a smooth parameterization $z = z(\theta)$ with Jacobian
 1068 $J_\theta = \partial z / \partial \theta$, Euclidean parameter flow instead induces $\dot{z} = c_G(P)J_\theta J_\theta^\top \nabla_z P$. The independently
 1069 trained logit theorem uses $J_\theta J_\theta^\top = I$ and does not automatically extend to arbitrary shared parameters
 1070 or another optimization metric. Equal binary reward alone does not force this winner selection: the
 1071 categorical natural-gradient reward flow can be represented by $\dot{z}_a = c_G(P)(R_a - P)$, which gives
 1072 identical velocities to correct logits and preserves their ratios. Here $R_a = \mathbf{1}[a \in \mathcal{C}]$. Parameter count
 1073 alone does not specify the optimization geometry. The empirical scale comparisons also change
 1074 pretrained models and learning-rate schedules; they neither measure the geometry nor identify the
 1075 cause of cross-scale differences. The theorem supplies a mechanism to test in controlled training; it
 1076 cannot identify the training history or cause of concentration in a hosted model.

1077 L.3 A replay signal when fresh groups supply none

1078 **Lemma L.4** (Fresh-group starvation and replay signal). *Fix the categorical policy above with*
 1079 *correctness probability P , and draw $G \geq 2$ fresh categories independently. The probability of a*
 1080 *mixed-reward group is*

$$h_G(P) = 1 - P^G - (1 - P)^G \leq G \min\{P, 1 - P\}.$$

1081 *Every all-equal group has zero fresh task component for Equation (11) and the specified centered*
 1082 *MaxRL estimator. A nonzero fresh task component therefore requires a mixed group, so its probability*
 1083 *is at most h_G . This gives no lower bound on gradient magnitude and says nothing about other loss*
 1084 *terms, clipping, or an optimizer step carrying momentum across successive updates.*

1085 *For a nonempty fixed verified bank $\mathcal{B} \subseteq \mathcal{C}$ with normalized positive weights w_b , let $R_w =$*
 1086 *$-\sum_b w_b \log p_b$ and $\rho_w > 0$. Extend w by zero outside the bank to obtain $\bar{w}$. The determinis-*
 1087 *tic replay ascent vector is*

$$v^{\text{replay}} = -\rho_w \nabla R_w = \rho_w (\bar{w} - p),$$

1088 *which vanishes only at $p = \bar{w}$. In particular, if $p_b \leq w_b/2$ for a banked mode, then $v_b^{\text{replay}} \geq \rho_w w_b/2$,*
 1089 *even when b is absent from the fresh group. For a uniform bank of $k \geq 2$ modes and fixed $\rho_w > 0$,*
 1090 *as $p \rightarrow \mathbf{e}_b$ at one banked correct mode, $h_G(P) \rightarrow 0$ whereas $\|v^{\text{replay}}\|_2 \rightarrow \rho_w \sqrt{1 - 1/k} > 0$. The*
 1091 *replay norm therefore stays positive as fresh task-signal opportunities vanish.*

1092 *Proof.* The all-correct and all-incorrect events are disjoint and have probabilities P^G and $(1 - P)^G$.
 1093 A mixed group requires both a correct and an incorrect sample; the union bound gives the two bounds
 1094 GP and $G(1 - P)$. For $R_w = -\sum_b w_b \log p_b$, the softmax score identity gives $\partial R_w / \partial z_a = p_a - \bar{w}_a$,
 1095 proving the replay formula and coordinate bound. Finally,

$$\|\bar{w} - \mathbf{e}_b\|_2^2 = (1 - 1/k)^2 + (k - 1)/k^2 = 1 - 1/k,$$

1096 which gives the strictly positive limiting norm for $k \geq 2$. □

1097 The displayed replay vector averages the fixed bank. A stochastic exemplar draw has this vector only
 1098 in expectation, and a nonzero logit coordinate can be annihilated by a degenerate neural Jacobian.
 1099 Replay also need not increase the next mixed-group probability: $h_G(P)$ is largest at $P = 1/2$. The
 1100 claim is that a banked target can supply a separate verified signal when the fresh task gradient is
 1101 unavailable.

1102 A bounded advantage based on fresh samples has a different boundary behavior. The next bound
 1103 includes group-dependent detached scores and distinguishes sampled updates from gradients of an
 1104 exactly evaluated entropy functional.

1105 **Lemma L.5** (Bounded sampled scores have no probability-independent boundary force). *In the same*
 1106 *independent-logit model, condition on the training history, draw $A_1, \dots, A_G$ independently from p ,*
 1107 *and let a_i be any detached, possibly group-dependent advantages with $|a_i| \leq B$. To interpret the*
 1108 *bound uniformly as $p_b \rightarrow 0$, assume a finite B independent of that limit. At the on-policy point define*

$$\hat{g} = \frac{1}{G} \sum_{i=1}^G a_i (\mathbf{e}_{A_i} - p).$$

1109 *For every category b ,*

$$|\mathbb{E}[\hat{g}_b]| \leq \mathbb{E}|\hat{g}_b| \leq 2Bp_b(1 - p_b).$$

1110 *On a group with no draw of b , exactly $\hat{g}_b = -p_b G^{-1} \sum_i a_i$; it is zero if the applied advantages*
 1111 *sum to zero. By comparison, the verified-replay component for a fixed target w has logit velocity*
 1112 *$\rho(w_b - p_b)$, which approaches $\rho w_b > 0$ for a banked category as $p_b \rightarrow 0$.*

1113 *Proof.* Use the triangle inequality and $|a_i| \leq B$. Each marginal draw satisfies $\mathbb{E}|\mathbf{1}\{A_i = b\} - p_b| =$
 1114 $2p_b(1 - p_b)$, irrespective of the dependence of a_i on the group. On the absence event all indicator
 1115 terms are zero. The replay formula follows by differentiating $-\sum_b w_b \log p_b$. □

1116 The bound compares the isolated logit-gradient components. It does not prove inevitable collapse,
 1117 rule out retention by a bounded estimator, or guarantee that replay dominates the full update. An
 1118 unsampled mode can still change through softmax normalization or shared parameters. Exact entropy
 1119 is outside the uniformly bounded-score premise because its surprisal is unbounded near zero; the
 1120 lemma is not an impossibility result for entropy regularization. Nor does the absence of a mode from
 1121 a finite sample establish zero probability.

1122 L.4 Fixed-bank replay and qualitative retention

1123 Let $\mathcal{B} \subseteq \mathcal{C}$ be a nonempty fixed bank of k modes that the policy has produced and the validator
 1124 has accepted. Uniform verified replay uses one exemplar per banked mode and, in the categorical
 1125 reduction, minimizes

$$R_{\mathcal{B}}(z) = -\frac{1}{k} \sum_{b \in \mathcal{B}} \log p_b. \quad (16)$$

1126 This is ordinary cross-entropy against the distribution that assigns mass $1/k$ to each banked mode
 1127 and zero elsewhere. It is not conditioned on the bank: decreasing (16) can raise total bank mass as
 1128 well as redistribute mass within the bank.

1129 In an averaged continuous-time model, absorb a fixed round-robin replay frequency and loss scale
 1130 into an effective dose $\rho > 0$. This assumes a fixed eligible-bank set and constant relative weighting;
 1131 it does not identify finite alternating updates with simultaneous gradient flow. The prompt-isolated
 1132 categorical mean flow for any of the three fresh binary-reward objectives above plus verified replay is

$$\dot{z} = c_G(P) \nabla_z P - \rho \nabla_z R_{\mathcal{B}}(z). \quad (17)$$

1133 Let $\Psi(P) = \int_0^P c_G(s) ds$. For the finite-group estimators in Lemmas L.1 and L.2, c_G has a
 1134 finite positive extension to $[0, 1]$; in particular, Dr.GRPO has $c_G = (G - 1)/G$ and MaxRL has
 1135 $c_G = [1 - (1 - P)^{G-1}]/P$. Define

$$F_{\mathcal{B}}(z) = \rho R_{\mathcal{B}}(z) - \Psi(P(z)). \quad (18)$$

1136 Then (17) is exactly gradient flow $\dot{z} = -\nabla_z F_{\mathcal{B}}$.

1137 **Theorem L.6** (Verified replay prevents banked-mode extinction). *Suppose logits are finite at time T*
 1138 *and thereafter the nonempty verified bank $\mathcal{B}$ is fixed, fits within replay capacity, and receives a fixed*
 1139 *positive effective replay dose ρ . Under (17), define*

$$C_T = R_{\mathcal{B}}(z(T)) + \frac{\Psi(1) - \Psi(P(T))}{\rho}.$$

1140 Then for every $b \in \mathcal{B}$ and every $t \geq T$,

$$p_b(t) \geq \exp(-kC_T) > 0. \quad (19)$$

1141 Thus each banked probability is uniformly bounded away from zero for all $t \geq T$, including
 1142 asymptotically, even when the bank does not cover every correct mode. This is stronger than
 1143 finite-time positivity, which already follows from finite softmax logits without replay.

1144 *Proof.* The smooth vector field has bounded velocity because c_G , $\nabla_z P$, and $\nabla_z R_{\mathcal{B}}$ are bounded, so
 1145 finite initial logits give a flow for every finite $t \geq T$. Along this flow,

$$\frac{d}{dt} F_{\mathcal{B}}(z(t)) = -\|\nabla_z F_{\mathcal{B}}(z(t))\|_2^2 \leq 0.$$

1146 Since $P(t) \leq 1$ and Ψ is increasing,

$$R_{\mathcal{B}}(z(t)) \leq R_{\mathcal{B}}(z(T)) + \frac{\Psi(1) - \Psi(P(T))}{\rho} = C_T.$$

1147 Every $-\log p_b$ is nonnegative and their sum is $kR_{\mathcal{B}}$. Therefore $-\log p_b(t) \leq kC_T$ for every banked
 1148 mode, which is exactly Equation (19) for every time after T . $\square$

1149 The floor is qualitative and can be numerically vacuous. For example, at $k = 16$ even a hypothetical
 1150 $C_T = 160$ gives e^{-2560} , far below any useful evaluation probability. The argument establishes a
 1151 uniform-in-time lower bound in the stated ideal flow, not practical visibility or a prediction for the
 1152 experimental replay loss. The effective dose also depends on relative fresh/replay frequencies, not
 1153 the replay coefficient alone.

1154 A common nonnegative, locally integrable rate $\eta(t)$ multiplying both terms of (17) preserves the
 1155 energy bound, since $\dot{F}_{\mathcal{B}} = -\eta(t)\|\nabla F_{\mathcal{B}}\|_2^2 \leq 0$. The convergence statements below additionally
 1156 require $\int_T^\infty \eta(t) dt = \infty$ and then follow by reparameterizing optimization time. Infinitely many
 1157 replay visits alone establish neither this condition nor a fixed replay-to-fresh weighting. Different
 1158 time-varying relative weights are outside the fixed-potential argument.

1159 Full coverage and the categorical target.

1160 **Corollary L.7** (Conditional full-coverage categorical limit). *Assume the conditions of Theorem L.6*
 1161 *and that by time T the bank covers the full correct support, $\mathcal{B} = \mathcal{C}$. Let $u_c = 1/m$ on $\mathcal{C}$. Then the*
 1162 *categorical mean flow converges to*

$$P(t) \longrightarrow 1, \quad q(t) \longrightarrow u.$$

1163 *Consequently*

$$\min_{c \in \mathcal{C}} q_c(t) \longrightarrow \frac{1}{m}, \quad \mathcal{H}(q(t)) \longrightarrow \log m,$$

1164 *and conditional correct-mode collapse is impossible.*

1165 *Proof.* For full coverage,

$$R_{\mathcal{C}} = \log m - \log P + \text{KL}(u||q). \quad (20)$$

1166 The potential is bounded below by $-\Psi(1)$, so its descent gives $\int_T^\infty \|\nabla F_{\mathcal{C}}\|_2^2 dt < \infty$. The finite-
 1167 group c_G is smooth on $[0, 1]$, and softmax derivatives and the cross-entropy gradient are bounded.
 1168 Thus the Hessian of $F_{\mathcal{C}}$ and the velocity are bounded; the gradient along the trajectory is uniformly
 1169 continuous. Square integrability then implies $\nabla F_{\mathcal{C}} \rightarrow 0$: recurring gradients of fixed positive size
 1170 would otherwise contribute energy over intervals of a uniform minimum length.

1171 Every limiting probability vector therefore satisfies the correct-coordinate equations

$$0 = \rho(p_c - 1/m) - c_G(P)p_c(1 - P).$$

1172 Summing them gives $0 = -(1 - P)(\rho + c_G(P)P)$, hence $P = 1$. Each correct coordinate is then
 1173 $1/m$ and every incorrect coordinate is zero. Compactness of the probability simplex and uniqueness
 1174 of this limit establish convergence of the probability vector. $\square$

1175 **Positive weights and the implemented length normalization.** Uniformity of the target is separate
 1176 from protection against extinction. For fixed weights $w_b > 0$ with $\sum_{b \in \mathcal{B}} w_b = 1$, define $R_w =$
 1177 $-\sum_b w_b \log p_b$ and use replay coefficient $\rho_w > 0$. The same potential argument gives, for all $t \geq T$,

$$p_b(t) \geq \exp(-C_{T,w}/w_b), \quad C_{T,w} = R_w(z(T)) + \frac{\Psi(1) - \Psi(P(T))}{\rho_w}.$$

1178 If $\mathcal{B} = \mathcal{C}$, then $R_w = H(w) - \log P + \text{KL}(w||q)$, and the independent-logit limit is $P \rightarrow 1, q \rightarrow w$.
 1179 The preceding stationarity proof applies with $1/m$ replaced by w_c : summing the correct-coordinate
 1180 equations forces $P = 1$, then each equation forces $p_c = w_c$.

1181 A partial bank has a stronger limitation in this same ideal flow. For any fixed nonempty $\mathcal{B} \subseteq \mathcal{C}$ and
 1182 fixed positive weights summing to one, extend w by zero to $\bar{w}$ on all categories. Then $p(t) \rightarrow \bar{w}$,
 1183 so even correct modes outside the frozen bank vanish asymptotically. For $F = \rho_w R_w - \Psi(P)$, the
 1184 preceding energy and bounded-Hessian argument still gives $\nabla F \rightarrow 0$; neither argument requires full
 1185 bank coverage. At every subsequential probability limit, the correct-coordinate equations are

$$0 = \rho_w(p_c - \bar{w}_c) - c_G(P)p_c(1 - P).$$

1186 Summing over correct categories gives $0 = -(1 - P)(\rho_w + c_G(P)P)$, hence $P = 1$; each equation
 1187 then gives $p_c = \bar{w}_c$, and all incorrect mass is zero. Compactness and uniqueness of this limiting
 1188 probability vector prove convergence. Thus protecting a fixed set of banked modes need not preserve

1189 unbanked correct alternatives. This conclusion concerns the fixed-bank categorical flow, not changing-
1190 bank or stochastic neural training.

1191 This distinction matters for the implemented mean-token exemplar loss. Even if each key were
1192 represented by exactly one categorical exemplar of fixed length L_b , its loss would be AR_w , where

$$A = \frac{1}{k} \sum_{b \in \mathcal{B}} \frac{1}{L_b}, \quad w_b = \frac{L_b^{-1}}{\sum_{j \in \mathcal{B}} L_j^{-1}}.$$

1193 Its effective categorical replay coefficient would be $\rho_w = \rho_{\text{LM}} A$. The full-coverage limit would
1194 therefore weight shorter exemplars more, and would be uniform only for equal lengths. For an actual
1195 language model, $\pi_\theta(e_b | x)$ is only one contribution to the total key probability. Equal averaging
1196 of per-token exemplar scores does not imply uniform learned key probabilities; the categorical
1197 uniformity corollary is not that claim.

1198 Full support is an additional premise, not a consequence of replay. In Python factors, prompts admit
1199 16–3,600 correct execution modes (mean 229.44), while the bank capacity is 16. The full-coverage
1200 corollary therefore cannot supply a domain-wide guarantee there. The retention theorem protects
1201 only admitted banked modes; it supplies no probability floor for an unbanked key.

1202 L.5 Complete responses and finite evaluation visibility

1203 The learner scores a particular response string, while evaluation groups strings by execution key.
1204 Event containment provides a conditional bridge: one complete verified response is one way of
1205 producing its key. This bridge also makes the loss’s length factor explicit. Here $p_\theta(b | x) =$
1206 $\Pr_{Y \sim \pi_\theta(\cdot | x)}[V(Y, s_x) = b]$ is the unconditional probability of producing key b , rather than the
1207 probability conditional on correctness.

1208 **Lemma L.8** (Retention from a joint exemplar potential). *Fix finitely many verified complete-response*
1209 *exemplars (x_j, e_j) , each realizing key b_j , with lengths $L_j > 0$ and fixed weights $\alpha_j > 0$. Let*

$$F(\theta) = \sum_j \alpha_j \frac{-\log \pi_\theta(e_j | x_j)}{L_j} - J(\theta), \quad J(\theta) \leq J_{\max} < \infty.$$

1210 *Suppose $F(\theta(T))$ is finite and $F(\theta(t)) \leq F(\theta(T))$ for every $t \geq T$. With $D = F(\theta(T)) + J_{\max}$,*
1211 *every protected exemplar satisfies*

$$p_\theta(b_j | x_j) \geq \pi_\theta(e_j | x_j) \geq \exp\left(-\frac{L_j D}{\alpha_j}\right) > 0 \quad (t \geq T).$$

1212 *Proof.* The weighted exemplar loss equals $F(\theta(t)) + J(\theta(t)) \leq D$. Each summand is nonnegative,
1213 so $-\log \pi_\theta(e_j | x_j) \leq L_j D / \alpha_j$. The complete-response event is a subset of the event producing key
1214 b_j , which proves the stated lower bound on every protected key’s probability. $\square$

1215 Exact descent of a smooth joint potential is sufficient for the energy inequality, even with shared
1216 parameters and multiple prompts. For example, the ideal categorical model permits $J(\theta) =$
1217 $\sum_x \nu_x \Psi_x(P_x(\theta))$ with fixed nonnegative prompt weights and $J_{\max} = \sum_x \nu_x \Psi_x(1)$. The lemma
1218 does not establish this descent condition for stochastic, clipped, alternating AdamW updates, and
1219 it neither implies uniform neural key probabilities nor requires them. Its bound applies only to the
1220 listed exemplar–prompt pairs. Shared parameters do not extend it to held-out prompts; transfer to
1221 those prompts remains an empirical question.

1222 The scored probability must describe the complete response, including its termination event or a fixed
1223 deterministic horizon, under the same prompt, temperature, masks, truncation rules, and verifier as
1224 the sampling law being bounded. A prefix score or a score under another decoding law does not
1225 supply this bound without a separate event or likelihood comparison. For a complete response of
1226 length L and mean-token log score s , its probability is e^{Ls} , not e^s . A change in this exemplar’s
1227 likelihood need not equal the change in its whole key’s probability.

1228 **Retention and finite evaluation visibility.** At a fixed checkpoint, suppose k distinct banked keys
 1229 for one prompt have probability floors $p_b \geq \delta_b > 0$. For K independent draws from that same policy,
 1230 the banked distinct count $D_{\mathcal{B},K}$ satisfies

$$\mathbb{E}[D_{\mathcal{B},K}] \geq \sum_{b \in \mathcal{B}} [1 - (1 - \delta_b)^K],$$

$$\Pr(\text{some banked key is unseen}) \leq \sum_{b \in \mathcal{B}} (1 - \delta_b)^K \leq k e^{-K \delta_{\min}},$$

1231 where $\delta_{\min} = \min_b \delta_b$. The first statement follows from indicator expectations and the second from
 1232 the union bound. Thus $K \geq \log(k/\varepsilon)/\delta_{\min}$ suffices to observe all banked keys with probability at
 1233 least $1 - \varepsilon$, for $0 < \varepsilon < 1$. The energy floors can make this budget impractically large; asymptotic
 1234 retention does not imply visibility in an eight-draw evaluation.

1235 Conversely, not observing a mode does not establish extinction. At a fixed checkpoint, zero sightings
 1236 of a prespecified key in N independent draws give a one-sided $(1 - \alpha)$ binomial upper bound $1 - \alpha^{1/N}$
 1237 and lower bound zero [Clopper and Pearson, 1934]. For $N = 32$ and $\alpha = .05$, the upper bound is
 1238 approximately 8.94%. These are single-key bounds; simultaneous statements need error allocation.
 1239 Counts from changing checkpoints cannot be pooled to certify a current policy, and teacher-forced
 1240 scores are not independent Bernoulli observations.

1241 L.6 Scope relative to entropy, admission, and neural optimization

1242 **Exact entropy is a different comparator.** For a known full correct support and uniform u , the
 1243 elementary identities

$$H(q) = \log m - \text{KL}(q\|u), \quad R_u(p) = \log m - \log P + \text{KL}(u\|q)$$

1244 show that exact conditional entropy and categorical replay both favor a uniform conditional target.
 1245 With strictly increasing correctness potential Ψ , both $\Psi(P) + \beta H(q)$ and $\Psi(P) - \rho R_u(p)$ have the
 1246 unique global optimum $P = 1, q = u$ in probability space on the simplex with $P > 0$, for positive
 1247 coefficients. This is a boundary limit for finite-logit softmax policies. The optimum follows from
 1248 nonnegativity of KL and increasing Ψ ; it does not assert convergence of either neural update. The KL
 1249 direction distinguishes their loss level sets [Pereyra et al., 2017, Agarwal et al., 2021]: replay loss
 1250 diverges when a protected coordinate vanishes, whereas conditional entropy can remain $\log(m - 1)$
 1251 with one missing mode.

1252 Exact full-output entropy is a third objective and also regularizes incorrect mass. Exact tabular
 1253 entropy-regularized policy gradient has its own positive probability and convergence guarantees under
 1254 specified assumptions [Mei et al., 2020]; our collapse theorem contains no entropy term. The sampled
 1255 semantic scores in the experiments are not exact gradients of full-output or conditional verified-key
 1256 entropy. Neither their empirical comparison nor the bounded-score lemma is an impossibility result
 1257 for entropy.

1258 **Discovery and optimization remain separate requirements.** Retention starts after a verified key
 1259 enters active memory. Actual insertion depends on generation, validation, capacity, and eligibility;
 1260 a positive sampling probability alone does not establish eventual admission. Once a non-evicting
 1261 capacity-16 bank fills, an unbanked key can remain ineligible. Changing membership or exemplar
 1262 weights changes the loss being bounded, so fixed-bank descent cannot simply be carried across those
 1263 changes. In particular, the fixed-weight argument does not cover a target whose weights continue to
 1264 change with fresh observation counts.

1265 For a fixed joint potential, a discrete bound $F(\theta_{n+1}) \leq F(\theta_n) + \epsilon_n$ with $\epsilon_n \geq 0$ and $\sum_{n \geq n_0} \epsilon_n =$
 1266 $E < \infty$ would extend Lemma L.8, using $D = F(\theta_{n_0}) + J_{\max} + E$ for steps $n \geq n_0$. This follows
 1267 by telescoping and bounds retention, not convergence. The noisy optimizer would need a separate
 1268 argument establishing such control of the joint loss’s upward drift. A positive replay coefficient or
 1269 small nominal learning rate does not establish it. Adam moments, clipping, momentum, prompt
 1270 scheduling, and bank changes are not covered by the exact-flow proofs. The experiments have not
 1271 established the required joint-energy or update-error assumptions. The mathematical contribution
 1272 is the stated categorical mechanism and conditional exemplar argument; empirical support and
 1273 longitudinal exemplar scores remain separate measurements.

1274 **M Source Integrity and Reproducibility**

1275 **M.1 Admissible sources and missing checkpoints**

1276 Falcon3-1B Countdown ReplayDr.GRPO seed 59 contains two complete four-draw terminal evalu-
1277 ation sequences with identical request identities but conflicting metrics. The integrity amendment
1278 excludes that source at every checkpoint and from every efficacy endpoint, paired contrast, and
1279 aggregate that reuses it; neither sequence is selected or averaged. The independently valid Dr.GRPO
1280 and GRPO seed-59 sources remain admissible. The core replay comparison therefore has 74 pairs:
1281 14 complete five-seed blocks and Falcon Countdown at $n = 4$ (seeds 55–58). Primary partial
1282 intersections are descriptive and receive no confidence interval.

1283 Trajectory records use registered sources and approved continuations or replacements, with superseded
1284 attempts excluded. Identical duplicates are deduplicated; conflicting records invalidate the affected
1285 checkpoint rather than being selected by metric value. Missing or ambiguous checkpoints remain
1286 gaps; curves do not interpolate or carry values forward across them. An ambiguous nonterminal
1287 checkpoint does not invalidate an independently admissible terminal endpoint. Each curve declares
1288 its fixed source-admissible seed cohort; paired core curves use the admitted intersection. Accuracy
1289 and mode-count views use the same populations and gap rules. Supporting methods retain their
1290 explicit method-specific cohorts; available checkpoint counts and gaps are recorded alongside the
1291 curves. Machine-readable records identify the sources, exclusions, exact seed sets, checkpoint rule,
1292 and source hashes.

1293 **M.2 Data, models, and scientific records**

1294 The ModeBench dataset provides separate level/domain configurations, fixed split roles, full row
1295 identities, and construction and admission records. Its loading guide and evaluation guide identify
1296 the reported splits and preserve auxiliary data separately. The model archive links each available
1297 model–domain–method–seed export to an immutable upload revision containing original weights,
1298 configuration, tokenizer, file hashes, completion provenance, and the applicable base-model license.
1299 The experiment index maps the core scale comparisons, MaxRL factorial, Level-2 factorial, replay-
1300 weighting ablation, and supporting controls to their model families; the scientific-record index and
1301 restore guide identify preserved analysis inputs and SHA-256 verification procedures.

1302 Each empirical figure has a machine-readable record identified in the figure manifest, containing
1303 its displayed values and frozen numerical inputs. `paper/FIGURE_MANIFEST.md` maps both paper
1304 versions to their builders, records, and reproduction commands. Detailed source and recovery
1305 inventories remain in those records and the repository’s archived audit documents. Model availability
1306 may extend beyond the frozen analysis population; it does not change a figure’s source-admission
1307 or seed-intersection rule. Terminal exports restore inference models, while intermediate trajectories
1308 require the recorded evaluation checkpoints.

1309 **M.3 Exact continuation and validation**

1310 Exact training continuation requires the atomic optimizer, bank, token, cursor, and request-stream
1311 state in Appendix D.4, in addition to model weights. State mismatch, non-finite values, missing
1312 completion evidence, changed dataset identities, and source or aggregate hash drift are hard failures.
1313 Placement-only recovery may change hardware or wall time but must preserve the scientific update.
1314 Figure reproduction uses frozen records; missing results are never replaced by shallower checkpoints
1315 or silently incorporated from a newer archive.

1316 **N Limitations and Disclosures**

1317 The categorical theory depends on its stated geometry, energy, and coverage assumptions; it does not
1318 certify the implemented neural optimizer. Uniform correct mass requires full verified coverage, which
1319 Python’s capacity-16 bank generally precludes. The benchmark measures task-defined executable
1320 support on synthetic tasks, not free-form reasoning diversity or downstream utility. Experiments
1321 fix $G = 16$, zero reference KL, no token-entropy bonus, and one decoding protocol. The current
1322 results state every partial seed intersection and its uncertainty limits; intervals condition on the frozen
1323 prompts and evaluation requests.

1324 **N.1 AI Use Statement**

1325 Generative AI assisted manuscript preparation, mathematical analysis, and supporting code. The
1326 authors reviewed and verified the resulting content and take responsibility for the final manuscript.

1327 **N.2 Ethics Statement**

1328 The benchmark and experiment-specific data are synthetic; this study collects no human-subject or
1329 personal data. More verified modes are not inherently better: a distinct execution can be costly or
1330 risky despite passing a validator. Applications should report correctness, invalidity, and relevant costs
1331 alongside mode counts.

1332 **N.3 Reproducibility Statement**

1333 Claims use versioned results and fixed source-admission rules. Missing required results and aggregate
1334 drift cause validation to fail. Appendix B specifies split identities and evaluation settings; Appendix M
1335 links frozen scientific records, model exports, and reproduction procedures.